

Functions and Their Graphs

39. Find the domain and range of each of the functions determined by the formulas (A) through (H) and draw the graphs of those functions.

(A) $H(x) = \sqrt{4 - x^2}$.

(B) $V(x) = |x - 1|$.

(C) $h(x) = \frac{1}{x}$.

(D) $F(x) = \frac{1}{x - 1}$.

(E) $K(x) = -\frac{1}{2}x^3$.

(F) $G(x) = \frac{x^2 - 4}{x + 2}$.

(G) $H(x) = \begin{cases} x & \text{if } x \leq 2 \\ 4 & \text{if } x > 2 \end{cases}$.

(H) $f(x) = \frac{|x|}{x}$.

40. (A) Is Figure 5.1 the graph of a function?

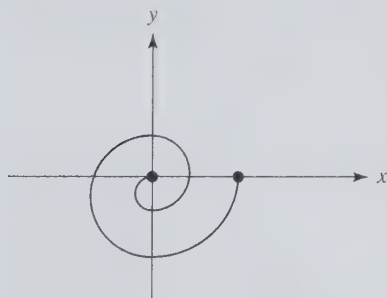


Figure 5.1

(B) Is Figure 5.2 the graph of a function?

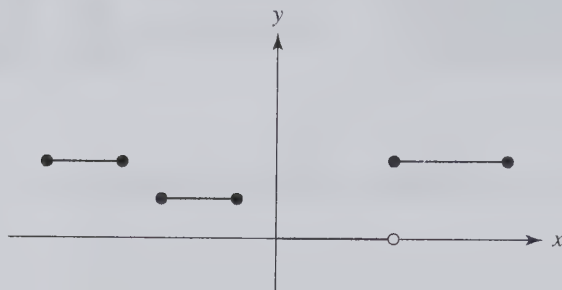


Figure 5.2

41. Specify the domain and range of each of the functions defined in (A) through (E) below.

(A) $f(x) = \frac{1}{(x-2)(x-3)}$.

(B) $g(x) = \frac{1}{\sqrt{1-x^2}}$.

(C) $h(x) = \begin{cases} x+1 & \text{if } -1 < x < 1 \\ 2 & \text{if } 1 \leq x \end{cases}$

(D) $F(x) = \begin{cases} x-1 & \text{if } 0 \leq x \leq 3 \\ x-2 & \text{if } 3 < x < 4 \end{cases}$

(E) $G(x) = |x| - x$.

42. Specify the domain and range of each of the functions defined in (A) through (C) below.

(A) $f(x) = \sqrt{x}$ for all nonnegative x .

(B) $f(x) = x^2$ for all x .

(C) $f(x) = \frac{1}{x}$ for all nonzero x .

43. For each of the functions $f(x)$ defined in (A) through (E) below, evaluate the expression $\frac{f(x+h)-f(x)}{h}$.
- (A) $f(x) = x^2 - 2x$.
 - (B) $f(x) = x + 4$.
 - (C) $f(x) = x^3 + 1$.
 - (D) $f(x) = \sqrt{x}$.
 - (E) $f(x) = \frac{1}{x}$.
44. For each of the one-one functions $f(x)$ in (A) and (B), find a formula for the inverse function $f^{-1}(y)$.
- (A) $f(x) = 2x + 1$.
 - (B) $f(x) = x^3$.
45. Find all real roots of each of the following polynomials.
- (A) $x^4 - 10x^2 + 9$.
 - (B) $x^3 + 2x^2 - 16x - 32$.
 - (C) $x^3 - 9x^2 + 26x + 24$.
 - (D) $x^3 - 5x - 2$.

CHAPTER 6

Limits

46. Define $\lim_{x \rightarrow a} f(x) = L$.

47. (A) Find $\lim_{u \rightarrow 5} \frac{u^2 - 25}{u - 5}$.

(B) Find $\lim_{x \rightarrow 1} \frac{x^3 - 1}{x - 1}$.

(C) Find $\lim_{u \rightarrow 0} \frac{5u^2 - 4}{u + 1}$.

(D) Find $\lim_{x \rightarrow 4} \frac{x^2 - x - 12}{x - 4}$.

(E) Find $\lim_{x \rightarrow 2} \frac{x^3 - 5x^2 + 2x - 4}{x^2 - 3x + 3}$.

(F) Find $\lim_{x \rightarrow 0} \frac{\sqrt{x+3} - \sqrt{3}}{x}$.

(G) Find $\lim_{x \rightarrow 3} \frac{1}{(x-3)^2}$.

(H) Find $\lim_{x \rightarrow 2} \frac{3}{x-2}$.

48. (A) Evaluate $\lim_{x \rightarrow +\infty} (2x^{11} - 5x^6 + 3x^2 + 1)$.

(B) Evaluate $\lim_{x \rightarrow -\infty} (2x^3 - 12x^2 + x - 7)$.

(C) Find $\lim_{x \rightarrow +\infty} \frac{2x+5}{x^2-7x+3}$.

(D) Find $\lim_{x \rightarrow +\infty} \frac{3x^3 - 4x + 2}{7x^3 + 5}$.

(E) Evaluate $\lim_{x \rightarrow +\infty} \frac{4x^5 - 1}{3x^3 + 7}$.

(F) Find $\lim_{x \rightarrow +\infty} \frac{4x - 1}{\sqrt{x^2 + 2}}$.

(G) Evaluate $\lim_{x \rightarrow +\infty} \frac{7x - 4}{\sqrt{x^3 + 5}}$.

(H) Evaluate $\lim_{x \rightarrow -\infty} \frac{3x^3 + 2}{\sqrt{x^4 - 2}}$.

49. (A) Find any vertical and horizontal asymptotes of the graph of the function $f(x) = (4x - 5)/(3x + 2)$.

(B) Find the vertical and horizontal asymptotes of the graph of the function $f(x) = (2x + 3)/\sqrt{x^2 - 2x - 3}$.

(C) Find the vertical and horizontal asymptotes of the graph of the function $f(x) = \sqrt{x + 1} - \sqrt{x}$.

50. Evaluate $\lim_{x \rightarrow 4^+} \frac{3}{x - 4}$ and $\lim_{x \rightarrow 4^-} \frac{3}{x - 4}$.

51. Evaluate $\lim_{x \rightarrow 3^+} \frac{1}{x^2 - 7x + 12}$ and $\lim_{x \rightarrow 3^-} \frac{1}{x^2 - 7x + 12}$.

52. Find $\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ when $f(x) = 4x^2 - x$.

53. Find $\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ when $f(x) = \sqrt{2x}$.

54. Evaluate $\lim_{x \rightarrow 2} \frac{\sqrt{x^2 + 5} - 3}{x^2 - 2x}$.

55. (A) Let $f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$ with $a_n > 0$. Prove that

$$\lim_{x \rightarrow +\infty} f(x) = +\infty.$$

(B) If $f(x) = \frac{a_n x^n + \cdots + a_1 x + a_0}{b_n x^n + \cdots + b_1 x + b_0}$, with $a_n \neq 0$ and $b_n \neq 0$, show that

$$\lim_{x \rightarrow \pm\infty} f(x) = \frac{a_n}{b_n}.$$

(C) If $f(x) = \frac{a_n x^n + \cdots + a_1 x + a_0}{b_k x^k + \cdots + b_1 x + b_0}$ with $a_n > 0$ and $b_k > 0$, prove that

$$\lim_{x \rightarrow +\infty} f(x) = +\infty \text{ if } n > k.$$

(D) If $f(x) = \frac{a_n x^n + \cdots + a_1 x + a_0}{b_k x^k + \cdots + b_1 x + b_0}$ with $a_n > 0$ and $b_k > 0$, and $n < k$, prove that

$$\lim_{x \rightarrow \pm\infty} f(x) = 0.$$

(E) Find $\lim_{x \rightarrow \pm\infty} \frac{7x^3 + 2x^2}{4x^3 - x}$.

(F) Find $\lim_{x \rightarrow +\infty} \frac{x^4 + 2x - 3}{x^2 + 100x}$.

(G) Find $\lim_{x \rightarrow \pm\infty} \frac{4x^3 + 20x^2}{x^4 - 1}$.

(H) Find $\lim_{x \rightarrow \pm\infty} \frac{3x - 4}{\sqrt[3]{x^2 - 1}}$.

(I) Find $\lim_{x \rightarrow +\infty} \frac{2x + 1}{\sqrt[3]{x^3 - 2}}$.

CHAPTER 7

Continuity

56. Define: $f(x)$ is continuous at $x = a$.
57. (A) Find the points of discontinuity (if any) of the function $f(x)$ whose graph is shown in Figure 7.1.

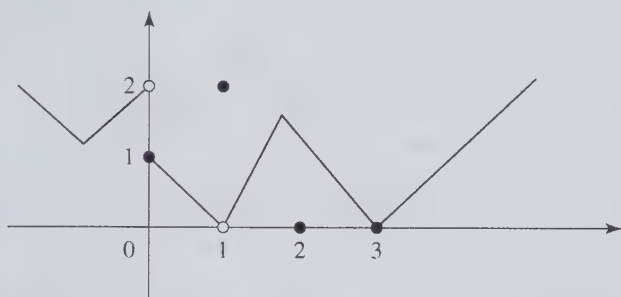


Figure 7.1

- (B) Determine the points of discontinuity (if any) of the function $f(x)$ such that $f(x) = x^2$ if $x \leq 0$ and $f(x) = x$ if $x > 0$.
- (C) Determine the points of discontinuity (if any) of the function $f(x)$ such that $f(x) = 1$ if $x \geq 0$ and $f(x) = -1$ if $x < 0$. (See Figure 7.2.)

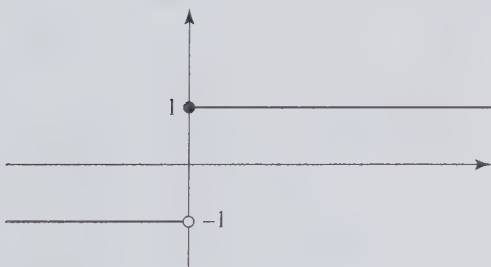


Figure 7.2

- (D) Determine the points of discontinuity (if any) of the function

$$f(x) \text{ such that } f(x) = \frac{x^2 - 4}{x + 2} \text{ if } x \neq -2 \text{ and } f(x) = 0 \text{ if } x = -2.$$

(See Figure 7.3.)

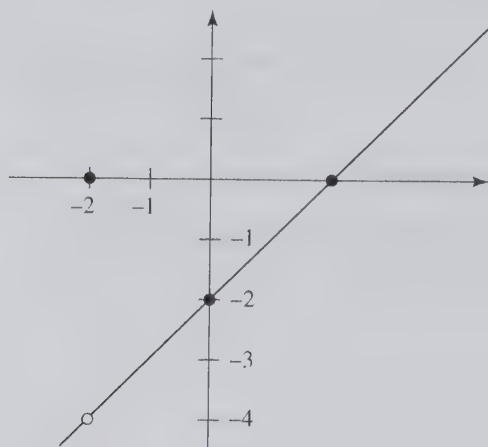


Figure 7.3

- (E) Find the points of discontinuity of the function $f(x) = \frac{x^2 - 1}{x - 1}$.

- (F) Find the points of discontinuity (if any) of the function $f(x)$ such

$$\text{that } f(x) = \frac{x^2 - 9}{x - 3} \text{ for } x \neq 3 \text{ and } f(x) = 6 \text{ for } x = 3.$$

- (G) Find the points of discontinuity (if any) of the function $f(x)$ such that

$$f(x) = \begin{cases} x + 1 & \text{if } x \geq 2 \\ 2x - 1 & \text{if } 1 < x < 2, \\ x - 1 & \text{if } x \leq 1 \end{cases}$$

(See Figure 7.4.)

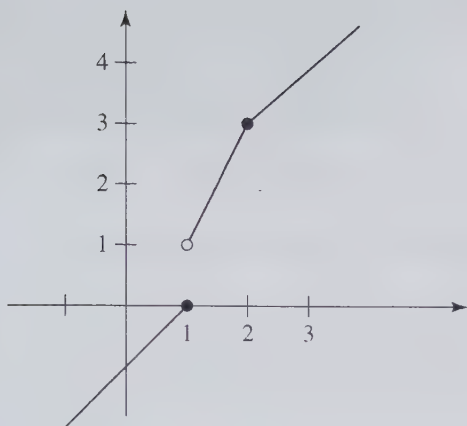


Figure 7.4

(H) Find the points of discontinuity (if any) of $f(x) = \frac{3x+3}{x^2-3x-4}$, and write an equation for each vertical and horizontal asymptote of the graph of f .

58. Define: (A) $f(x)$ is continuous on the left at $x = a$. (B) $f(x)$ is continuous on the right at $x = a$.
59. Consider the function $f(x)$ graphed in Figure 7.5. At all points of discontinuity, determine whether $f(x)$ is continuous on the left and whether $f(x)$ is continuous on the right.

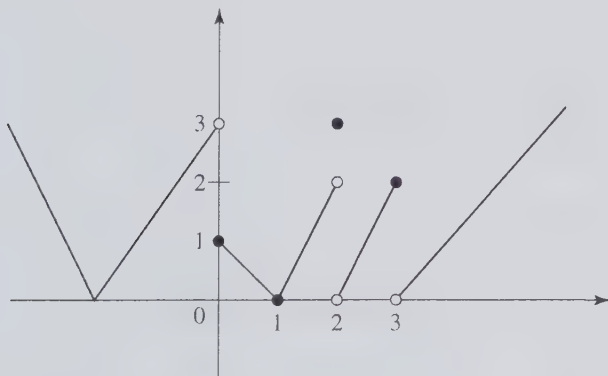


Figure 7.5

60. Show that the function $f(x) = 2x^3 - 4x^2 + 5x - 4$ has a zero between $x = 1$ and $x = 2$.
61. Verify the intermediate value theorem in the case of the function $f(x) = \sqrt{16 - x^2}$, the interval $[-4, 0]$, and the intermediate value $\sqrt{7}$.
62. (A) Is the function f such that $f(x) = \frac{1}{x}$ for $x > 0$ and $f(0) = 0$ continuous over $[0, 1]$?
 (B) Consider the function f such that $f(x) = 2x$ if $0 \leq x \leq 1$ and $f(x) = x - 1$ if $x > 1$. Is f continuous over $[0, 1]$?
 (C) Is the function of part (B) continuous over $[1, 2]$?

63. Let

$$f(x) = \begin{cases} 3x^2 - 1 & \text{if } x < 0 \\ cx + d & \text{if } 0 \leq x \leq 1 \\ \sqrt{x+8} & \text{if } x > 1 \end{cases}$$

Determine c and d so that f is continuous everywhere (as indicated in Figure 7.6).

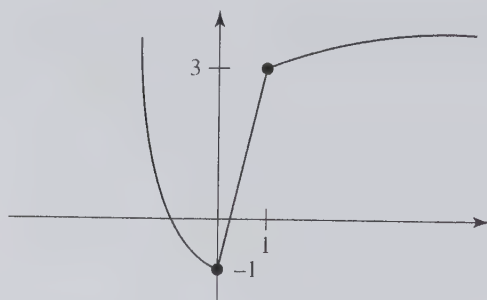


Figure 7.6

CHAPTER 9

The Chain Rule

74. If $f(x) = x^2 + 2x - 5$ and $g(x) = x^3$, find formulas for the composite functions $f \circ g$ and $g \circ f$.
75. Write the function $\sqrt{3x-5}$ as the composition of two functions.
76. If $f(x) = 2x$ and $g(x) = 1/(x-1)$, find all solutions of the equation $(f \circ g)(x) = (g \circ f)(x)$.
77. (A) Write the chain rule formula for the derivative of $f \circ g$.
(B) If $y = F(u)$ and $u = G(x)$, then we can write $y = F(G(x))$. Write the chain rule formula for dy/dx , where we think of y as a function of x .
(C) Find the derivative of $(x^3 - 2x^2 + 7x - 3)^4$.
(D) Find the derivatives of $\frac{1}{(3x^2 + 5)^4}$.
(E) Find the derivative of $\sqrt{2x+7}$.
(F) Find the derivative of $\left(\frac{x+2}{x-3}\right)^3$.
78. Find the derivative of $(4x^2 - 3)^2(x+5)^3$.
79. Find the slope-intercept equation of the tangent line to the graph of $y = \frac{\sqrt{x-1}}{x^2+1}$ at the point $\left(2, \frac{1}{5}\right)$.
80. Find the slope-intercept equation of the normal line to the curve $y = \sqrt{x^2+16}$ at the point $(3, 5)$.

81. If $y = x^3 - 2$ and $x = 3z^2 + 5$, then y can be considered a function of z . Express $\frac{dy}{dz}$ in terms of z .
82. Let F and G be differentiable functions such that $F(3) = 5$, $G(3) = 7$, $F'(3) = 13$, $G'(3) = 6$, $F'(7) = 2$, $G'(7) = 0$. If $H(x) = F(G(x))$, find $H'(3)$.
83. Find the derivations of $F(x) = \sqrt[3]{(1+x^2)^4}$.
84. A point moves along the curve $y = x^3 - 3x + 5$ so that $x = \frac{1}{2}\sqrt{t} + 3$, where t is time. At what rate is y changing when $t = 4$?

CHAPTER 10

Trigonometric Functions and Their Derivatives

85. Define radian measure, that is, describe an angle of 1 radian.
86. Give the equations relating degree measure and radian measure of angles.
87. Give the radian measure of angles of 30° , 45° , 60° , 90° , 120° , 135° , 180° , 270° , and 360° .
88. Give the degree measure of angles of $3\pi/5$ radians and $5\pi/6$ radians.
89. In a circle of radius 10 inches, what arc length along the circumference is intercepted by a central angle of $\pi/5$ radians?
90. Draw a picture of the rotation determining an angle of $-\pi/3$ radians.
91. Give the definition of $\sin \theta$ and $\cos \theta$.
92. State the values of $\cos \theta$ and $\sin \theta$ for $\theta = 0, \pi/6, \pi/4, \pi/3, \pi/2, \pi, 3\pi/2, 2\pi, 9\pi/4$.
93. Evaluate:
 - (A) $\cos(-\pi/6)$.
 - (B) $\sin(-\pi/6)$.
 - (C) $\cos(2\pi/3)$.
 - (D) $\sin(2\pi/3)$.
94. Sketch the graph of the cosine and sine functions.
95. Sketch the graph of $y = \cos 3x$.

96. Sketch the graph of $y = 1.5 \sin 4x$.
97. (A) Calculate $\lim_{x \rightarrow 0} \frac{\sin 3x}{x}$.
(B) Calculate $\lim_{x \rightarrow 0} \frac{\sin 2x}{\sin 3x}$.
(C) Calculate $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x}$.
98. (A) Using the Δ -definition, calculate $\frac{d}{dx}(\sin x)$.
(B) Calculate $\frac{d}{dx}(\cos x)$ from the known derivative of $\sin x$.
(C) Calculate $\frac{d}{dx}(\sin 3x)$.
(D) Calculate $\frac{d}{dx}(\cos^2 x)$.
(E) Calculate $\frac{d}{dx}(\sqrt{\sin x})$.
99. Find an equation of the tangent line to the graph of $y = \sin^2 x$ at the point where $x = \pi/3$.
100. (A) Derive the formula $\frac{d}{dx}(\tan x) = \sec^2 x$.
(B) Find an equation of the tangent line to the curve $y = \tan^2 x$ at the point $(\pi/3, 3)$.
(C) Show that $\frac{d}{dx}(\sec x) = \sec x \tan x$.
101. Find an equation of the normal line to the curve $y = 3 \sec^2 x$ at the point $(\pi/6, 4)$.
102. Evaluate $\lim_{x \rightarrow 0} \frac{\tan x}{x}$.
103. At what values of x does the graph of $y = \sec x$ have a horizontal tangent?

Rolle's Theorem, the Mean Value Theorem, and the Sign of the Derivative

104. State Rolle's theorem.

In Questions 105 through 107, determine whether the hypotheses of Rolle's theorem hold for the function $f(x)$ in the given interval, and, if they do, verify the conclusion.

105. $f(x) = x^2 - 2x - 3$ on $[-1, 3]$.

106. $f(x) = \frac{x^2 - x - 6}{x - 1}$ on $[-2, 3]$.

107. $f(x) = x^{2/3} - 2x^{1/3}$ on $[0, 8]$.

108. (A) State the mean value theorem.

For parts (B), (C), and (D), determine whether the hypotheses of the mean value theorem hold for the function $f(x)$ on the given interval, and, if they do, find a value c satisfying the conclusion of the theorem.

(B) $f(x) = 2x + 3$ on $[1, 4]$.

(C) $f(x) = 3x^2 - 5x + 1$ on $[2, 5]$.

(D) $f(x) = x^{3/4}$ on $[0, 16]$.

109. Prove that, if $f'(x) > 0$ for all x in the open interval (a, b) , then $f(x)$ is an increasing function on (a, b) .

110. For each function $f(x)$, determine where $f(x)$ is increasing and where it is decreasing.
- (A) $f(x) = 3x + 1$.
 - (B) $f(x) = x^2 - 4x + 7$.
 - (C) $f(x) = x^3 - 9x^2 + 15x - 3$.
 - (D) $f(x) = x + 1/x$.
111. Consider the polynomial $f(x) = 5x^3 - 2x^2 + 3x - 4$. Prove that $f(x)$ has a zero between 0 and 1 that is the only zero of $f(x)$.
112. Show that $x^3 + 2x - 5 = 0$ has exactly one real root.
113. (A) If $f'(x) = 0$ throughout an interval $[a, b]$, prove that $f(x)$ is constant on that interval.
- (B) If $f''(x) = g'(x)$ for all x in an interval $[a, b]$, show that there is a constant K such that $f(x) = g(x) + K$ for all x in $[a, b]$.

Higher-Order Derivatives and Implicit Differentiation

114. Find all derivatives $y^{(n)}$ of the function $y = \pi x^3 - 7x$.

115. Find all derivatives $y^{(n)}$ of the function $y = 1/(3 + x)$.

116. Find all derivatives $y^{(n)}$ of the function $y = \sin x$.

117. On the circle $x^2 + y^2 = a^2$, find y'' .

118. If $x^2 + 2xy + 3y^2 = 2$, find y' and y'' when $y = 1$.

119. Evaluate y'' on the ellipse $b^2x^2 + a^2y^2 = a^2b^2$.

Maxima and Minima

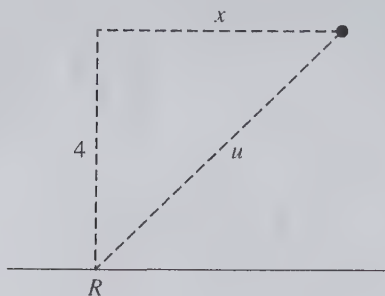
120. State the second-derivative test for relative extrema.
121. State the first-derivative test for relative extrema.
122. (A) Find the critical numbers of $f(x) = 5 - 2x + x^2$, and determine whether they yield relative maxima, relative minima, or inflection points.
- (B) Find the critical numbers of $f(x) = x^3 - 5x^2 - 8x + 3$, and determine whether they yield relative maxima, relative minima, or inflection points.
- (C) Find the critical numbers of $f(x) = x(x-1)^3$, and determine whether they yield relative maxima, relative minima, or inflection points.
- (D) Find the critical numbers of $f(x) = \sin x - x$, and determine whether they yield relative maxima, relative minima, or inflection points.
123. Find the critical numbers of $f(x) = (x-1)^{2/3}$, and determine whether they yield relative maxima, relative minima, or inflection points.
124. (A) Describe a procedure for finding the absolute maximum and absolute minimum values of a continuous function $f(x)$ on a closed interval $[a, b]$.
- (B) Find the absolute maximum and minimum of the function $f(x) = 4x^2 - 7x + 3$ on the interval $[-2, 3]$.
- (C) Find the absolute maximum and minimum of $f(x) = 4x^3 - 8x^2 + 1$ on the closed interval $[-1, 1]$.
- (D) Find the absolute maximum and minimum of $f(x) = x^4 - 2x^3 - x^2 - 4x + 3$ on the interval $[0, 4]$.
- (E) Find the absolute maximum and minimum of $f(x) = x^3/(x+2)$ on the interval $[-1, 1]$.
- (F) Find the absolute extrema of $f(x) = \sin x + x$ on $[0, 2\pi]$.

125. Show that $f(x) = (x - a_1)^2 + (x - a_2)^2 + \cdots + (x - a_n)^2$ has an absolute minimum when $x = (a_1 + a_2 + \cdots + a_n)/n$. (In words: the least-squares estimate of a finite set of numbers is their arithmetic mean.)
126. (A) Find the absolute maximum and minimum (if they exist) of $f(x) = x/(x^2 + 1)^{3/2}$ on $[0, +\infty)$.
- (B) Find the absolute maximum and minimum of $f(x) = 2 \sin x + \sin 2x$ on $[0, 2\pi]$.
- (C) Find the absolute maximum and minimum (if they exist) of $f(x) = (x + 2)/(x - 1)$.

CHAPTER 14

Related Rates

127. The top of a 25-foot ladder leaning against a vertical wall is slipping down the wall at the rate of 1 foot per second. How fast is the bottom of the ladder slipping along the ground when the bottom of the ladder is 7 feet away from the base of the wall?
128. A cylindrical tank of radius 10 feet is being filled with wheat at the rate of 314 cubic feet per minute. How fast is the depth of the wheat increasing? (The volume of a cylinder is $\pi r^2 h$, where r is its radius and h is its height.)
129. A 5-foot girl is walking toward a 20-foot lamppost at the rate of 6 feet per second. How fast is the tip of her shadow (cast by the lamp) moving?
130. A rocket is shot vertically upward with an initial velocity of 400 feet per second. Its height s after t seconds is $s = 400t - 16t^2$. How fast is the distance changing from the rocket to an observer on the ground 1,800 feet away from the launching site, when the rocket is still rising and is 2,400 feet above the ground?
131. A plane flying parallel to the ground at a height of 4 kilometers passes over a radar station R (Figure 14.1). A short time later, the radar equipment reveals that the distance between the plane and the station is 5 kilometers and that the distance between the plane and the station is increasing at a rate of 300 kilometers per hour. At that moment, how fast is the plane moving horizontally?

**Figure 14.1**

132. A boat passes a fixed buoy at 9 A.M. heading due west at 3 miles per hour. Another boat passes the same buoy at 10 A.M. heading due north at 5 miles per hour. How fast is the distance between the boats changing at 11:30 A.M.?
133. A revolving beacon is situated 3,600 feet off a straight shore. If the beacon turns at 4π radians per minute, how fast does its beam sweep along the shore at its nearest point A ?

CHAPTER 15

Curve Sketching (Graphs)

When sketching a graph, show all relative extrema, inflection points, and asymptotes. Indicate concavity, and suggest the behavior at infinity.

In Questions 134 and 135, determine the intervals where the graphs of the following functions are concave upward and where they are concave downward. Find all inflection points and draw the graphs.

134. $f(x) = x^3 + 15x^2 + 6x + 1.$

135. $f(x) = x^4 + 18x^3 + 120x^2 + x + 1.$

In Questions 136 through 141, find the critical numbers and determine whether they yield relative maxima, relative minima, or inflection points, or none of these.

136. $f(x) = x^3 - 5x^2 - 8x + 3.$

137. $f(x) = x^2/(x^2 + 1).$

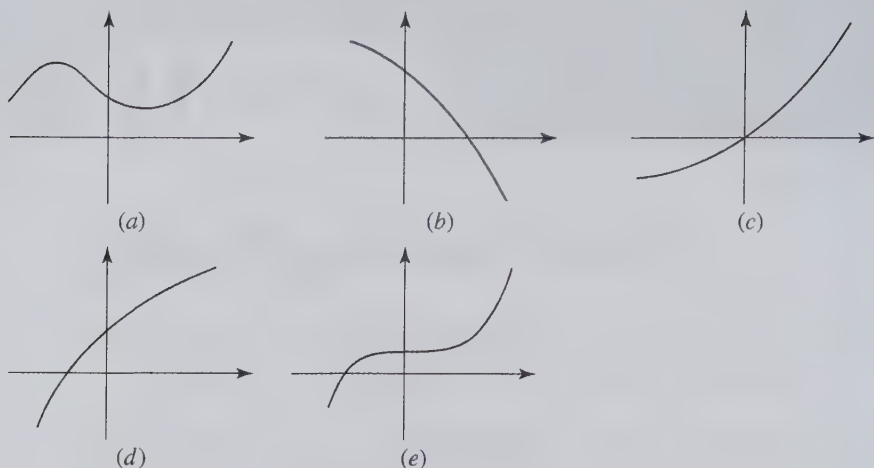
138. $f(x) = x^4 + 4x^3.$

139. $f(x) = 3x^5 - 20x^3.$

140. $f(x) = x^2 + 2/x.$

141. $f(x) = (x^2 - 3)/x^3.$

142. If for all x , $f'(x) > 0$ and $f''(x) < 0$, which of the curves in Figure 15.1 could be part of the graph of f ?

**Figure 15.1**

In Questions 143 through 147, sketch the graphs of the given functions.

143. $f(x) = \sin^2 x$.

144. $f(x) = \sin x + \cos x$.

145. $f(x) = \frac{x}{x^2 - 1}$.

146. $f(x) = \frac{3}{5}x^{5/3} - 3x^{2/3}$.

147. $f(x) = 3x^5 - 5x^3 + 1$.

Applied Maximum and Minimum Problems

148. Find the point(s) on the hyperbola $x^2 - y^2 = 2$ closest to the point $(0, 1)$ (Figure 16.1).

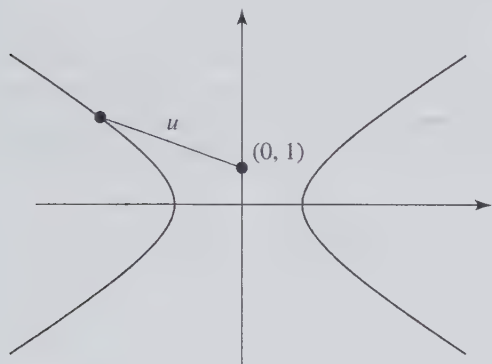


Figure 16.1

149. Find the dimensions of the closed cylindrical can that will have a capacity of k units of volume and will use the minimum amount of material. Find the ratio of the height h to the radius r of the top and bottom.
150. A man at a point P on the shore of a circular lake of radius 1 mile wants to reach the point Q on the shore diametrically opposite P (Figure 16.2). He can row 1.5 miles per hour and walk 3 miles per hour. At what angle θ ($0 \leq \theta \leq \pi/2$) to the diameter PQ should he row in order to minimize the time required to reach Q ?

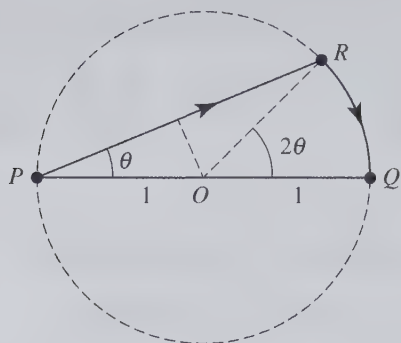


Figure 16.2

151. A wire 16 feet long has to be formed into a rectangle. What dimensions should the rectangle have to maximize the area?
152. A rectangular box with an open top is to be formed from a rectangular piece of cardboard that is 3 inches $\times$ 8 inches (Figure 16.3). What size square should be cut from each corner to form the box with maximum volume? (The cardboard is folded along the dotted lines to form the box.)

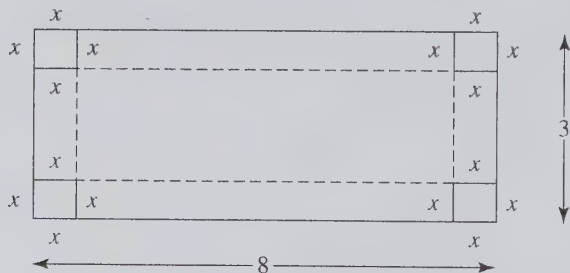


Figure 16.3

153. A telephone company has to run a line from a point A on one side of a river to another point B that is on the other side, 5 miles down from the point opposite A (Figure 16.4). The river is uniformly 12 miles wide. The company can run the line along the shoreline to a point C and then run the line under the river to B . The cost of laying the line along the shore is \$1000 per mile, and the cost of laying it underwater is twice as great. Where should the point C be located to minimize the cost?

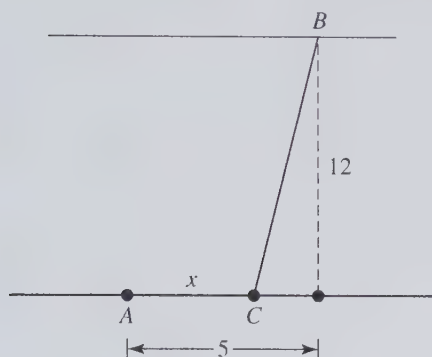


Figure 16.4

Rectilinear Motion

154. (A) The equation of the free fall of an object (under the influence of gravity alone) is $s = s_0 + v_0 t - 16t^2$, where s_0 is the initial position and v_0 is the initial velocity at time $t = 0$. (We assume that the s -axis is directed upward away from the earth, along the vertical line on which the object moves, with $s = 0$ at the earth's surface. s is measured in feet and t in seconds.) Show that, if an object is released from rest at any given height, it will have dropped $16t^2$ feet after t seconds.
- (B) How many seconds does it take the object released from rest to fall 64 feet?
- (C) A rock is dropped down a well that is 256 feet deep. When will it hit the bottom of the well?
- (D) Assuming that one story of a building is 10 feet, with what speed, in miles per hour, does an object dropped from the top of a 40-story building hit the ground?
155. An automobile moves along a straight highway, with its position s given by $s = 12t^3 - 18t^2 + 9t - 1.5$ (s in feet, t in seconds). When is the car moving to the right, when to the left, and where and when does it change direction?
156. A particle moves along the x -axis according to the equation $x = 10t - 2t^2$. What is the total distance covered by the particle between $t = 0$ and $t = 3$?
157. A rocket was shot straight up from the ground. What must its initial velocity have been if it returned to earth in 20 seconds?

158. Two particles move along the x -axis. Their positions $f(t)$ and $g(t)$ are given by $f(t) = 6t - t^2$ and $g(t) = t^2 - 4t$.
- (A) When do they have the same position?
 - (B) When do they have the same velocity?
 - (C) When they have the same position, are they moving in the same direction?
159. A particle moving on a line is at position $s = t^3 - 6t^2 + 9t - 4$ at time t . At which time(s) t , if any, does it change direction?
160. (A) A ball is thrown vertically upward. Its height s (in feet) after t seconds is given by $s = 40t - 16t^2$. Find (i) when the ball hits the ground, (ii) the instantaneous velocity at $t = 1$, and (iii) the maximum height.
- (B) An object is thrown straight up from the ground with an initial velocity v_0 feet per second. Show that the time taken on the upward flight is equal to the time taken on the way down.
- (C) Under the conditions of Question 160(B), show that the object hits the ground with the same speed at which it was initially thrown.
161. A stone is dropped from the roof of a building 256 feet high. Two seconds later a second stone is thrown downward from the roof of the same building with an initial velocity of v_0 feet per second. If both stones hit the ground at the same time, what is v_0 ?

CHAPTER 18

Approximation by Differentials

162. (A) State the approximation principle for a differentiable function $f(x)$.
In Questions 162(B) and (C), estimate the value of the given quantity.
(B) $\sqrt{51}$
(C) $\sqrt[3]{123}$
163. If the side of a cube is measured with an error of at most 3 percent, estimate the percentage error in the volume of the cube.
164. Assume, contrary to fact, that the earth is a perfect sphere, with a radius of 4,000 miles. The volume of ice at the north and south poles is estimated to be about 8,000,000 cubic miles. If this ice were melted and if the resulting water were distributed uniformly over the globe, approximately what would be the depth of the added water at any point on the earth?
165. Let $y = x^{3/2}$. When $x = 4$ and $dx = 2$, find the value of dy .
166. A cubical box is to be built so that it holds 125 cm^3 . How precisely should the edge be made so that the volume will be correct to within 3 cm^3 ?
167. Show that the relative error in the n th power of a number is about n times the relative error in the number.

CHAPTER 19

Antiderivatives (Indefinite Integrals)

168. (A) Evaluate $\int (g(x))^r g'(x) dx$.
(B) Evaluate $\int x^r dx$ for $r \neq -1$.
(C) Evaluate $\int (3 \sin x + 5 \cos x) dx$.
(D) Find $\int (7 \sec^2 x - \sec x \tan x) dx$.
(E) Evaluate $\int (\csc^2 x + 3x^2) dx$.
169. (A) Evaluate $\int \sqrt{7x+4} dx$.
(B) Evaluate $\int \frac{\cos 3x}{\sin^2 3x} dx$.
170. A particle moves along the x -axis with acceleration $a = 2t - 3$ feet per second square. At time $t = 0$ it is at the origin and moving with a speed of 4 ft/s in the positive direction. Find formulas for its velocity v and position s , and determine where it changes direction and where it is moving to the left.
171. A motorist applies the brakes on a car moving at 45 miles per hour on a straight road, and the brakes cause a constant deceleration of 22 feet per second square. In how many seconds will the car stop, and how many feet will the car have traveled after the time the brakes were applied?

172. Suppose that a particle moves along the x -axis, and its velocity at time t is given by $v = t^2 - t - 2$ for $1 \leq t \leq 4$. Find the total distance traveled in the period from $t = 1$ to $t = 4$.
173. Evaluate $\int \sec^5 x \tan x \, dx$.
174. Compute $\int \cos^2 x \, dx$.

CHAPTER 20

The Definite Integral and the Fundamental Theorem of Calculus

175. Show that $\int_0^b x \, dx = \frac{b^2}{2}$ by the direct definition of the integral.
176. For the function f graphed in Figure 20.1, express $\int_0^5 f(x) \, dx$ in terms of the areas A_1 , A_2 , and A_3 .

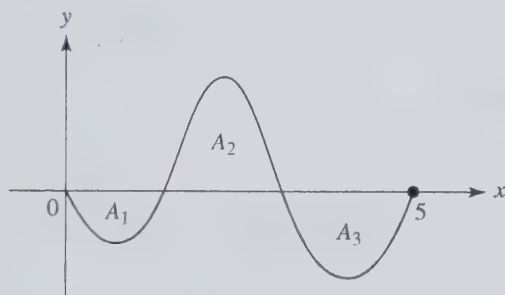


Figure 20.1

In Questions 177 and 179, evaluate the indicated definite integrals. In Question 178, find the definite integrals of the indicated functions on the given intervals.

177. (A) $\int_{-1}^3 (3x^2 - 2x + 1) \, dx$.
- (B) $\int_0^{\pi/4} \cos x \, dx$.

178. (A) $f(x) = \sin x$, $a = \pi/6$, $b = \pi/3$.
(B) $f(x) = x^2 + 4x$, $a = 0$, $b = 3$.
(C) $f(x) = \sin^2 x \cos x$, $a = 0$, $b = \pi/2$.
179. (A) $\int_0^{\pi/2} \cos x \sin x \, dx$.
(B) $\int_0^{\pi/4} \tan x \sec^2 x \, dx$.
(C) $\int_{-1}^2 \sqrt{x+2} \, x^2 \, dx$.
180. Find the average value of $f(x) = \sqrt[3]{x}$ on $[0, 1]$.
181. (A) State the mean-value theorem for integrals.
(B) Verify the mean-value theorem for integrals for the function $f(x) = x + 2$ on $[1, 2]$.
182. If, in a period of time T , an object moves along the x -axis from x_1 to x_2 , find a formula for its average velocity.
183. (A) Prove that, if f is continuous on $[a, b]$, $D_x \left[\int_a^x f(t) \, dt \right] = f(x)$.
(B) Find $D_x \left[\int_x^b f(t) \, dt \right]$.
184. The region above the x -axis and under the curve $y = \sin x$, between $x = 0$ and $x = \pi$, is divided into two parts by the line $x = c$. If the area of the left part is one-third the area of the right part, find c .
185. The velocity v of an object moving on the x -axis is $\cos 3t$, and the object is at the origin at $t = 0$. Find the average value of the position x over the interval $0 \leq t \leq \pi/3$.
186. (A) State the trapezoidal rule for approximation of integrals.
(B) State Simpson's rule for approximation of integrals.
(C) Apply Simpson's rule with $n = 4$ to approximate $\int_0^1 x^4 \, dx$.
(D) Use the trapezoidal rule with $n = 10$ to approximate $\int_0^1 x^2 \, dx$.

187. Use geometric reasoning to calculate $\int_{-a}^a \sqrt{a^2 - x^2} dx$.
188. Find the area inside the ellipse $x^2/a^2 + y^2/b^2 = 1$.
189. Find the volume of the solid generated when the region between the semicircle $y = 1 - \sqrt{1 - x^2}$ and the line $y = 1$ is rotated around the x -axis (see Figure 20.2).

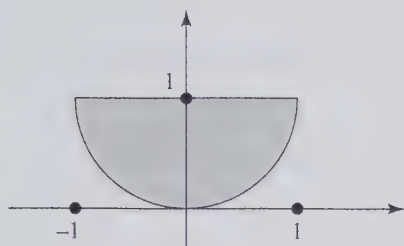


Figure 20.2

CHAPTER 21

Area and Arc Length

190. Sketch and find the area of the region to the left of the parabola $x = 2y^2$, to the right of the y -axis, and between $y = 1$ and $y = 3$.
191. Sketch and find the area of the region above the line $y = 3x - 2$ in the first quadrant, and below the line $y = 4$.
192. Find the area of the regions in parts (A) and (B).
(A) The bounded region between the parabola $x = -y^2$ and the line $y = x + 6$.
(B) The bounded region between the parabola $y = x^2 - x - 6$ and the line $y = -4$.

Find the arc lengths of the curves in Questions 193 and 194.

193. $y = \frac{x^4}{8} + \frac{1}{4x^2}$ from $x = 1$ to $x = 2$.
194. $x^{2/3} + y^{2/3} = 4$ from $x = 1$ to $x = 8$.
195. Find the area under the arch of $y = \sin x$ between $x = 0$ and $x = \pi$.

CHAPTER 22

Volume

196. Derive the formula $V = \frac{4}{3}\pi r^3$ for the volume of a sphere of radius r .
197. Derive the formula $V = \frac{1}{3}\pi r^2 h$ for the volume of a right circular cone of height h and radius of base r .
198. Find the volume generated by revolving the given region about the given axis.
- (A) The region above the curve $y = x^3$, under the line $y = 1$, and between $x = 0$ and $x = 1$, about the x -axis.
 - (B) The region of part (A), about the y -axis.
 - (C) The region inside the circle $x^2 + y^2 = r^2$ with $0 \leq x \leq a < r$, about the y -axis. (This gives the volume cut from a sphere of radius r by a pipe of radius a whose axis is a diameter of the sphere.)
 - (D) The region below the quarter-circle $x^2 + y^2 = r^2$ ($x \geq 0, y \geq 0$) and above the line $y = a$, where $0 < a < r$, about the y -axis. (This gives the volume of a polar cap of a sphere.)
 - (E) The region inside the circle $x^2 + (y - b)^2 = a^2$ ($0 < a < b$), about the x -axis. (This yields the volume of a doughnut.)

199. Use the cross-section formula to find the volume of the given solid.
- (A) The solid has a base which is a circle of radius r . Each cross-section perpendicular to a fixed diameter of the circle is an isosceles triangle with altitude equal to one-half of its base.
 - (B) The solid is a wedge, cut from a perfectly round tree of radius r by two planes, one perpendicular to the axis of the tree and the other intersecting the first plane at an angle of 30° along a diameter. (See Figure 22.1.)

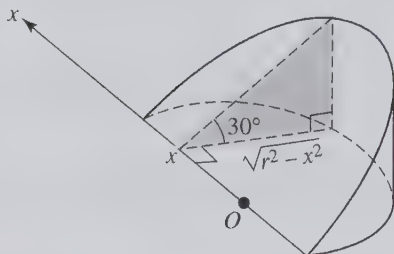


Figure 22.1

- (C) A square pyramid with a height of h units and a base of side r units.
 - (D) The tetrahedron formed by three mutually perpendicular edges of lengths a , b , c .
200. Let $\mathcal{R}$ be the region bounded by the curves $y = x^2 - 4x + 6$ and $y = x + 2$. Find the volume of the solid generated when $\mathcal{R}$ is rotated about the x -axis.
201. Find the volume of the solid generated when the region in the first quadrant under the hyperbola $xy = 1$, between $x = 1$ and $x = b > 1$, is rotated about the x -axis.
202. Find the volume of the ellipsoid obtained when the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is rotated
- (A) about the x -axis, and
 - (B) about the y -axis.

203. Find the volume of the solid obtained by rotating about the x -axis the region in the first quadrant under the line segment from $(0, r_1)$ to (h, r_2) , where $0 < r_1 < r_2$ and $0 < h$. (See Figure 22.2. Note that this is the volume of a frustum of a cone with height h and radii r_1 and r_2 of the bases.)

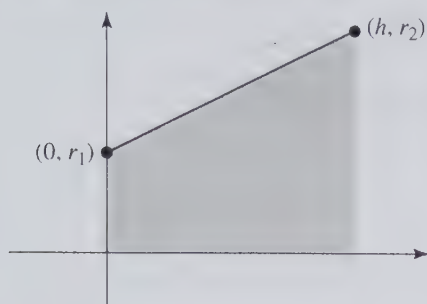


Figure 22.2

The Natural Logarithm

204. (A) State the definition of $\ln x$, and show that $D_x(\ln x) = \frac{1}{x}$.
 (B) Show that $\int \frac{1}{x} dx = \ln|x| + C$ for $x \neq 0$.
205. Find the derivative of the given function.
 (A) $\ln(4x - 1)$.
 (B) $\ln(\ln x)$.
 (C) $\ln\left(\frac{x-1}{x+1}\right)$.
206. Find the indicated antiderivative.
 (A) $\int \frac{1}{3x} dx$.
 (B) $\int \frac{1}{7x-2} dx$.
 (C) $\int \frac{1}{x \ln x} dx$.
 (D) $\int \frac{\ln x}{x} dx$.
207. (A) Show that $\int \frac{g'(x)}{g(x)} dx = \ln|g(x)| + C$.
 (B) Find $\int \frac{x}{3x^2+1} dx$.
 (C) Find $\int \tan x dx$.
208. Use logarithmic differentiation to find the derivative of $y = x^3 \sqrt{4-x^2}$.

209. Express the given numbers in terms of $\ln 2$ and $\ln 5$.

(A) $\ln 10$.

(B) $\ln \frac{1}{2}$.

(C) $\ln \frac{1}{5}$.

(D) $\ln 25$.

(E) $\ln \sqrt{2}$.

(F) $\ln \sqrt[3]{5}$.

(G) $\ln \frac{1}{20}$.

(H) $\ln 2^{12}$.

210. (A) Show that $1 - \frac{1}{x} \leq \ln x \leq x - 1$.

(B) Show that $\frac{\ln x}{x} < \frac{2}{\sqrt{x}}$.

(C) Prove that $\lim_{x \rightarrow +\infty} \frac{\ln x}{x} = 0$.

(D) Prove that $\lim_{x \rightarrow 0^+} x \ln x = 0$.

(E) Prove that $\lim_{x \rightarrow +\infty} (x - \ln x) = \infty$.

211. Use the trapezoidal rule, with $n = 10$, to approximate $\ln 2 = \int_1^2 \frac{1}{x} dx$.

212. Prove that $|\ln b - \ln a| < |b - a|$ for any distinct a, b in $[1, +\infty)$.

213. Prove the basic property of logarithms: $\ln uv = \ln u + \ln v$.

Exponential Functions

214. (A) Evaluate $e^{-\ln x}$.
 (B) Evaluate $\ln e^{-x}$.
 (C) Find $(e^2)^{\ln x}$.

In Question 215, find the derivative of the given function.

215. Find the derivative of the given function.
 (A) e^{-x} .
 (B) $e^{1/x}$.
 (C) $e^{\cos x}$.
 (D) x^π .
 (E) π^x .
 (F) $\ln e^{2x}$.
 (G) $e^x - e^{-x}$.

In Problem 216, evaluate the given antiderivative.

216. Evaluate the given antiderivative.
 (A) $\int e^{3x} dx$.
 (B) $\int e^{-x} dx$.
 (C) $\int e^{\cos x} \sin x dx$.
 (D) $\int a^x dx$, for $a \neq 1$.
 217. Find the derivative of $y = x^{\ln x}$.
 218. (A) Solve $e^{3x} = 2$ for x .
 (B) Solve $\ln x^3 = -1$ for x .
 (C) Solve $e^x - 2e^{-x} = 1$ for x .
 (D) Solve $\ln(\ln x) = 1$ for x .
 (E) Solve $\ln(x - 1) = 0$ for x .

219. (A) Let $\mathcal{R}$ be the region under the curve $y = e^x$, above the x -axis, and between $x = 0$ and $x = 1$. Find the area of $\mathcal{R}$.
 (B) Find the volume of the solid generated by rotating the region of Part (A) around the x -axis.

220. Find the absolute extrema of $y = e^{\sin x}$ on $[-\pi, \pi]$.

221. (A) Graph $y = e^{-x^2}$.

(B) Graph $y = x \ln x$.

(C) Graph $y = \frac{\ln x}{x}$.

(D) Sketch the graph of $y = e^{-x}$.

Question 222 refers to the function $\log_a x \equiv \frac{\ln x}{\ln a}$, the so-called *logarithm of x to the base a* . (Assume $a > 0$ and $a \neq 1$.)

222. (A) Show that $D_x(\log_a x) = \frac{1}{x \ln a}$.

(B) Show that $a^{\log_a x} = x$.

(C) Show that $\log_a a^x = x$.

(D) Show that $\log_e x = \ln x$.

(E) Show that $\log_a uv = \log_a u + \log_a v$.

(F) Show that $\log_a \frac{u}{v} = \log_a u - \log_a v$.

(G) Show that $\log_a u^r = r \log_a u$.

(H) Prove that $\ln x = \frac{\log_a x}{\log_a e}$.

223. Prove that the only solutions of the differential equation $f'(x) = f(x)$ are the functions Ce^x , where C is a constant.

224. Prove $e^x = \lim_{u \rightarrow +\infty} \left(1 + \frac{x}{u}\right)^u$.

225. Prove that, for any positive n , $\lim_{x \rightarrow +\infty} \frac{x^n}{e^x} = 0$.

226. Evaluate $\lim_{x \rightarrow 0^+} x^{\sin x}$.

227. Evaluate $\lim_{x \rightarrow 0} (\sin x)^{\cos x}$.

228. Show that $e = \lim_{u \rightarrow +\infty} \left(1 + \frac{1}{u}\right)^u$.

229. Graph $y = x^2 e^x$.

230. (A) Find $D_x(\sinh x)$ and $D_x(\cosh x)$.

(B) Find $D_x^2(\sinh x)$ and $D_x^2(\cosh x)$.

CHAPTER 25

L'Hôpital's Rule

231. State L'Hôpital's rule.

In Questions 232–240, evaluate the given limit.

232. $\lim_{x \rightarrow 0} \frac{\sin x}{x}.$

233. $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x}.$

234. $\lim_{x \rightarrow +\infty} \frac{5x^3 - 4x + 3}{2x^2 - 1}.$

235. $\lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{1}{\sin x} \right).$

236. $\lim_{x \rightarrow 0} \frac{\tan x}{x}.$

237. $\lim_{x \rightarrow 0} \frac{3^x - 2^x}{x}.$

238. $\lim_{x \rightarrow 0^+} x^{\sin x}.$

239. $\lim_{x \rightarrow 0} \frac{e^{3x} - 1}{\tan x}.$

240. $\lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3}.$

Exponential Growth and Decay

241. (A) A quantity y is said to *grow* or *decay exponentially* in time if $D_t y = Ky$ for some constant K . (K is called the *growth constant* or *decay constant*, depending on whether it is positive or negative.) Show that $y = y_0 e^{Kt}$, where y_0 is the value of y at time $t = 0$.
- (B) A bacteria culture grows exponentially so that the initial number has doubled in 3 hours. How many times the initial number will be present after 9 hours?
- (C) A certain chemical decomposes exponentially. Assume that 200 grams becomes 50 grams in 1 hour. How much will remain after 3 hours?
- (D) If a quantity y grows exponentially with a growth constant K and if during each unit of time there is an increase in y of r percent, find the relationship between K and r .
242. If a population is increasing exponentially at the rate of 2 percent per year, what will be the percentage increase over a period of 10 years?
243. (A) If an amount of money y_0 is invested at a rate of r percent per year, compounded n times per year, what is the amount of money that will be available after k years?
- (B) An amount of money y_0 earning r percent per year is compounded continuously (that is, assume that it is compounded n times per year, and then let n approach infinity). How much is available after k years?
- (C) If an amount of money earning 8 percent per year is compounded quarterly, what is the equivalent yearly rate of return?
- (D) If money is invested at 5 percent, compounded continuously, in how many years will it double in value?

244. (A) Assume that a quantity y decays exponentially, with a decay constant K . The *half-life* T is defined to be the time interval after which half of the original quantity remains. Find the relationship between K and T .
- (B) The half-life of radium is 1,690 years. If 10 percent of an original quantity of radium remains, how long ago was the radium created?
- (C) If radioactive carbon-14 has a half-life of 5,750 years, what will remain of 1 gram after 3,000 years?
- (D) If 20 percent of a radioactive element disappears in 1 year, compute its half-life.
245. If y represents the amount by which the temperature of a body exceeds that of the surrounding air, then the rate at which y decreases is proportional to y (Newton's law of cooling). If y was initially 8 degrees and was 7 degrees after 1 minute, what will it be after 2 minutes?
246. A tank initially contains 400 gallons of brine in which 100 pounds of salt are dissolved. Pure water is running into the tank at the rate of 20 gallons per minute, and the mixture (which is kept uniform by stirring) is drained off at the same rate. How many pounds of salt remain in the tank after 30 minutes?

Inverse Trigonometric Functions

247. Draw the graph of $y = \sin^{-1} x$.
248. Show that $D_x(\sin^{-1} x) = 1/\sqrt{1-x^2}$.
249. Draw the graph of $y = \tan^{-1} x$.
250. Show that $D_x(\tan^{-1} x) = 1/(1+x^2)$.
251. Find the indicated numbers.
- (A) $\cos^{-1}(-\sqrt{3}/2)$.
 - (B) $\sin^{-1}(\sqrt{2}/2)$.
 - (C) $\sin^{-1}(-\sqrt{2}/2)$.
 - (D) $\tan^{-1} 1$.
 - (E) $\tan^{-1}(\sqrt{3}/3)$.
252. Compute the indicated functional values.
- (A) $\sin\left(\cos^{-1}\frac{4}{5}\right)$.
 - (B) $\cos\left(\sin^{-1}\frac{3}{5} + \sec^{-1} 3\right)$.
 - (C) $\sin^{-1}(\sin \pi)$.
253. Evaluate the indicated antiderivatives.
- (A) $\int \frac{1}{4+x^2} dx$.
 - (B) $\int \frac{dx}{\sqrt{25-x^2}}$.

254. A person is viewing a painting hung high on a wall. The vertical dimension of the painting is 2 feet and the bottom of the painting is 2 feet above the eye level of the viewer. Find the distance x that the viewer should stand from the wall in order to maximize the angle θ subtended by the painting.

CHAPTER 28

Integration by Parts

In Questions 255 through 261, find the indicated antiderivative.

255. $\int x^2 e^{-x} dx.$

256. $\int e^x \sin x dx.$

257. $\int \sin^{-1} x dx.$

258. $\int x \sin x dx.$

259. $\int \sin^2 x dx.$

260. $\int x e^{3x} dx.$

261. Find $\int x^n \ln x dx$ for $n \neq -1$.

Trigonometric Integrands and Substitutions

In Questions 262 through 268, find the indicated antiderivatives.

262. $\int \sin x \cos^2 x \, dx.$

263. $\int \sin^4 x \cos^5 x \, dx.$

264. $\int \sin \pi x \cos 3\pi x \, dx.$

265. $\int \sin 5x \sin 7x \, dx.$

266. $\int \frac{x^2}{\sqrt{4-x^2}} \, dx.$

267. $\int \frac{\sqrt{1+x^2}}{x} \, dx.$

268. $\int \frac{dx}{x^2 \sqrt{x^2-9}}.$

269. Find the arc length of the curve $y = \ln \cos x$ from $(0, 0)$ to $(\pi/3, -\ln 2)$.

Integration of Rational Functions: The Method of Partial Fractions

In Questions 270 through 275, evaluate the indicated antiderivative.

270. $\int \frac{dx}{x^2 - 9}.$

271. $\int \frac{x - 5}{x^2(x + 1)} dx.$

272. $\int \frac{dx}{x(x^2 + 5)}.$

273. $\int \frac{dx}{(x^2 + 1)(x^2 + 4)}.$

274. $\int \frac{dx}{x(x^2 + 1)^2}.$

275. $\int \frac{dx}{1 + e^x}.$

276. Find $\int \frac{x^2 + 3}{(x - 1)^3(x + 1)} dx.$

Integrals for Surface Area, Work, Centroids

Surface Area of a Solid of Revolution

277. If the region under a curve $y = f(x)$, above the x -axis, and between $x = a$ and $x = b$, is revolved about the x -axis, state a formula for the surface area S of the resulting solid.
278. Find the surface area of a sphere of radius r .
279. Find the surface generated when the given arc is revolved about the given axis.
- (A) $y = x^3$, $0 \leq x < 1$; about the x -axis.
- (B) $y = \frac{1}{4}x^4 + \frac{1}{8x^2}$, $1 \leq x \leq 2$; about the y -axis.
280. Find the surface area of a right circular cone of height h and radius of base r .

Work

281. A spring with a natural length of 10 inches is stretched $\frac{1}{2}$ inch by a 12-pound force. Find the work done in stretching the spring from 10 to 18 inches.

Centroid of a Planar Region

In Questions 282 and 283, find the centroid of the given planar region.

282. The region bounded by $y = x^2$, $y = 0$, and $x = 1$ (See Figure 31.1).

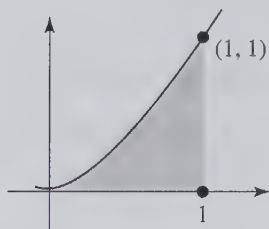


Figure 31.1

283. The region bounded by the semicircle $y = \sqrt{a^2 - x^2}$, and $y = 0$.

Improper Integrals

284. Determine whether the area in the first quadrant under the curve $y = 1/x$, for $x \geq 1$, is finite.
285. (A) Determine whether $\int_1^{\infty} (1/x^2) dx$ converges.
 (B) For what values of p is $\int_1^{\infty} (1/x)^p dx$ convergent?
286. (A) For $p > 1$, is $\int_1^{\infty} \frac{\ln x}{x^p} dx$ convergent?
 (B) For $p \leq 1$, is $\int_1^{\infty} \frac{\ln x}{x^p} dx$ convergent?
287. Evaluate $\int_a^{\infty} x e^{-x} dx$.
288. (A) Is $\int_e^{\infty} \frac{dx}{(\ln x)^p}$ convergent when $p \geq 1$?
 (B) If $\int_a^{\infty} f(x) dx = +\infty$ and $g(x) \geq f(x)$ for all $x \geq x_0$, show that $\int_a^{\infty} g(x) dx$ is divergent.
289. Show that $\int_e^{\infty} \frac{dx}{(\ln x)^p}$ is divergent for $p < 1$.
290. Evaluate $\int_0^{\infty} \frac{dx}{x^2 + a^2}$.
291. Evaluate $\int_0^{\infty} e^{-x} dx$.

292. (A) Investigate $\int_0^1 \frac{1}{x} dx$.

(B) Investigate $\int_0^1 \frac{1}{\sqrt{x}} dx$.

(C) Investigate $\int_0^1 \frac{1}{x^2} dx$.

(D) For what values of k , with $k \neq 1$ and $k > 0$, does $\int_0^1 \frac{1}{x^k} dx$ converge?

293. Evaluate $\int_1^3 \frac{dx}{x-2}$.

294. Evaluate $\int_{-1}^8 \frac{1}{x^{1/3}} dx$.

295. Evaluate $\int_0^1 \ln x dx$.

Planar Vectors

296. Find the vector from the point $A(1, -2)$ to the point $B(3, 7)$.
297. Given $\mathbf{A} = 3\mathbf{i} + 4\mathbf{j}$ and $\mathbf{C} = 2\mathbf{i} - \mathbf{j}$, find the magnitude and direction of $\mathbf{A} + \mathbf{C}$.
298. Describe a method for resolving a vector $\mathbf{A}$ into components $\mathbf{A}_1$ and $\mathbf{A}_2$ that are, respectively, parallel and perpendicular to a given nonzero vector $\mathbf{B}$.
299. Use vector methods to find the distance from $P(2, 3)$ to the line $3x + 4y - 12 = 0$. See Figure 33.1.

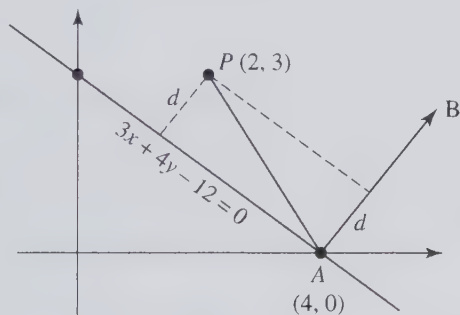


Figure 33.1

300. Generalize the method of Question 299 to find a formula for the distance from a point $P(x_1, y_1)$ to the line $ax + by + c = 0$.
301. Write the vector of length 2 and direction 150° in the form $a\mathbf{i} + b\mathbf{j}$.

302. Given $O(0, 0)$, $A(3, 1)$, and $B(1, 5)$ as vertices of the parallelogram $OAPB$, find the coordinates of P (see Figure 33.2).

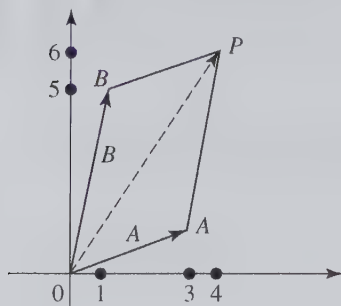


Figure 33.2

303. Find a vector perpendicular to the vector $(2, 5)$.

Parametric Equations, Vector Functions, Curvilinear Motion

Parametric Equations of Plane Curves

304. (A) Sketch the curve given by the parametric equations $x = a \cos \theta$, $y = a \sin \theta$.
 (B) Sketch the curve with the parametric equations $x = 2 \cos \theta$, $y = 3 \sin \theta$.
 (C) Sketch the curve with the parametric equations $x = t$, $y = t^2$.
305. Find the arc length of the curve $x = e^t \cos t$, $y = e^t \sin t$, from $t = 0$ to $t = \pi$.

Vector-Valued Functions

306. $\mathbf{F}(u) = (f(u), g(u))$ is a two-dimensional vector function, $\lim_{u \rightarrow a} \mathbf{F}(u) = (\lim_{u \rightarrow a} f(u), \lim_{u \rightarrow a} g(u))$, where the limit on the left exists if and only if the limits on the right exist. Taking this as the definition of vector convergence, show that $\mathbf{F}'(u) = (f'(u), g'(u))$.
307. Show that, if $\mathbf{R}(u)$ traces out a curve, then $\mathbf{R}'(u)$ is a tangent vector pointing in the direction of motion along the curve.
308. Show that, if $\mathbf{R}(t)$ traces out a curve and the parameter t represents time, then $\mathbf{R}'(t)$ is the velocity vector—that is, its direction is the direction of motion and its length is the speed.
309. If $h(u) = \mathbf{F}(u) \cdot \mathbf{G}(u)$, show that $h'(u) = \mathbf{F}(u) \cdot \mathbf{G}'(u) + \mathbf{F}'(u) \cdot \mathbf{G}(u)$, another analogue of the product formula for derivatives.

310. If $|\mathbf{R}(t)|$ is a constant $c > 0$, show that the tangent vector $\mathbf{R}'(t)$ is perpendicular to the position vector $\mathbf{R}(t)$.
311. For any vector function $\mathbf{F}(u)$ and scalar function $h(u)$, prove a chain rule:
$$\frac{d}{du}\mathbf{F}(h(u)) = h'(u)\mathbf{F}'(h(u)).$$
312. Assume that an object moves on a circle of radius r with constant speed $v > 0$. Show that the acceleration vector is directed toward the center of the circle and has length v^2/r .
313. (A) Let ϕ be the angle between the velocity vector and the positive x -axis. Show that $|d\mathbf{T}/d\phi| = 1$.
(B) Define the *curvature* κ and *radius of curvature* ρ of a curve $\mathbf{R}(t)$.
(C) For a circle of radius a , traced out in the counterclockwise direction, show that the curvature is $1/a$, and the radius of curvature is a , the radius of the circle.
(D) Find the curvature of a straight line $\mathbf{R}(t) = \mathbf{A} + t\mathbf{B}$.
(E) Show that, for a curve $y = f(x)$, the curvature is given by the formula $\kappa = y''/[1 + (y')^2]^{3/2}$. (We assume that $ds/dx > 0$ —that is, the arc length increases with x .)

Polar Coordinates

314. (A) Write the relations between polar coordinates (r, θ) and rectangular coordinates (x, y) .
(B) Give all possible polar representations of the point with rectangular coordinates $(1, 0)$.
315. (A) Describe the graph of the polar equation $r = 2$.
(B) Describe the graph of the polar equation $\theta = \pi/4$.
(C) Describe the graph of the polar equation $\theta = 0$.
316. Write a polar equation for the y -axis.
317. Describe the graph of the polar equation $r = 2 \sin \theta$.
318. Transform the rectangular equation $x = 3$ into a polar equation.
319. (A) Sketch the graph of $r = 1 + \cos \theta$.
(B) Sketch the graph of $r = 1 + 2 \cos \theta$.
(C) Sketch the graph of $r^2 = \cos 2\theta$.
(D) Sketch the graph of $r = \sin 2\theta$.
320. (A) Sketch the graph of $r = \sin 3\theta$.
(B) Sketch the graph of $r = \sin 4\theta$.
321. Find all points of intersection of the curves $r = 1 + \sin^2 \theta$ and $r = -1 - \sin^2 \theta$.
322. Find the area enclosed by the cardioid $r = 1 + \cos \theta$.
323. (A) Find the arc length of the spiral $r = \theta$ from $\theta = 0$ to $\theta = 1$.
(B) Find the arc length of the spiral $r = e^\theta$ from $\theta = 0$ to $\theta = \ln 2$.

CHAPTER 36

Infinite Sequences

In Questions 324 through 332, write a formula for the n th term a_n of the sequence and determine its limit (if it exists). It is understood that $n = 1, 2, 3, \dots$

$$324. \quad \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \frac{5}{6}, \dots$$

$$325. \quad \frac{2}{1}, \left(\frac{3}{2}\right)^2, \left(\frac{4}{3}\right)^3, \left(\frac{5}{4}\right)^4, \dots$$

$$326. \quad \ln \frac{2}{1}, \ln \frac{3}{2}, \ln \frac{4}{3}, \dots$$

$$327. \quad \frac{2}{1}, \frac{4}{2}, \frac{8}{6}, \frac{16}{24}, \frac{32}{120}, \dots$$

$$328. \quad 1, \sqrt{2}, \sqrt[3]{3}, \sqrt[4]{4}, \dots$$

$$329. \quad a_n = \frac{4n+5}{n^3-2n+3}.$$

$$330. \quad a_n = \frac{2n^5-3n+20}{5n^4+2}.$$

$$331. \quad a_n = \sqrt{n+1} - \sqrt{n}.$$

$$332. \quad a_n = 2n \sin \frac{\pi}{n}.$$

333. Show that $a_n = 2n/(3n + 1)$ is an increasing sequence.

334. (A) Determine whether the sequence $a_n = \frac{1 \cdot 3 \cdot 5 \cdot \dots \cdot (2n-1)}{2 \cdot 4 \cdot 6 \cdot \dots \cdot (2n)}$ is increasing, decreasing, or neither.

(B) Show that the sequence (a) of Question 334(A) is convergent.

CHAPTER 37

Infinite Series

335. Prove that, if $\sum a_n$ converges, then $\lim_{n \rightarrow +\infty} a_n = 0$.
336. (A) Show that the harmonic series $\sum 1/n = 1 + \frac{1}{2} + \frac{1}{3} + \dots$ diverges.
 (B) Does $\lim_{n \rightarrow +\infty} a_n = 0$ imply that $\sum a_n$ converges?
337. (A) Let $S_n = a + ar + \dots + ar^{n-1}$, with $r \neq 1$. Show that $S_n = \frac{a(r^n - 1)}{r - 1}$.
 (B) Let $a \neq 0$. Show that the infinite geometric series $\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r}$ if $|r| < 1$ and diverges if $|r| \geq 1$.
 (C) Evaluate $\sum_{n=0}^{\infty} \frac{1}{2^n} = 1 + \frac{1}{2} + \frac{1}{4} + \dots$.
 (D) Evaluate the infinite repeating decimal $d = 0.215626262 \dots$.
338. Investigate the series $\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \dots + \frac{1}{n(n+1)} + \dots$.
339. Find the sum of the series $4 - 1 + \frac{1}{4} - \frac{1}{16} + \dots$.
340. Test the convergence of $3 + \frac{5}{2} + \frac{7}{3} + \frac{9}{4} + \dots$.
341. Evaluate $\sum_{n=0}^{\infty} \frac{1}{n+100} = \frac{1}{100} + \frac{1}{101} + \dots$.

342. Zeno's paradox: Achilles (A) and a tortoise (T) have a race. T gets a 1000-ft head start, but A runs at 10 ft/s while the tortoise only does 0.01 ft/s. When A reaches T 's starting point, T has moved a short distance ahead. When A reaches that point, T again has moved a short distance ahead, etc. Zeno claimed that A would never catch T . Show that is not so.
343. Investigate $\sum_{n=1}^{\infty} \frac{3}{5^n + 2}$.
344. If $0 < p \leq 1$, show that the series $\sum_{n=1}^{\infty} \frac{1}{n^p} = 1 + \frac{1}{2^p} + \frac{1}{3^p} + \cdots$ is divergent.
345. Determine whether $\sum_{n=0}^{\infty} \frac{1}{n!}$ is convergent.
346. (A) State the integral test.
 (B) For $p > 1$, show that the so-called p -series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges.
 (Compare with Problem 344.)
 (C) Determine whether $\sum_{n=2}^{\infty} \frac{1}{n \ln n}$ converges.
347. (A) State the limit comparison test.
 (B) Determine whether $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n^3 + 3}}$ converges.
 (C) Determine whether $\sum_{n=1}^{\infty} \frac{n^3}{2n^4 + 1}$ is convergent.
348. Determine whether $\sum_{n=1}^{\infty} \frac{1}{n^n}$ is convergent.
349. Determine whether $\sum_{n=1}^{\infty} \frac{\ln n}{n}$ converges.

350. Give an example of a series that is conditionally convergent (that is, convergent but not absolutely convergent).
351. (A) State the ratio test for a series $\sum a_n$.
 (B) Determine whether $\sum_{n=1}^{\infty} \frac{n^2}{e^n}$ is convergent.
352. Find the error if the sum of the first three terms is used as an approximation to the sum of the alternating series $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n}$.
353. Study the convergence of $\sum_{n=0}^{\infty} (-1)^{n+1} \frac{2^n}{n!}$.
354. Determine whether $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{\sin(\pi/n)}{n^2}$ converges.
355. (A) Prove the *root test*: A series of positive terms $\sum a_n$ converges if $\lim_{n \rightarrow +\infty} \sqrt[n]{a_n} < 1$ and diverges if $\lim_{n \rightarrow +\infty} \sqrt[n]{a_n} > 1$.
 (B) Test $\sum_{n=2}^{\infty} \frac{1}{(\ln n)^n}$ for convergence.
356. Determine the n th term of and test for convergence the series $2 - \frac{2^3}{3!} + \frac{2^5}{5!} - \frac{2^7}{7!} + \cdots$.
357. Show how to rearrange the terms of the conditionally convergent series $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots$ so as to obtain a series whose sum is 1.
358. Show that the ratio test gives no information when $\lim |a_{n+1}/a_n| = 1$.

359. Show that, in Figure 37.1, the areas in the rectangles and above $y = 1/x$ add up to a number γ between $\frac{1}{2}$ and 1. (γ is called *Euler's constant*.)

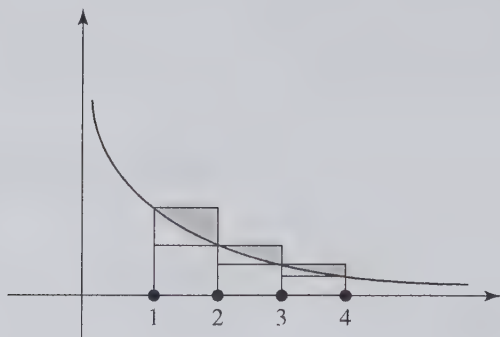


Figure 37.1

360. (A) Find the values of x for which the series $1 + x + x^2 + \cdots$ converges, and express the sum as a function of x .
- (B) Find the values of x for which the series $\ln x + (\ln x)^2 + (\ln x)^3 + \cdots$ converges, and express the sum as a function of x .

CHAPTER 38

Power Series

In Question 361, find the interval of convergence of the given power series. Use the ratio test unless otherwise instructed.

361. (A) $\sum x^n/n$.
 (B) $\sum x^n/n^2$.
 (C) $\sum x^n/n!$.
 (D) $\sum n!x^n$.
 (E) $\sum x^n/2^n$.
 (F) $\sum x^n/(n \cdot 2^n)$.
 (G) $\sum nx^n$.
362. Find the radius of convergence of the power series $\sum_{n=1}^{\infty} \frac{(n!)^2}{(2n)!} x^n$.
363. Prove that, if a power series $\sum a_n x^n$ converges for $x = b$, then it converges absolutely for all x such that $|x| < |b|$.
364. If $\lim_{n \rightarrow +\infty} \sqrt[n]{|a_n|} = L > 0$, show that the radius of convergence of $\sum a_n x^n$ is $1/L$.
365. Find the radius of convergence of the binomial series

$$1 + \frac{m}{1}x + \frac{m(m-1)}{1 \cdot 2}x^2 + \frac{m(m-1)(m-2)}{1 \cdot 2 \cdot 3}x^3 + \cdots$$
366. If $\sum a_n x^n$ has a radius of convergence r_1 and if $\sum b_n x^n$ has a radius of convergence $r_2 > r_1$, what is the radius of convergence of the sum $\sum (a_n + b_n)x^n$?

367. (A) Show that $\frac{1}{1+x} = 1 - x + x^2 - \cdots = \sum_{n=0}^{\infty} (-1)^n x^n$ for $|x| < 1$.
- (B) Show that $\frac{1}{1+x^2} = 1 - x^2 + x^4 - \cdots = \sum_{n=0}^{\infty} (-1)^n x^{2n}$ for $|x| < 1$.
- (C) Show that $\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1}$ for $|x| < 1$.
- (D) Find a power series representation for $\frac{1}{(1-x)^2}$.
- (E) Find a power series representation for $\ln(1+x)$ for $|x| < 1$.
368. (A) Show that $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$ for all x .
- (B) Find a power series representation for e^{-x} .
- (C) Find a power series representation for $e^{-x^2/2}$.
369. Approximate $1/e$ correct to two decimal places.
370. (A) Find a power series representation for $\ln(1-x)$.
- (B) Find a power series representation for $\ln \frac{1+x}{1-x}$.
- (C) Use the power series for $\ln \frac{1+x}{1-x}$ to approximate $\ln 2$.
371. (A) Show directly that, if $y'' = -y$, and $y' = 1$, and $y = 0$ when $x = 0$, then $y = \sin x$.
- (B) Show that $\sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}$.
- (C) Show that $\cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}$.
372. (A) Show that $\ln 2 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$.
- (B) Show that $\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1}$.
373. Find a power series for $\sin^2 x$.

374. For what values of x can $\sin x$ be replaced by x if the allowable error is 0.0005?
375. Find the first five terms of the power series for $e^x \cos x$ by multiplication of power series.
376. Let $f(x) = \sum_{n=0}^{\infty} a_n x^n$. Show that $\frac{1}{1-x} f(x) = \sum_{n=0}^{\infty} (a_0 + a_1 + \cdots + a_n) x^n$.
377. Use the result of Question 376 to evaluate $\sum_{n=0}^{\infty} (n+1) x^n$.
378. Evaluate $x/2! + x^2/3! + x^3/4! + x^4/5! + \cdots$.
379. (A) For the binomial series (Question 365), $f(x) = 1 + mx + \frac{m(m-1)}{2!} x^2 + \cdots + \frac{m(m-1)\cdots(m-n+1)}{n!} x^n + \cdots$, which is convergent for $|x| < 1$, show that $(1+x)f'(x) = mf(x)$.
- (B) Prove that the binomial series $f(x)$ of Question 379(A) is equal to $(1+x)^m$.
- (C) Show that $\sqrt{1-x} = 1 - \frac{1}{2}x - \frac{1}{2 \cdot 4}x^2 - \frac{1 \cdot 3}{2 \cdot 4 \cdot 6}x^3 - \cdots - \frac{1 \cdot 3 \cdots (2n-3)}{2^n \cdot n!}x^n - \cdots$.
- (D) Derive the series $\frac{1}{\sqrt{1-x}} = 1 + \frac{1}{2}x + \frac{1 \cdot 3}{2 \cdot 4}x^2 + \frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6}x^3 + \cdots$.
- (E) Obtain the series $\sin^{-1} x = x + \frac{1}{2} \frac{x^3}{3} + \frac{1 \cdot 3}{2 \cdot 4} \frac{x^5}{5} + \frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6} \frac{x^7}{7} + \cdots$, $|x| < 1$.
- (F) By means of the binomial series, approximate $\sqrt[5]{33}$ correct to three decimal places.
380. If infinitely many coefficients of a power series are nonzero integers, show that the radius of convergence $r \leq 1$.

Taylor and Maclaurin Series

381. (A) Find the Maclaurin series of e^x .
(B) Find the Maclaurin series for $\sin x$.
(C) Find the Maclaurin series for $\ln(1-x)$.
(D) Compute the first three nonzero terms of the Maclaurin series for $e^{\cos x}$.
382. (A) Find the Taylor series for $\sin x$ about $\pi/4$.
(B) Calculate the Taylor series for $1/x$ about 1.
383. (A) If $f(x) = \sum_{n=0}^{\infty} a_n(x-a)^n$ for $|x-a| < r$, prove that $a_k = \frac{f^{(k)}(a)}{k!}$. In other words, if $f(x)$ has a power series expansion about a , that power series must be the Taylor series for $f(x)$ about a .
(B) Find the Maclaurin series for $\frac{1}{1+x}$.
(C) Obtain the Maclaurin series for $\cos^2 x$.
384. State Taylor's formula with Lagrange's form of the remainder and indicate how it is used to show that a function is represented by its Taylor series.
385. Show that $e = \sum_{n=0}^{\infty} \frac{1}{n!}$.
386. Estimate the error when $\sqrt{e} = e^{1/2}$ is approximated by the first four terms of the Maclaurin series for e^x .

387. Use the Maclaurin series to estimate e to within two-decimal-place accuracy.
388. If $f(x) = \sum_{n=0}^{\infty} 2^n x^n$, find $f^{(33)}(0)$.
389. Approximate $\int_0^1 e^{-x^2} dx$ to within two-decimal-place accuracy.

Vectors in Space; Lines and Planes

390. (A) Find the equation of a sphere $\mathcal{S}$ of radius r and center (a, b, c) .
(B) Describe the graph of the equation $x^2 + 4x + y^2 + z^2 - 8z = 5$.
(C) When does an equation $x^2 + y^2 + z^2 + Ax + By + Cz + D = 0$ represent a sphere?
391. Show that the three points $P(1, 2, 3)$, $Q(4, -5, 2)$, and $R(0, 0, 0)$ are the vertices of a right triangle.
392. Show that the points $P(2, -1, 5)$, $Q(6, 0, 6)$, and $R(14, 2, 8)$ are collinear.
393. Describe the intersection of the graphs of $x^2 + y^2 = 1$ and $z = 2$.
394. Find the direction cosines of $\mathbf{A} = 3\mathbf{i} + 12\mathbf{j} + 4\mathbf{k}$. (Recall that $\mathbf{i} = (1, 0, 0)$, $\mathbf{j} = (0, 1, 0)$, and $\mathbf{k} = (0, 0, 1)$.)
395. Find the angle θ between the vectors $\mathbf{A} = (1, 2, 3)$ and $\mathbf{B} = (2, -3, -1)$.
396. Find a value of c for which $\mathbf{A} = 3\mathbf{i} - 2\mathbf{j} + 5\mathbf{k}$ and $\mathbf{B} = 2\mathbf{i} + 4\mathbf{j} + c\mathbf{k}$ will be perpendicular.
397. Write the formula for the cross product $\mathbf{A} \times \mathbf{B}$, where $\mathbf{A} = (a_1, a_2, a_3)$ and $\mathbf{B} = (b_1, b_2, b_3)$.
398. Find a vector $\mathbf{N}$ that is perpendicular to the plane of the three points $P(1, -1, 4)$, $Q(2, 0, 1)$, and $R(0, 2, 3)$.
399. Find the volume of the parallelepiped formed by the vectors $\overrightarrow{PQ}$ and $\overrightarrow{PR}$ of Question 398 and the vector $\overrightarrow{PS}$ where $S = (3, 5, 7)$.

400. Establish the formula

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

where $\mathbf{A} = (a_1, a_2, a_3)$, $\mathbf{B} = (b_1, b_2, b_3)$, $\mathbf{C} = (c_1, c_2, c_3)$.

401. Verify that $\mathbf{B} \times \mathbf{A} = -(\mathbf{A} \times \mathbf{B})$.

402. Show that the points $(0, 0, 0)$, (a_1, a_2, a_3) , (b_1, b_2, b_3) , and (c_1, c_2, c_3) are coplanar if and only if

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = 0.$$

403. (A) Find the vector representation, the parametric equations, and the rectangular equations for the line through the points $P(1, -2, 5)$ and $Q(3, 4, 6)$.

(B) Find the points at which the line of Problem 403(A) cuts the coordinate planes.

404. Write equations for the line through the point $(1, 2, -6)$ and parallel to the vector $(4, 1, 3)$.

405. By the methods of calculus, find the point $P_1(x, y, z)$ on the line $x = 3 + t$, $y = 2 + t$, and $z = 1 + t$ that is closest to the point $P_0(1, 2, 1)$, and verify that $\overline{P_0P_1}$ is perpendicular to the line.

406. Find an equation of the plane containing the point $P_1(3, -2, 5)$ and perpendicular to the vector $\mathbf{N} = (4, 2, -7)$.

407. Find an equation for the plane through the points $P(1, 3, 5)$, $Q(-1, 2, 4)$, and $R(4, 4, 0)$.

408. Find the cosine of the angle θ between the planes $4x + 4y - 2z = 9$ and $2x + y + z = -3$.

409. Find an equation of the plane containing the point $P(1, 3, 1)$ and the line $\mathcal{L}: x = t, y = t, z = t + 2$.
410. (A) Show that the distance D from the point $P(x_1, y_1, z_1)$ to the plane $ax + by + cz + d = 0$ is given by $D = |ax_1 + by_1 + cz_1 + d| / \sqrt{a^2 + b^2 + c^2}$.
- (B) Find the distance D from the point $(3, -5, 2)$ to the plane $8x - 2y + z = 5$.
411. Consider the sphere of radius 3 and center at the origin. Find the coordinates of the point P where the plane tangent to this sphere at $(1, 2, 2)$ cuts the x -axis.
412. Check that the planes $x - 2y + 2z = 1$ and $3x - y - z = 2$ intersect, and find their line of intersection.
413. Find an equation of a plane containing the intersection of the planes $3x - 2y + 4z = 5$ and $2x + 4y - z = 7$ and passing through the point $(2, 1, 2)$.
414. Find the coordinates of the point P at which the line $\frac{x+8}{9} = \frac{y-10}{-4} = \frac{z-9}{-2}$ cuts the plane $3x + 4y + 5z = 76$.
415. Find an equation of the line through $P(4, 2, -1)$ and perpendicular to the plane $6x - 3y + z = 5$.

Functions of Several Variables

Multivariate Functions and Their Graphs

416. (A) Describe and sketch the graph of $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ where $a, b, c > 0$.
- (B) Describe and sketch the graph of $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$ where $a, b, c > 0$.
- (C) Describe and sketch the graph of $\frac{z^2}{c^2} - \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ where $a, b, c > 0$.
- (D) Describe and sketch the graph of $\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{z^2}{c^2}$ where $a, b, c > 0$.
- (E) Describe and sketch the graph of the function $f(x, y) = x^2 + y^2$.
- (F) Describe and sketch the graph of the function $f(x, y) = 2x + 5y - 10$.
- (G) Describe and sketch the graph of $z = y^2 - x^2$.
417. Find the volume of the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$.
418. (A) Find $\lim_{(x, y) \rightarrow (0, 0)} \frac{2xy^2}{x^2 + y^2}$, if it exists.
- (B) Find $\lim_{(x, y) \rightarrow (0, 0)} \frac{2xy^2}{x^2 + y^4}$, if it exists.
- (C) Is it possible to define $f(x, y) = \frac{x^3 + y^3}{x^2 + y^2}$ at $(0, 0)$ so that $f(x, y)$ is continuous?

(D) Determine whether the function

$$f(x, y) = \begin{cases} \frac{xy}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

is continuous at the origin.

Cylindrical and Spherical Coordinates

419. (A) Give the equations connecting rectangular and cylindrical coordinates of a point in space.
 (B) Describe the surface with the cylindrical equation $r = k$.
 (C) Describe the surface with the cylindrical equation $\theta = k$.
 (D) Find cylindrical coordinates for the point with rectangular coordinates $(2, 2\sqrt{3}, 8)$.
 (E) Find a rectangular equation for the surface with the cylindrical equation $\theta = \pi/3$.
 (F) Find a rectangular equation for the surface with the cylindrical equation $r = 2 \sin \theta$.
420. (A) Write down the equations connecting spherical coordinates (ρ, ϕ, θ) with rectangular and cylindrical coordinates.
 (B) Describe the surface with the equation $\rho = k$ in spherical coordinates.
 (C) Describe the surface with the equation $\phi = k$ ($0 < k < \pi/2$) in spherical coordinates.
 (D) Find a set of spherical coordinates for the point whose rectangular coordinates are $(1, 1, \sqrt{6})$.
 (E) Find the rectangular coordinates of the point with spherical coordinates $(4, 2\pi/3, \pi/3)$.
421. Find a spherical equation for the surface whose rectangular equation is $x^2 + y^2 + z^2 + 6z = 0$.
422. Describe the surface whose equation in spherical coordinates is $\rho \sin \phi = 3$.

Partial Derivatives

423. If $f(x, y) = 4x^3 - 3x^2y^2 + 2x + 3y$, find the partial derivatives f_x and f_y .
424. Give an example of a function $f(x, y)$ such that $f_x(0, 0) = f_y(0, 0) = 0$, but f is not continuous at $(0, 0)$. Hence, the existence of the first partial derivatives does not ensure continuity.
425. (A) State a set of conditions under which the mixed partial derivatives $f_{xy}(x_0, y_0)$, and $f_{yx}(x_0, y_0)$ are equal.
(B) For $f(x, y) = 3x^2y - 2xy + 5y^2$, verify that $f_{xy} = f_{yx}$.
426. (A) For $f(x, y) = e^x \cos y$, verify that $f_{xy} = f_{yx}$.
(B) If $f(x, y) = 3x^2 - 2xy + 5y^3$, verify that $f_{xy} = f_{yx}$.
(C) If $f(x, y) = x^2 \cos y + y^2 \sin x$, verify that $f_{xy} = f_{yx}$.
427. (A) For $f(x, y) = 3x^4 - 2x^3y^2 + 7y$, find f_{xx} , f_{xy} , f_{yx} , and f_{yy} .
(B) If $f(x, y) = e^{xy}y^2 + \frac{x}{y}$, find f_{xx} , f_{xy} , f_{yx} , and f_{yy} .
(C) If $f(x, y, z) = x^2y + y^2z - 2xyz$, find f_{xy} , f_{yx} , f_{xz} , f_{zx} , f_{yz} , and f_{zy} .
428. Give an example to show that the equation $f_{xy} = f_{yx}$ is not always valid.
429. If $z = e^{ax} \sin ay$, show that $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 0$.
430. If $z = e^{-1}(\sin x + \cos y)$, show that $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = \frac{\partial z}{\partial t}$.
431. If $f(x, y) = g(x)h(y)$, show that $f_{xy} = f_{yx}$.

432. (A) Show that $f(x, t) = (x + at)^3$ satisfies the wave equation $a^2 f_{xx} = f_{tt}$.
 (B) Show that $f(x, t) = \sin(x + at)$ satisfies the wave equation $a^2 f_{xx} = f_{tt}$.
 (C) Show that $f(x, t) = e^{x-at}$ satisfies the wave equation $a^2 f_{xx} = f_{tt}$.
433. (A) If $u = f(x_1, \dots, x_n)$ and $x_1 = h_1(t), \dots, x_n = h_n(t)$, state the chain rule for du/dt .
 (B) If $u = x^2 - 2y^2 + z^3$ and $x = \sin t, y = e^t, z = 3t$, find du/dt .
 (C) If $w = \Phi(x, y, z)$ and $x = f(u, v), y = g(u, v), z = h(u, v)$, state the chain rule for $\partial w/\partial u$ and $\partial w/\partial v$.
434. How fast is the volume V of a rectangular box changing when its length l is 10 feet and increasing at the rate of 2 feet per second, its width w is 5 feet and decreasing at the rate of 1 foot per second, and its height h is 3 feet and increasing at the rate of 2 feet per second?
435. (A) Prove Euler's theorem: if $f(x, y)$ is homogeneous of degree n , then $xf_x + yf_y = nf$. (Recall that $f(x, y)$ is homogeneous of degree n if and only if $f(tx, ty) = t^n f(x, y)$ for all x, y and for all $t > 0$.)
 (B) Verify Euler's theorem (Question 435(A)) for the function $f(x, y) = xy^2 + x^2y - y^3$.
 (C) Verify Euler's theorem (Question 435(A)) for the function $f(x, y) = \sqrt{x^2 + y^2}$.
 (D) Verify Euler's theorem (Question 435(A)) for the function $f(x, y, z) = 3xz^2 - 2xyz + y^2z$.
436. If $z = 2x^2 - 3xy + 7y^2, x = \sin t, y = \cos t$, find dz/dt .
437. If $z = \ln(x^2 + y^2), x = e^{-t}, y = e^t$, find dz/dt .
438. If $z = f(x, y) = x^4 + 3xy - y^2$ and $y = \sin x$, find dz/dx .
439. If $z = f(x, y) = xy^2 + x^2y$ and $y = \ln x$, find dz/dx and dz/dy .
440. If the radius r of a right circular cylinder is increasing at the rate of 3 inches per second and the altitude h is increasing at the rate of 2 inches per second, how fast is the surface area S changing when $r = 10$ inches and $h = 5$ inches?
441. If a point is moving on the curve of intersection of $x^2 + 3xy + 3y^2 = z^2$ and the plane $x - 2y + 4 = 0$, how fast is it moving when $x = 2$, if x is increasing at the rate of 3 units per second?

442. (A) Prove Leibniz's formula for differentiable functions $u(x)$ and $v(x)$,
(B) Verify Leibniz's formula (Question 442(A)) for $u = x$, $v = x^2$, and $f(x, y) = x^3y^2 + x^2y^3$.
443. (A) Show that the tangent plane to a surface $z = f(x, y)$ at a point (x_0, y_0, z_0) has a normal vector $(f_x(x_0, y_0), f_y(x_0, y_0), -1)$.
(B) If a surface has the equation $F(x, y, z) = 0$, show that a normal vector to the tangent plane at (x_0, y_0, z_0) is $(F_x(x_0, y_0, z_0), F_y(x_0, y_0, z_0), F_z(x_0, y_0, z_0))$.
444. Find an equation of the tangent plane to the sphere $x^2 + y^2 + z^2 = 1$ at the point $\left(\frac{1}{2}, \frac{1}{2}, 1/\sqrt{2}\right)$.
445. Find an equation of the tangent plane to $z^3 + xyz - 2 = 0$ at $(1, 1, 1)$.
446. Find an equation of the tangent plane to the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ at a point (x_0, y_0, z_0) .

Directional Derivatives and the Gradient; Extreme Values

447. Given a function $f(x, y)$ and a unit vector $\mathbf{u}$, define the *derivative of f in the direction $\mathbf{u}$* , at a point (x_0, y_0) , and state its connection with the gradient $\nabla f = (f_x, f_y)$.
448. Show that the direction of the gradient ∇f is the direction in which the derivative achieves its maximum, $|\nabla f|$, and the direction of $-\nabla f$ is the direction in which the derivative achieves its minimum, $-|\nabla f|$.
449. Find the derivative of $f(x, y) = 2x^2 - 3xy + 5y^2$ at the point $(1, 2)$ in the direction of the unit vector $\mathbf{u}$ making an angle of 45° with the positive x -axis.
450. Find the derivative of $z = x \ln y$ at the point $(1, 2)$ in the direction making an angle of 30° with the positive x -axis.
451. If the temperature is given by $f(x, y, z) = 3x^2 - 5y^2 + 2z^2$ and you are located $\left(\frac{1}{3}, \frac{1}{5}, \frac{1}{2}\right)$ and want to get cool as soon as possible, in which direction should you set out?
452. Prove that the gradient ∇F of a function $F(x, y, z)$ at a point $P(x_0, y_0, z_0)$ is perpendicular to the level surface of F going through P .
453. In what direction should one initially travel, starting at the origin, to obtain the most rapid rate of increase of the function $f(x, y, z) = (3 - x + y)^2 + (4x - y + z + 2)^3$?
454. If $z = f(x, y)$ has a relative maximum or minimum at a point (x_0, y_0) , show that $\nabla z = \mathbf{0}$ at (x_0, y_0) .

455. (A) Assume that $f(x, y)$ has continuous second partial derivatives in a disk containing the point (x_0, y_0) inside it. Assume also that (x_0, y_0) is a critical point of f —that is, $f_x = f_y = 0$ at (x_0, y_0) . Let $\Delta \equiv f_{xx}f_{yy} - (f_{xy})^2$ (the Hessian determinant). State sufficient conditions for f to have a relative maximum or minimum at (x_0, y_0) .
- (B) Using the assumptions and notation of Question 455(A), give examples to show that Case 3, where $\Delta = 0$, allows no conclusions to be drawn.
456. Find the relative maxima and minima of the function $f(x, y) = 2x + 4y - x^2 - y^2 - 3$.
457. Determine the values of p and q so that the sum S of the squares of the vertical distances of the points $(0, 2)$, $(1, 3)$, and $(2, 5)$ from the line $y = px + q$ shall be a minimum (method of least squares).
458. Find the absolute maximum and minimum of $f(x, y) = 4x^2 + 2xy - 3y^2$ on the unit square $0 \leq x \leq 1$, $0 \leq y \leq 1$.
459. Find the absolute extrema of $f(x, y) = \sin x + \sin y + \sin(x + y)$.
460. Find the point(s) on the sphere $x^2 + y^2 + z^2 = 1$ farthest from the point $(2, 1, 2)$.
461. Find the point(s) on the cone $x^2 = y^2 + z^2$ nearest the point $(0, 1, 3)$.

Multiple Integrals and Their Applications

462. (A) Evaluate the iterated integral $I = \int_0^{\pi/2} \int_0^{\cos \theta} \rho^2 \sin \theta \, d\rho \, d\theta$.
 (B) Evaluate $I = \int_0^1 \int_y^1 e^{x^2} \, dx \, dy$.
463. Evaluate $I = \int_0^1 \int_{y^4}^{y^2} \sqrt{y/x} \, dx \, dy$.
464. (A) Evaluate $I = \iint_{\mathcal{R}} x \, dA$, where $\mathcal{R}$ is the region bounded by $y = x$ and $y = x^2$.
 (B) Evaluate $I = \iint_{\mathcal{R}} y^2 \, dA$, where $\mathcal{R}$ is the region bounded by $y = 2x$, $y = 5x$, and $x = 2$.
465. Find the volume V under the plane $z = 3x + 4y$ and over the rectangle $\mathcal{R}$ $1 \leq x \leq 2$, $0 \leq y \leq 3$.
466. Find the volume V in the first octant bounded by $z = y^2$, $x = 2$, and $y = 4$.
467. Find $I = \iint_{\mathcal{R}} \sin \theta \, dA$, if $\mathcal{R}$ is the region outside the circle $r = 1$ and inside the cardioid $r = 1 + \cos \theta$ (see Figure A44.4 in the Answers section).
468. Use cylindrical coordinates to calculate the volume of a sphere of radius a .

469. Show that $\int_0^{\infty} e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$.
470. Use spherical coordinates to find the volume of a sphere of radius a .
471. Find the mass of a circular plate $\mathcal{R}$ of radius a whose density is numerically equal to the distance from the center.
472. Find the center of mass of the first-quadrant part of the disk of radius a with center at the origin, if the density function is y .
473. Find the moments of inertia of the triangle bounded by $3x + 4y = 24$, $x = 0$, and $y = 0$, and having density 1.
474. Find the centroid $(\bar{x}, \bar{y})$ of the region in the first quadrant bounded by $y^2 = 6x$, $y = 0$, and $x = 6$ (Figure A44.5 in the Answers section).

CHAPTER 45

Vector Functions in Space; Divergence and Curl; Line Integrals

475. (A) For the space curve $\mathbf{R}(t) = (t, t^2, t^3)$, find a tangent vector and the tangent line at the point $(1, 1, 1)$.
 (B) Find the speed of a particle tracing the curve of Question 475(A) at time $t = 1$. (The parameter t is usually, but not necessarily, interpreted as the time.)
476. Find a tangent vector, the tangent line, the speed, and the normal plane to the helical curve $\mathbf{R}(t) = (a \cos 2\pi t, a \sin 2\pi t, bt)$ at $t = 1$.
477. Prove that the angle θ between a tangent vector and the positive z -axis is the same for all points of the helix of Question 476.
478. If $\mathbf{F}(t) = (\sin t, \cos t, t)$ and $\mathbf{G}(t) = (t, 1, \ln t)$, find $\frac{d}{dt}[\mathbf{F}(t) \cdot \mathbf{G}(t)]$.
479. If $\mathbf{G}(3) = (1, 1, 2)$ and $\mathbf{G}'(3) = (3, -2, 5)$, find $\frac{d}{dt}[t^2 \mathbf{G}(t)]$ at $t = 3$.
480. Let $\mathbf{F}(t) = \mathbf{G}(t) \times \mathbf{H}'(t)$. Prove $\mathbf{F}' = (\mathbf{G} \times \mathbf{H}') + (\mathbf{G}' \times \mathbf{H})$.
481. Prove that $\frac{d}{dt}[\mathbf{R}(t) \times \mathbf{R}'(t)] = \mathbf{R}(t) \times \mathbf{R}''(t)$.
482. If $\mathbf{G}'(t)$ is perpendicular to $\mathbf{G}(t)$ for all t , show that $|\mathbf{G}(t)|$ is constant—that is, the point $\mathbf{G}(t)$ lies on the surface of a sphere with center at the origin.

483. Derive the converse of Question 482: if $|\mathbf{G}(t)|$ is constant, then $\mathbf{G}(t) \cdot \mathbf{G}'(t) = 0$.
484. Find the principal unit normal vector $\mathbf{N}$ to the curve $\mathbf{R}(t) = (t, t^2, t^3)$ when $t = 1$.
485. Find a formula for the binormal vector $\mathbf{B} = \mathbf{T} \times \mathbf{N}$ in the case of the helix $\mathbf{R}(t) = (3 \cos 2t, 3 \sin 2t, 8t)$.
486. Find an equation of the *osculating plane* to the curve $\mathbf{R}(t) = (2t - t^2, t^2, 2t + t^2)$ at $t = 1$.
487. Show that a normal vector to the osculating plane of a curve $\mathbf{R}(t)$ is given by $\mathbf{R}' \times \mathbf{R}''$.
488. Find the curvature κ of the curve $\mathbf{R}(t) = \left(\sin t, \cos t, \frac{1}{2}t^2 \right)$ at $t = 0$.
489. Prove the following formula for the curvature: $\kappa = \frac{|\mathbf{R}' \times \mathbf{R}''|}{|\mathbf{R}'|^3}$.
490. Compute the divergence, $\text{div } \mathbf{F}$, for the vector field $\mathbf{F} = (xy, yz, xz)$.
491. For any vector field $\mathbf{F}(x, y, z) = (f(x, y, z), g(x, y, z), h(x, y, z))$, define $\text{curl } \mathbf{F}$.
492. Find $\text{curl } \mathbf{F}$ when $\mathbf{F} = (yz, xz, xy)$.
493. Show that, for any scalar function $f(x, y, z)$, $\text{div } \nabla f = f_{xx} + f_{yy} + f_{zz}$. (The latter sum, called the *Laplacian* of f , is often notated as $\nabla^2 f$.)
494. For any scalar function $f(x, y, z)$ with continuous mixed second partial derivatives, show that $\text{curl } \nabla f = 0$.
495. For a vector field $\mathbf{F}(x, y, z) = (f(x, y, z), g(x, y, z), h(x, y, z))$, where f, g , and h have continuous mixed second partial derivatives, show that $\text{div } \text{curl } \mathbf{F} = 0$.
496. For a scalar field f and a vector field $\mathbf{F}$, prove the product rule $\text{div } (f\mathbf{F}) = f \text{div } \mathbf{F} + \nabla f \times \mathbf{F}$.

497. For a scalar field f and a vector field $\mathbf{F} = (\phi, \psi, \eta)$, prove the product rule $\text{curl}(f\mathbf{F}) = f \text{curl } \mathbf{F} + \nabla f \times \mathbf{F}$.
498. For vector fields $\mathbf{F} = (f, g, h)$ and $\mathbf{G} = (\phi, \psi, \eta)$, prove $\text{div}(\mathbf{F} \times \mathbf{G}) = \text{curl } \mathbf{F} \cdot \mathbf{G} - \mathbf{F} \cdot \text{curl } \mathbf{G}$.
499. Find the work W done by a force $\mathbf{F} = (xy, yz, xz)$ acting on an object moving along the curve $\mathbf{R}(t) = (t, t^2, t^3)$ for $0 \leq t \leq 1$.
500. Let $\mathcal{C}$ be any curve in the xy -plane. Show that $\int_{\mathcal{C}} y \, dx + x \, dy$ depends only on the endpoints of $\mathcal{C}$.

ANSWERS

Chapter 1: Inequalities

1. x may be positive or negative. **Case 1.** $x > 0$. $2/x < 3$. $2 < 3x$ [Multiply by x .], $\frac{2}{3} < x$ [Divide by 3.] **Case 2.** $x < 0$. $2/x < 3$. $2 > 3x$ [Multiply by x . Reverse the inequality.], $\frac{2}{3} > x$ [Divide by 3.] Notice that this condition $\frac{2}{3} > x$ is satisfied whenever $x < 0$. Hence, in the case where $x < 0$, the inequality is satisfied by all such x . Hence, $\frac{2}{3} < x$ or $x < 0$. As shown in Figure A1.1, the solution is the union of the intervals $\left(\frac{2}{3}, \infty\right)$ and $(-\infty, 0)$.



Figure A1.1

2. We cannot simply multiply both sides by $x - 3$, because we do not know whether $x - 3$ is positive or negative. **Case 1.** $x - 3 > 0$ [This is equivalent to $x > 3$.] Multiplying the given inequality (1) by the positive quantity $x - 3$ preserves the inequality: $x + 4 < 2x - 6$, $4 < x - 6$ [Subtract x .], $10 < x$ [Add 6.] Thus, when $x > 3$, the given inequality holds when and only when $x > 10$. **Case 2.** $x - 3 < 0$ [This is equivalent to $x < 3$.] Multiplying the given inequality (1) by the negative quantity $x - 3$ reverses the inequality: $x + 4 > 2x - 6$, $4 > x - 6$ [Subtract x .], $10 > x$ [Add 6.] Thus, when $x < 3$, the inequality (1) holds when and only when $x < 10$. But $x < 3$ implies $x < 10$, and, therefore, the inequality (1) holds for all $x < 3$. Thus, $x > 10$ or $x < 3$. As shown in Figure A1.2, the solution is the union of the intervals $(10, \infty)$ and $(-\infty, 3)$.

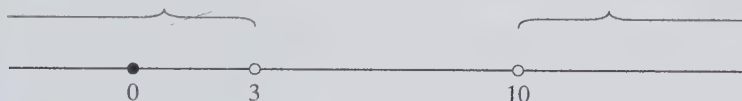


Figure A1.2

3. Factor: $x^2 - 6x + 5 = (x - 1)(x - 5)$. Let $h(x) = (x - 1)(x - 5)$. To the left of $x = 1$, both $x - 1$ and $x - 5$ are negative and, therefore, $h(x)$ is positive. When we pass through $x = 1$, $x - 1$ changes sign and $h(x)$ becomes negative. When we move further to the right and pass through $x = 5$, $x - 5$ changes sign and $h(x)$ becomes positive again. Thus, $h(x)$ is

positive for $x < 1$ and for $x > 5$. Hence $x > 5$ or $x < 1$. This is the union of the intervals $(5, \infty)$ and $(-\infty, 1)$.

4. $(x-1)^2$ is always positive except when $x = 1$ (when it is 0). So, the only solutions occur when $x + 4 < 0$ and $x \neq 1$. $x < -4$ [In interval notation, $(-\infty, -4)$.]

5. See Figure A1.3. The key points for the function $K(x) = (2x+1)(x-3)(x+7)$ are $x = -7$, $x = -\frac{1}{2}$, and $x = 3$. For x to the left of $x = -7$, all three factors are negative and, therefore, $K(x)$ is negative. When we pass from left to right through $x = -7$, $x+7$ changes sign, and, therefore, $K(x)$ becomes positive. When we later pass through $x = -\frac{1}{2}$, $2x+1$ changes sign, and, therefore, $K(x)$ becomes negative again. Finally, as we pass through $x = 3$, $x-3$ changes sign, and $K(x)$ becomes and remains positive. Hence, $K(x)$ is negative when and only when $x < -7$ or $-\frac{1}{2} < x < 3$.

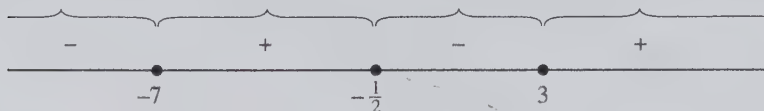


Figure A1.3

6. $x > x^2$ is equivalent to $x^2 - x < 0$, $x(x-1) < 0$; hence one of x and $(x-1)$ is positive and the other negative. Since $x > x-1$, x must be positive and $x-1$ must be negative (that is, $x-1 < 0$, which implies $x < 1$). Thus, $0 < x < 1$.

7. $x^2 > x^3$ is equivalent to $x^3 - x^2 < 0$, $x^2(x-1) < 0$, $x < 1$, and $x \neq 0$.

8. This is clearly true when x is negative and y positive, and false when x is positive and y negative. When x and y are both positive, or x and y are both negative, multiplication by the positive quantity xy yields the equivalent inequality $y < x$.

Chapter 2: Absolute Value

9. (A) $\frac{7}{3} > x > 1$ (Divide by -3 and reverse the inequalities.) In interval notation, the solution is the set $\left(1, \frac{7}{3}\right)$. $|5-3x| < 2$ if and only if $-2 < 5-3x < 2$, $-7 < -3x < -3$ [Subtract 5.]

(B) Let us solve the negation of the given relation: $|3x-2| < 1$. This is equivalent to $-1 < 3x-2 < 1$, $1 < 3x < 3$ [Add 2.], $\frac{1}{3} < x < 1$ [Divide by 3].

The points not satisfying this condition correspond to x such that $x \leq \frac{1}{3}$ or $x \geq 1$.

(C) $|u| = -u$ when and only when $u \leq 0$. So, $|3 - x| = x - 3$ when and only when $3 - x \leq 0$; that is, $3 \leq x$.

(D) If $c > 0$, $|u| = c$ if and only if $u = \pm c$. So, $|2x + 3| = 4$ when and only when $2x + 3 = \pm 4$. There are two cases: **Case 1.** $2x + 3 = 4$, $2x = 1$, $x = \frac{1}{2}$. **Case 2.** $2x + 3 = -4$, $2x = -7$, $x = -\frac{7}{2}$.

So, either $x = \frac{1}{2}$ or $x = -\frac{7}{2}$.

10. (A) There are two cases: **Case 1.** $2x - 3 = x + 2$, $x - 3 = 2$, $x = 5$. **Case 2.** $2x - 3 = -(x + 2)$, $2x - 3 = -x - 2$, $3x - 3 = -2$, $3x = 1$, $x = \frac{1}{3}$.

So, either $x = 5$ or $x = \frac{1}{3}$.

(B) Since an absolute value is never negative, $2x - 1 \geq 0$. There are two cases: **Case 1.** $x + 7 \geq 0$, $2x - 1 = x + 7$, $x - 1 = 7$, $x = 8$. **Case 2.** $x + 7 < 0$, $2x - 1 = -(x + 7)$, $2x - 1 = -x - 7$, $3x - 1 = -7$, $3x = -6$, $x = -2$. But then, $2x - 1 = -5 < 0$.

So, the only solution is $x = 8$.

(C) This is equivalent to $-|x + 2| < 2x - 3 < |x + 2|$. There are two cases: **Case 1.** $x + 2 \geq 0$, $-(x + 2) < 2x - 3 < x + 2$, $-x - 2 < 2x - 3 < x + 2$, $1 < 3x$ and $x < 5$, $\frac{1}{3} < x < 5$. **Case 2.** $x + 2 < 0$, $-(x + 2) > 2x - 3 > x + 2$, $-x - 2 > 2x - 3 > x + 2$, $1 > 3x$ and $x > 5$, $\frac{1}{3} > x$ and $x > 5$ (Impossible). So, $\frac{1}{3} < x < 5$ is the solution.

(D) First solve $|3x + 1| < \frac{1}{3}$. This is equivalent to $-\frac{1}{3} < 3x + 1 < \frac{1}{3}$, $-\frac{4}{3} < 3x < -\frac{2}{3}$ (Subtract 1.), $-\frac{4}{9} < x < -\frac{2}{9}$ (Divide by 3.) The inequality $0 < |3x + 1|$ excludes the case where $0 = |3x + 1|$, that is, where $x = -\frac{1}{3}$.

All x for which $-\frac{4}{9} < x < -\frac{2}{9}$ except $x = -\frac{1}{3}$.

11. (A) The case $n = 2$ is the triangle inequality. Assume the result true for some n . By the triangle inequality and the inductive hypothesis,

$$|u_1 + u_2 + \cdots + u_n + u_{n+1}| \leq |u_1 + u_2 + \cdots + u_n| + |u_{n+1}| \leq (|u_1| + |u_2| + \cdots + |u_n|) + |u_{n+1}|$$

and, therefore, the result also holds for $n + 1$.

(B) $|u| = |v + (u - v)| \leq |v| + |u - v|$ (Triangle inequality.) Hence, $|u - v| \geq |u| - |v|$. Similarly, $|v - u| \geq |v| - |u|$. But, $|v - u| = |u - v|$. So, $|u - v| \geq (\text{maximum of } |u| - |v| \text{ and } |v| - |u|) = ||u| - |v||$.

12. (A) This is equivalent to $\left| \frac{x^2 + 1}{x} \right| > 2$, $\frac{x^2 + 1}{|x|} > 2$ [Since $x^2 + 1 > 0$], $x^2 + 1 > 2|x|$, $x^2 - 2|x| + 1 > 0$, $|x|^2 - 2|x| + 1 > 0$ [Since $x^2 = |x|^2$], $(|x| - 1)^2 > 0$, $|x| \neq 1$.

All x except $x = 1$ and $x = -1$.

(B) This is equivalent to $\left| \frac{x^2 + 1}{x} \right| < 4$, $x^2 + 1 < 4|x|$, $x^2 - 4|x| + 1 < 0$, $|x|^2 - 4|x| + 1 < 0$, $(|x| - 2)^2 < 3$ [completing the square], $||x| - 2| < \sqrt{3}$, $-\sqrt{3} < |x| - 2 < \sqrt{3}$, $2 - \sqrt{3} < |x| < 2 + \sqrt{3}$.

When $x > 0$, $2 - \sqrt{3} < x < 2 + \sqrt{3}$, and, when $x < 0$, $-2 - \sqrt{3} < x < -2 + \sqrt{3}$.

(C) When $x \geq 0$, this reduces to $x + 1 < x$, which is impossible. When $x < 0$, the inequality becomes $x + 1 < -x$, which is equivalent to $2x + 1 < 0$, or $2x < -1$, or $x < -\frac{1}{2}$.

13. From the definition of absolute value, $|a| = \pm a$ and $|b| = \pm b$. Hence, $|a| \cdot |b| = (\pm a) \cdot (\pm b) = \pm(ab)$. Since $|a| \cdot |b|$ is nonnegative, $|a| \cdot |b|$ must be $|ab|$.

14. (A) This is equivalent to $-|x - 1| \leq 3x - 2 \leq |x - 1|$. **Case 1.** $x - 1 \geq 0$. Then $-(x - 1) \leq 3x - 2 \leq x - 1$, $-x + 1 \leq 3x - 2 \leq x - 1$; the first inequality is equivalent to $\frac{3}{4} \leq x$ and the second to $x \leq \frac{1}{2}$. But this is impossible. **Case 2.** $x - 1 < 0$. $-x + 1 \geq 3x - 2 \geq x - 1$; the first inequality is equivalent to $x \leq \frac{3}{4}$ and the second to $x \geq \frac{1}{2}$. Hence, we have $\frac{1}{2} \leq x \leq \frac{3}{4}$.

(B) **Case 1.** $x \geq 5$. Then $x - 2 + x - 5 = 9$, $2x - 7 = 9$, $2x = 16$, $x = 8$. **Case 2.** $2 \leq x < 5$. Then $x - 2 + 5 - x = 9$, $3 = 9$, which is impossible. **Case 3.** $x < 2$. Then $2 - x + 5 - x = 9$, $7 - 2x = 9$, $2x = -2$, $x = -1$. So, the solutions are 8 and -1.

(C) **Case 1.** $5x + 1 \geq 0$, that is, $x \geq -\frac{1}{5}$. Then $4 - x \geq 5x + 1$, $3 \geq 6x$, $\frac{1}{2} \geq x$. Thus, we obtain the solutions $-\frac{1}{5} \leq x \leq \frac{1}{2}$. **Case 2.** $5x + 1 < 0$; that is, $x < -\frac{1}{5}$. Then $4 - x \geq -5x - 1$, $4x \geq -5$, $x \geq -\frac{5}{4}$. Thus, we obtain the solutions $-\frac{5}{4} \leq x < -\frac{1}{5}$. Hence, the set of solutions is $\left[-\frac{1}{5}, \frac{1}{2}\right] \cup \left[-\frac{5}{4}, -\frac{1}{5}\right] = \left[-\frac{5}{4}, \frac{1}{2}\right]$.

15. We argue geometrically from Figure A2.1. $|x - 1|$ is the distance of x from 1, and $|x - 3|$ is the distance of x from 3. The point $x = 2$ is equidistant from 1 and 3. Hence, the solutions consist of all $x \geq 2$.

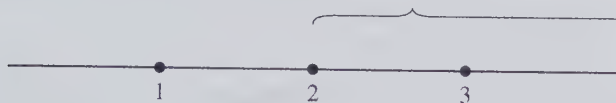


Figure A2.1

Chapter 3: Lines

16. (A) Remember that the slope m of the line through two points (x_1, y_1) and (x_2, y_2) is given by the equation $m = \frac{y_2 - y_1}{x_2 - x_1}$. Hence, the slope of the given line is

$$\frac{1 - 5}{7 - (-2)} = \frac{-4}{9} = -\frac{4}{9}.$$

(B) The slope m of the given line is $(6 - 3)/(3 - 1) = \frac{3}{2}$. Recall that the point-slope equation of the line through point (x_1, y_1) and with slope m is $y - y_1 = m(x - x_1)$. Hence, one point-slope equation of the given line using the point $(1, 3)$ is $y - 3 = \frac{3}{2}(x - 1)$.

Another point-slope equation, using the point $(3, 6)$, is $y - 6 = \frac{3}{2}(x - 3)$.

(C) The line through $(1, 2)$ and $(1, 3)$ is vertical and, therefore, does not have a slope. Thus, there is no point-slope equation of the line.

(D) $y - 3 = 5(x - 1)$.

(E) Remember that the slope-intercept equation of a line is $y = mx + b$, where m is the slope and b is the y -intercept (that is, the y -coordinate of the point where the line cuts the y -axis). In this case, the slope $m = (8 - 4)/(4 - 2) = \frac{4}{2} = 2$.

Method 1. A point-slope equation of the line is $y - 8 = 2(x - 4)$. This is equivalent to $y - 8 = 2x - 8$, or, finally, to $y = 2x$.

Method 2. The slope-intercept equation has the form $y = 2x + b$. Since $(2, 4)$ lies on the line, we may substitute 2 for x and 4 for y . So, $4 = 2 \cdot 2 + b$, and, therefore, $b = 0$. Hence, the equation is $y = 2x$.

17. If a given line is vertical it has an equation $x = C$. In this case, we can let $A = 1$ and $B = 0$. If the given line is not vertical, it has a slope-intercept equation $y = mx + b$, or, equivalently, $-mx + y = b$. So, let $A = -m$, $B = 1$, and $C = b$. Conversely, assume that we are given an equation $Ax + By = C$, with A and B not both 0. If $B = 0$, the equation is

equivalent to $x = C/A$, which is the equation of a vertical line. If $B \neq 0$, solve the equation for y : $y = -\frac{A}{B}x + \frac{C}{B}$. This is the slope-intercept equation of the line with slope $-\frac{A}{B}$ and y -intercept $\frac{C}{B}$.

18. Remember that two lines are parallel if and only if their slopes are equal. If we solve $3x + 4y = 2$ for y , namely, $y = -\frac{3}{4}x + \frac{1}{2}$, we obtain the slope-intercept equation for M . Hence, the slope of M is $-\frac{3}{4}$ and, therefore, the slope of the parallel line L also is $-\frac{3}{4}$. So, L has a slope-intercept equation of the form $y = -\frac{3}{4}x + b$. Since L goes through $(-1, 4)$, $4 = -\frac{3}{4}(-1) + b$, and, therefore, $b = 4 - \frac{3}{4} = \frac{13}{4}$. Thus, the equation of L is $y = -\frac{3}{4}x + \frac{13}{4}$.

19. If we solve $Ax + By = C$ for y , we obtain the slope-intercept equation $y = -\frac{A}{B}x + \frac{C}{B}$. So, the slope is $-A/B$. Given a parallel line, it must also have slope $-A/B$ and, therefore, has a slope-intercept equation $y = -\frac{A}{B}x + b$, which is equivalent to $\frac{A}{B}x + y = b$, and, thence to $Ax + By = bB$. Conversely, a line with equation $Ax + By = E$ must have slope $-A/B$ (obtained by putting the equation in slope-intercept form) and is, therefore, parallel to the line with equation $Ax + By = C$.

20. In Question 17, set $C/A = a$ and $C/B = b$. Notice that, when $y = 0$, the equation yields the value $x = a$, and, therefore, a is the x -intercept of the line, similarly for the y -intercept.

21. Solve $2x - 6y = 5$ for y , obtaining $y = \frac{1}{3}x - \frac{5}{6}$. So, the slope of L is $\frac{1}{3}$. Recall that two lines with slopes m_1 and m_2 are perpendicular if and only if $m_1 m_2 = -1$, or, equivalently, $m_1 = -1/m_2$. Hence, the slope of M is the negative reciprocal of $\frac{1}{3}$, that is, -3 . The slope-intercept equation of M has the form $y = -3x + b$. Since $(1, 4)$ is on M , $4 = -3 \cdot 1 + b$. Hence, $b = 7$, and the required equation is $y = -3x + 7$.

22. Assume that $A_2 = rA_1$ and $B_2 = rB_1$, with $r \neq 0$. Then the equation of M is $rA_1x + rB_1y = C_2$, which is equivalent to $A_1x + B_1y = \frac{1}{r}C_2$. Then, by Question 19, M is parallel to L . Conversely, assume M is parallel to L . By solving the equations of L and M for y , we see

that the slope of L is $-(A_1/B_1)$ and the slope of M is $-(A_2/B_2)$. Since M and L are parallel, their slopes are equal:

$$-\frac{A_1}{B_1} = -\frac{A_2}{B_2} \quad \text{or} \quad \frac{A_2}{A_1} = \frac{B_2}{B_1} = r$$

(In the special case where the lines are vertical, $B_1 = B_2 = 0$, and we can set $r = A_2/A_1$.)

23. (A) The slope m_1 of line $\overline{AB}$ is $(3-1)/(7-4) = \frac{2}{3}$. The slope m_2 of line $\overline{BC}$ is $(9-3)/(3-7) = -\frac{6}{4} = -\frac{3}{2}$.

Since m_2 is the negative reciprocal of m_1 , the lines $\overline{AB}$ and $\overline{BC}$ are perpendicular. Hence, $\triangle ABC$ has a right angle at B .

(B) The slope of line $\overline{AB}$ is $(5-2)/[7-(-1)] = \frac{3}{8}$. The slope of line $\overline{BC}$ is $(2-0)/(-1-k) = -2/(1+k)$. The condition for $\triangle ABC$ to have a right angle at B is that lines $\overline{AB}$ and $\overline{BC}$ are perpendicular, which holds when and only when the product of their slopes is -1 ; that is, $\left(\frac{3}{8}\right)[-2/(1+k)] = -1$. This is equivalent to $6 = 8(1+k)$, or $8k = -2$, or $k = -\frac{1}{4}$.

24. (A) By the midpoint formula, the coordinates of the midpoint are the averages of the coordinates of the endpoints. In this case, the midpoint (x, y) is given by $[(2+(-1))/2, (5+3)/2] = \left(\frac{1}{2}, 4\right)$.

(B) The midpoint M of segment $\overline{BC}$ is $((8+2)/2, (1+3)/2) = (5, 2)$. So, $\overline{AM}$ is horizontal, with equation $y = 2$.

(C) The slope of $\overline{AC}$ is $(3-2)/(2-1) = 1$. Hence, the slope of the altitude is the negative reciprocal of 1, namely, -1 . Thus, its slope-intercept equation has the form $y = -x + b$. Since $B(8, 1)$ is on the altitude, $1 = -8 + b$, and, so, $b = 9$. Hence, the equation is $y = -x + 9$.

25. Solving for y , we obtain the equation $y = -\frac{3}{2}x + 2$. For any fixed x , the vertical line with that x -coordinate cuts the line at the point Q where the y -coordinate is $-\frac{3}{2}x + 2$ (see Figure A3.1). The points along that vertical line and above Q have y -coordinates $y > -\frac{3}{2}x + 2$. This is equivalent to $2y > -3x + 4$, and thence to $3x + 2y > 4$.

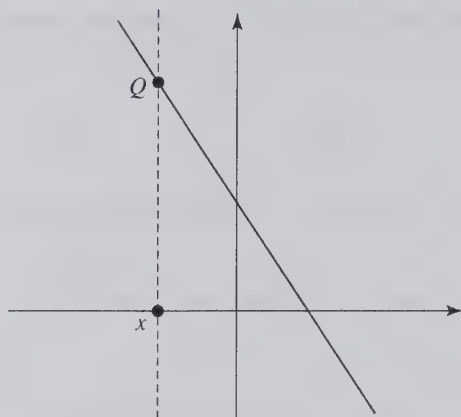


Figure A3.1

26. (A) The set of all nonvertical lines through $(0, 2)$.

(B) The family of mutually parallel lines of slope 3.

27. Given $\triangle ABC$, choose the coordinate system so that A and B lie on the x -axis and C lies on the y -axis (Figure A3.2). Let the coordinates of A , B , and C be $(u, 0)$, $(v, 0)$, and $(0, w)$.

(i) The altitude from C to $\overline{AB}$ is the y -axis. (ii) The slope of $\overline{BC}$ is $-w/v$. So, the altitude from A to $\overline{BC}$ has slope v/w . Its slope-intercept equation has the form $y = (v/w)x + b$. Since $(u, 0)$ lies on the line, $0 = (v/w)(u) + b$; hence, its y -intercept $b = -vu/w$. Thus, this altitude intersects the altitude from C (the y -axis) at the point $(0, -vu/w)$. (iii) The slope of $\overline{AC}$ is $-w/u$. So, the altitude from B to $\overline{AC}$ has slope u/w , and its slope-intercept equation is $y = (u/w)x + b$. Since $(v, 0)$ lies on the altitude, $0 = (u/w)(v) + b$, and its y -intercept $b = -uv/w$. Thus, this altitude also goes through the point $(0, -vu/w)$.

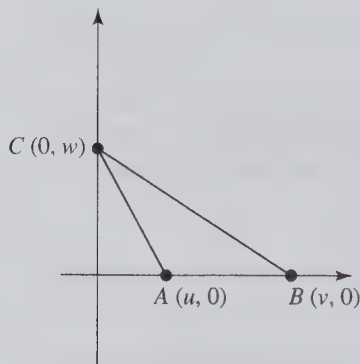


Figure A3.2

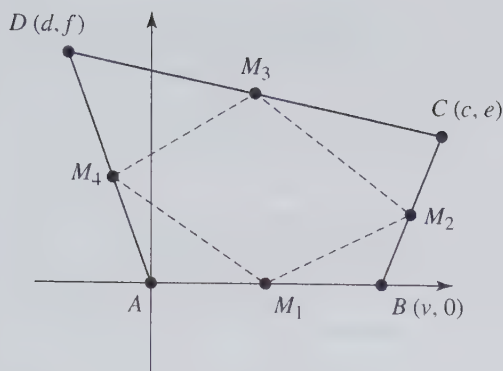


Figure A3.3

28. Refer to Figure A3.3. Let A be the origin and let B be on the x -axis, with coordinates $(v, 0)$. Let C be (c, e) and let D be (d, f) . The midpoint M_1 of $\overline{AB}$ has coordinates $(v/2, 0)$, the midpoint M_2 of $\overline{BC}$ is $((c+v)/2, e/2)$, the midpoint M_3 of $\overline{CD}$ is $((c+d)/2, (e+f)/2)$, and the midpoint M_4 of $\overline{AD}$ is $(d/2, f/2)$.

$$\text{Slope of line } \overline{M_1M_2} = \frac{e/2 - 0}{(c+v)/2 - v/2} = \frac{e}{c} \quad \text{Slope of line } \overline{M_3M_4} = \frac{(e+f)/2 - f/2}{(c+d)/2 - d/2} = \frac{e}{c}$$

Thus, $\overline{M_1M_2}$ and $\overline{M_3M_4}$ are parallel. Similarly, the slopes of $\overline{M_2M_3}$ and $\overline{M_1M_4}$ both turn out to be $f/(d-v)$, and, therefore, $\overline{M_2M_3}$ and $\overline{M_1M_4}$ are parallel. Thus, $M_1M_2M_3M_4$ is a parallelogram. (Note two special cases. When $c=0$, both $\overline{M_1M_2}$ and $\overline{M_3M_4}$ are vertical and, therefore, parallel. When $d=v$, both $\overline{M_1M_4}$ and $\overline{M_2M_3}$ are vertical and, therefore, parallel.)

29.

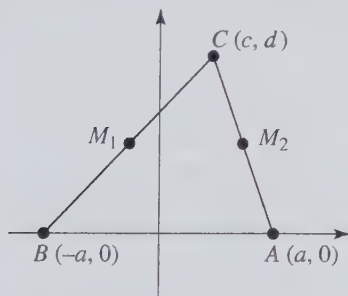


Figure A3.4

See Figure A3.4. Choose the x -axis so that it goes through A and B and let the origin be halfway between A and B . Let A be $(a, 0)$. Then B is $(-a, 0)$. Let C be (c, d) . Then M_1 is $((c-a)/2, d/2)$, and M_2 is $((c+a)/2, d/2)$. By the distance formula,

$$\overline{AM}_1 = \sqrt{\left(a - \frac{c-a}{2}\right)^2 + \left(0 - \frac{d}{2}\right)^2} = \sqrt{\left(\frac{3a-c}{2}\right)^2 + \left(\frac{d}{2}\right)^2}$$

and

$$\overline{BM}_2 = \sqrt{\left[\frac{c+a}{2} - (-a)\right]^2 + \left(\frac{d}{2} - 0\right)^2} = \sqrt{\left(\frac{3a+c}{2}\right)^2 + \left(\frac{d}{2}\right)^2}$$

Setting $\overline{AM}_1 = \overline{BM}_2$ and squaring both sides, we obtain $[(3a-c)/2]^2 + (d/2)^2 = [(3a+c)/2]^2 + (d/2)^2$, and, simplifying, $(3a-c)^2 = (3a+c)^2$. So, $(3a+c)^2 - (3a-c)^2 = 0$, and, factoring the left-hand side, $[(3a+c) + (3a-c)] \cdot [(3a+c) - (3a-c)] = 0$; that is, $(6a) \cdot (2c) = 0$. Since $a \neq 0$, $c = 0$. Now the distance formula gives

$$\overline{CA} = \sqrt{(0-a)^2 + (d-0)^2} = \sqrt{[0-(-a)]^2 + (d-0)^2} = \overline{CB}$$

as required.

30. Remember that the distance from a point (x_1, y_1) to a line $Ax + By + C = 0$ is $|Ax_1 + By_1 + C|/\sqrt{A^2 + B^2}$. In our case, $A = 3$, $B = -4$, $C = 10$, and $\sqrt{A^2 + B^2} = \sqrt{25} = 5$. So the distance is $|3(1) - 4(2) - 10|/5 = \frac{15}{5} = 3$.

31. A point-slope equation of a line through $(4, -2)$ with slope m is $y + 2 = m(x - 4)$ or $mx - y - (4m + 2) = 0$. The distance of $(0, 0)$ from this line is $|4m + 2|/\sqrt{m^2 + 1}$. Hence, $|4m + 2|/\sqrt{m^2 + 1} = 2$. So $(4m + 2)^2 = 4(m^2 + 1)$, or $(2m + 1)^2 = m^2 + 1$. Simplifying, $m(3m + 4) = 0$, and, therefore, $m = 0$ or $m = -\frac{4}{3}$. The required equations are $y = -2$ and $4x + 3y - 10 = 0$.

32. Since the equation is linear, we can write it as $y = mx + b$. From the information about the freezing point of water, we see that $b = 32$. From the information about the boiling point, we have $212 = 100m + 32$, $180 = 100m$, $m = \frac{9}{5}$. So $y = \frac{9}{5}x + 32$.

Chapter 4: Circles

33. By the distance formula, a point (x, y) is on the circle if and only if

$\sqrt{(x-a)^2 + (y-b)^2} = r$. Squaring both sides, we obtain the standard equation: $(x-a)^2 + (y-b)^2 = r^2$.

34. Complete the square in x and in y : $(x-6)^2 + (y+10)^2 + 15 = 36 + 100$. (Here the -6 in $(x-6)$ is half of the coefficient, -12 , of x in the original equation, and the $+10$ in $(y+10)$ is half of the coefficient, 20 , of y . The 36 and 100 on the right balance the squares of -6 and $+10$ that have in effect been added on the left.) Simplifying, we obtain $(x-6)^2 + (y+10)^2 = 121$, the standard equation of a circle with center at $(6, -10)$ and radius 11 .

35. Let the center be $(-12, b)$. The distance formula yields $13 = \sqrt{(-12-0)^2 + (b-0)^2} = \sqrt{144 + b^2}$. So $144 + b^2 = 169$, $b^2 = 25$, and $b = \pm 5$. Hence, there are two circles, with equations $(x+12)^2 + (y-5)^2 = 169$ and $(x+12)^2 + (y+5)^2 = 169$.

36. The radius is the perpendicular distance from the center $(1, 3)$ to the line:

$$\frac{|5(1) - 12(3) - 8|}{13} = 3. \text{ So the standard equation is } (x-1)^2 + (y-3)^2 = 9.$$

37. Clearly, the indicated equation yields the equation of a circle that contains the intersection points. Conversely, given a circle $\mathcal{C} \neq \mathcal{C}_2$ that goes through those intersection points, take a point (x_0, y_0) of $\mathcal{C}$ that does not lie on $\mathcal{C}_2$ and substitute x_0 for x and y_0 for y in the indicated equation. By choice of (x_0, y_0) the coefficient of k is nonzero, so we can solve for k . If we then put this value of k in the indicated equation, we obtain an equation of a circle that is satisfied by (x_0, y_0) and by the intersection points of $\mathcal{C}_1$ and $\mathcal{C}_2$. Since three noncollinear points determine a circle, we have an equation for $\mathcal{C}$. [Again, it is the choice of (x_0, y_0) that makes the three points noncollinear, i.e., $k \neq -1$.]

38. Using Question 37, substitute $(3, 1)$ for (x, y) in the equation $(x^2 + y^2 - x - y - 2) + k(x^2 + y^2 + 4x - 4y - 8) = 0$. Then $4 + 10k = 0$, $k = -\frac{2}{5}$. So the desired equation can be written as

$$5(x^2 + y^2 - x - y - 2) - 2(x^2 + y^2 + 4x - 4y - 8) = 0 \text{ or } 3x^2 + 3y^2 - 13x + 3y + 6 = 0.$$

Chapter 5: Functions and Their Graphs

39. (A) The domain is the closed interval $[-2, 2]$, since $\sqrt{4-x^2}$ is defined when and only when $x^2 \leq 4$. The graph (Figure A5.1) is the upper half of the circle $x^2 + y^2 = 4$ with center at the origin and radius 2. The range is the closed interval $[0, 2]$.

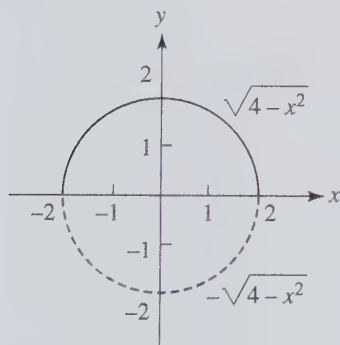


Figure A5.1

(B) The domain is the set of all real numbers. The range is the set of all nonnegative real numbers. The graph (Figure A5.2) is the graph of $y = |x|$ shifted one unit to the right.

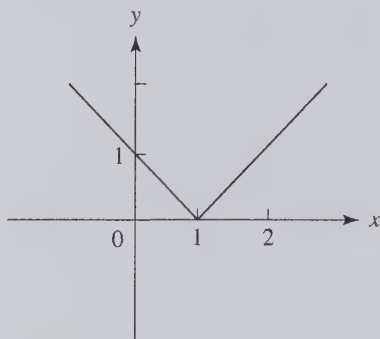


Figure A5.2

(C) The domain is the set of all nonzero real numbers, and the range is the same set. The graph (Figure A5.3) is the hyperbola $xy = 1$.

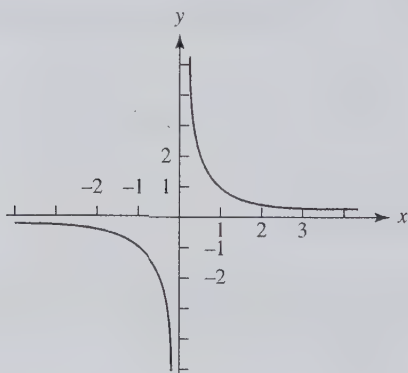


Figure A5.3

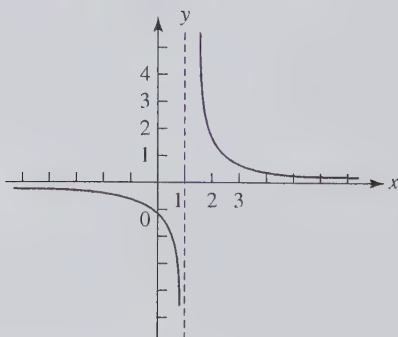


Figure A5.4

(D) The domain is the set of all real numbers $\neq 1$. The graph (Figure A5.4) is Figure A5.3 shifted one unit to the right. The range consists of all nonzero real numbers.

(E) The domain is the set of all real numbers, and the range is the same set. See Figure A5.5.

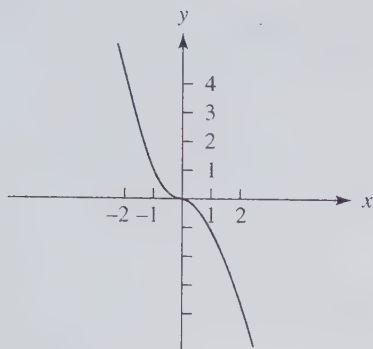


Figure A5.5

(F) The domain is the set of all real numbers except -2 . Since $G(x) = \frac{(x+2)(x-2)}{x+2} = x-2$ for $x \neq -2$, the graph (Figure A5.6) consists of all points on the line $y = x - 2$ except the point $(-2, -4)$. The range is the set of all real numbers except -4 .

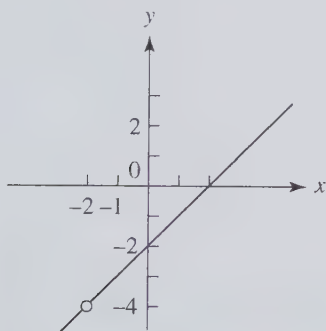


Figure A5.6

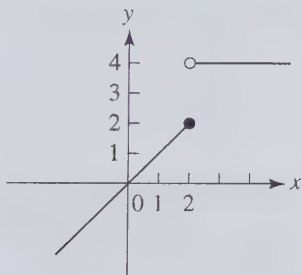


Figure A5.7

(G) The domain is the set of all real numbers. The graph (Figure A5.7) is made up of the left half of the line $y = x$ for $x \leq 2$ and the right half of the line $y = 4$ for $x > 2$. The range consists of all real numbers ≤ 2 , plus the number 4.

(H) The domain is the set of all nonzero real numbers. The graph (Figure A5.8) is the right half of the line $y = 1$ for $x > 0$, plus the left half of the line $y = -1$ for $x < 0$. The range is $\{1, -1\}$.

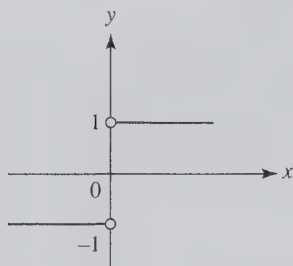


Figure A5.8

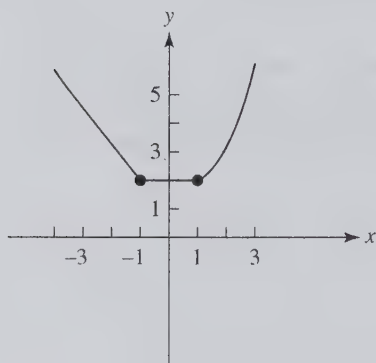


Figure A5.9

The domain is the set of all real numbers. The graph (Figure A5.9) is a continuous curve consisting of three pieces: the half of the line $y = 1 - x$ to the left of $x = -1$, the horizontal segment $y = 2$ between $x = -1$ and $x = 1$, and the part of the parabola $y = x^2 + 1$ to the right of $x = 1$. The range consists of all real numbers ≥ 2 .

40. (A) Since some vertical lines cut the graph in more than one point, this cannot be the graph of a function.

(B) Since each vertical line cuts the graph in at most one point, this is the graph of a function.

41. (A) The domain consists of all real numbers except 2 and 3. To determine the range, set $y = \frac{1}{(x-2)(x-3)} = \frac{1}{x^2 - 5x + 6}$ and solve for x . $x^2 - 5x + \left(6 - \frac{1}{y}\right) = 0$. This has a

solution when and only when $b^2 - 4ac = (-5)^2 - 4(1)\left(6 - \frac{1}{y}\right) = 25 - 24 + \frac{4}{y} = 1 + \frac{4}{y} \geq 0$.

This holds if and only if $4/y \geq -1$. This holds when $y > 0$, and, if $y < 0$, when $y \leq -4$. Hence the range is $(0, +\infty) \cup (-\infty, -4]$.

(B) x is in the domain if and only if $x^2 < 1$. Thus, the domain is $(-1, 1)$.

To find the range, first note that $g(x) > 0$. Then set $y = 1/\sqrt{1-x^2}$ and solve for x . $y^2 = 1/(1-x^2)$, $x^2 = 1-1/y^2 \geq 0$, $1 \geq 1/y^2$, $y^2 \geq 1$, $y \geq 1$. Thus, the range is $[1, +\infty)$.

(C) The domain is $(-1, +\infty)$. The graph consists of the open segment from $(-1, 0)$ to $(1, 2)$ plus the half-line of $y = 2$, with $x \geq 1$. Hence, the range is the half-open interval $(0, 2]$.

(D) The domain is $[0, 4)$. Inspection of the graph shows that the range is $[-1, 2]$.

(E) The domain is the set of all real numbers. To determine the range, note that $G(x) = 0$ if $x \geq 0$, and $G(x) = -2x$ if $x < 0$. Hence, the range consists of all nonnegative real numbers.

42. (A) Assume $f(u) = f(v)$. Then $\sqrt{u} = \sqrt{v}$. Square both sides; $u = v$. Thus, f is one-one.

(B) $f(-1) = 1 = f(1)$. Hence, f is not one-one.

(C) Assume $f(u) = f(v)$. Then $\frac{1}{u} = \frac{1}{v}$. Hence, $u = v$. Thus, f is one-one.

43. (A) $f(x+h) = (x+h)^2 - 2(x+h) = (x^2 + 2xh + h^2) - 2x - 2h$. So, $f(x+h) - f(x) = [(x^2 + 2xh + h^2) - 2x - 2h] - (x^2 - 2x) = 2xh + h^2 - 2h = h(2x + h - 2)$. Hence, $\frac{f(x+h) - f(x)}{h} = 2x + h - 2$.

(B) $f(x+h) = x+h+4$. So, $f(x+h) - f(x) = (x+h+4) - (x+4) = h$. Hence, $\frac{f(x+h) - f(x)}{h} = 1$.

(C) $f(x+h) = (x+h)^3 + 1 = x^3 + 3x^2h + 3xh^2 + h^3 + 1$. So, $f(x+h) - f(x) = (x^3 + 3x^2h + 3xh^2 + h^3 + 1) - (x^3 + 1) = 3x^2h + 3xh^2 + h^3 = h(3x^2 + 3xh + h^2)$. Hence, $\frac{f(x+h) - f(x)}{h} = 3x^2 + 3xh + h^2$.

(D)

$$\begin{aligned} f(x+h) - f(x) &= \sqrt{x+h} - \sqrt{x} = (\sqrt{x+h} - \sqrt{x}) \frac{\sqrt{x+h} + \sqrt{x}}{\sqrt{x+h} + \sqrt{x}} = \frac{(x+h) - x}{\sqrt{x+h} + \sqrt{x}} \\ &= \frac{h}{\sqrt{x+h} + \sqrt{x}} \end{aligned}$$

So, $\frac{f(x+h)-f(x)}{h} = \frac{1}{\sqrt{x+h}+\sqrt{x}}$

(E) $f(x+h)-f(x) = \frac{1}{x+h} - \frac{1}{x} = \frac{x-(x+h)}{x(x+h)} = \frac{-h}{x(x+h)}$

Hence, $\frac{f(x+h)-f(x)}{h} = -\frac{1}{x(x+h)}$

44. (A) Let $y = 2x + 1$ and solve for x . $x = \frac{1}{2}(y-1)$. Thus, $f^{-1}(y) = \frac{1}{2}(y-1)$.

(B) Let $y = x^3$. Then $x = \sqrt[3]{y}$. So, $f^{-1}(y) = \sqrt[3]{y}$.

45. (A) $x^4 - 10x^2 + 9 = (x^2 - 9)(x^2 - 1) = (x-3)(x+3)(x-1)(x+1)$. Hence, the roots are 3, -3, 1, -1.

(B) Inspection of the divisors of the constant term 32 reveals that -2 is a root. Division by $x + 2$ yields the factorization

$$(x+2)(x^2-16) = (x+2)(x^2+4)(x^2-4) = (x+2)(x^2+4)(x-2)(x+2).$$

So, the roots are 2 and -2.

(C) Testing the divisors of 24 reveals the root -2. Dividing by $x + 2$ yields the factorization $(x+2)(x^2+7x+12) = (x+2)(x+3)(x+4)$. Thus, the roots are -2, -3, and -4.

(D) -2 is a root. Dividing by $x + 2$ yields the factorization $(x+2)(x^2-2x-1)$. The quadratic formula applied to x^2-2x-1 gives the additional roots $1 \pm \sqrt{2}$.

Chapter 6: Limits

46. Intuitively, this means that, as x gets closer and closer to a , $f(x)$ gets closer and closer to L . We can state this in more precise language as follows: For any $\varepsilon > 0$, there exists $\delta > 0$ such that, if $|x - a| < \delta$, then $|f(x) - L| < \varepsilon$. Here, we assume that, for any $\delta > 0$, there exists at least one x in the domain of $f(x)$ such that $|x - a| < \delta$.

47. (A) The numerator and denominator both approach 0. However, $u^2 - 25 = (u + 5)(u - 5)$. Hence, $\frac{u^2 - 25}{u - 5} = u + 5$. Thus, $\lim_{u \rightarrow 5} \frac{u^2 - 25}{u - 5} = \lim_{u \rightarrow 5} (u + 5) = 10$.

(B) Both the numerator and denominator approach 0. However,

$x^3 - 1 = (x-1)(x^2 + x + 1)$. Hence, $\lim_{x \rightarrow 1} \frac{x^3 - 1}{x - 1} = \lim_{x \rightarrow 1} (x^2 + x + 1) = 1 + 1 + 1 = 3$.

(C) $\lim_{u \rightarrow 0} 5u^2 - 4 = -4$ and $\lim_{u \rightarrow 0} (u + 1) = 1$. Hence, by the quotient law for limits,

$$\lim_{u \rightarrow 0} \frac{5u^2 - 4}{u + 1} = \frac{-4}{1} = -4.$$

(D) Both the numerator and denominator approach 0. However, $x^2 - x - 12 = (x + 3)(x - 4)$. Hence, $\lim_{x \rightarrow 4} \frac{x^2 - x - 12}{x - 4} = \lim_{x \rightarrow 4} (x + 3) = 7$.

(E) In this case, neither the numerator nor the denominator approaches 0. In fact, $\lim_{x \rightarrow 2} (x^3 - 5x^2 + 2x - 4) = -12$ and $\lim_{x \rightarrow 2} (x^2 - 3x + 3) = 1$. Hence, our limit is $\frac{-12}{1} = -12$.

(F) Both the numerator and denominator approach 0. "Rationalize" the numerator by multiplying both numerator and denominator by $\sqrt{x+3} + \sqrt{3}$:

$$\frac{\sqrt{x+3} - \sqrt{3}}{x} \cdot \frac{\sqrt{x+3} + \sqrt{3}}{\sqrt{x+3} + \sqrt{3}} = \frac{(x+3) - 3}{x(\sqrt{x+3} + \sqrt{3})} = \frac{1}{\sqrt{x+3} + \sqrt{3}}$$

So, we obtain $\lim_{x \rightarrow 0} \frac{1}{\sqrt{x+3} + \sqrt{3}} = \frac{1}{\sqrt{3} + \sqrt{3}} = \frac{1}{2\sqrt{3}}$.

(G) As $x \rightarrow 3$, from either the right or the left, $(x-3)^2$ remains positive and approaches 0. Hence, $1/(x-3)^2$, becomes larger and larger without bound and is positive. Hence,

$\lim_{x \rightarrow 3} \frac{1}{(x-3)^2} = +\infty$, an improper limit.

(H) As $x \rightarrow 2$ from the right (that is, with $x > 2$), $x - 2$ approaches 0 and positive; therefore, $3/(x-2)$ approaches $+\infty$. However, as $x \rightarrow 2$ from the left (that is, with $x < 2$), $x - 2$ approaches 0 and is negative; therefore, $3/(x-2)$ approaches $-\infty$. Hence, nothing can be said about $\lim_{x \rightarrow 2} \frac{3}{x-2}$.

(Some people prefer to write $\lim_{x \rightarrow 2} \frac{3}{x-2} = \infty$ to indicate that the magnitude $\left| \frac{3}{x-2} \right|$

approaches $+\infty$.)

48. (A) $2x^{11} - 5x^6 + 3x^2 + 1 = x^{11} \left(2 - \frac{5}{x^5} + \frac{3}{x^9} + \frac{1}{x^{11}} \right)$. But $\frac{5}{x^5}$ and $\frac{3}{x^9}$ and $\frac{1}{x^{11}}$ all approach 0 as $x \rightarrow +\infty$. Thus, $2 - \frac{5}{x^5} + \frac{3}{x^9} + \frac{1}{x^{11}}$ approaches 2. At the same time x^{11} approaches $+\infty$. Hence, the limit is $+\infty$.

(B) $2x^3 - 12x^2 + x - 7 = x^3 \left(2 - \frac{12}{x} + \frac{1}{x^2} - \frac{7}{x^3} \right)$. As $x \rightarrow -\infty$, $\frac{12}{x}$, $\frac{1}{x^2}$, and $\frac{7}{x^3}$ all approach 0. Hence, $2 - \frac{12}{x} + \frac{1}{x^2} - \frac{7}{x^3}$ approaches 2. But x^3 approaches $-\infty$. Therefore, the limit is $-\infty$. (Note that the limit will always be $-\infty$ when $x \rightarrow -\infty$ and the function is a polynomial of odd degree with a positive leading coefficient.)

(C) The numerator and denominator both approach ∞ . Hence, we divide the numerator and denominator by x^2 , the highest power of x in the denominator. We obtain

$$\lim_{x \rightarrow +\infty} \frac{2/x + 5/x^2}{1 - 7/x + 3/x^2} = \frac{\lim_{x \rightarrow +\infty} (2/x) + \lim_{x \rightarrow +\infty} (5/x^2)}{\lim_{x \rightarrow +\infty} 1 - \lim_{x \rightarrow +\infty} (7/x) + \lim_{x \rightarrow +\infty} (3/x^2)} = \frac{0 + 0}{1 - 0 + 0} = \frac{0}{1} = 0$$

(D) Both the numerator and denominator approach 0. So we divide both of them by x^3 , the highest power of x in the denominator.

$$\lim_{x \rightarrow +\infty} \frac{3 - 4/x^2 + 2/x^3}{7 + 5/x^3} = \frac{\lim_{x \rightarrow +\infty} 3 - \lim_{x \rightarrow +\infty} (4/x^2) + \lim_{x \rightarrow +\infty} (2/x^3)}{\lim_{x \rightarrow +\infty} 7 + \lim_{x \rightarrow +\infty} (5/x^3)} = \frac{3 - 0 + 0}{7 + 0} = \frac{3}{7}$$

(For a generalization, see Question 55(B).)

(E) Both the numerator and denominator approach ∞ . So we divide both of them by x^3 , the highest power of x in the denominator.

$$\lim_{x \rightarrow +\infty} \frac{4x^2 - 1/x^3}{3 + 7/x^3} = \frac{\lim_{x \rightarrow +\infty} 4x^2 - \lim_{x \rightarrow +\infty} (1/x^3)}{\lim_{x \rightarrow +\infty} 3 + \lim_{x \rightarrow +\infty} (7/x^3)} = \frac{\lim_{x \rightarrow +\infty} 4x^2 - 0}{3 + 0} = \frac{\lim_{x \rightarrow +\infty} 4x^2}{3} = +\infty$$

(For a generalization, see Question 55(C).)

(F) When $f(x)$ is a polynomial of degree n , it is useful to think of the degree of $\sqrt{f(x)}$ as being $n/2$. Thus, in this question, the denominator has a degree of 1, and, therefore, in line with the procedure that has worked before, we divide the numerator and denominator by x . Notice that, when $x > 0$, as it is when $x \rightarrow +\infty$, $x = \sqrt{x^2}$. So, we obtain

$$\lim_{x \rightarrow +\infty} \frac{4 - (1/x)}{\sqrt{(x^2 + 2)/x^2}} = \lim_{x \rightarrow +\infty} \frac{4 - (1/x)}{\sqrt{1 + (2/x^2)}} = \frac{\lim_{x \rightarrow +\infty} 4 - \lim_{x \rightarrow +\infty} (1/x)}{\sqrt{\lim_{x \rightarrow +\infty} 1 + \lim_{x \rightarrow +\infty} (2/x^2)}} = \frac{4 - 0}{\sqrt{1 + 0}} = 4$$

(G) We divide the numerator and denominator by $x^{3/2}$. Note that $x^{3/2} = \sqrt{x^3}$ when $x > 0$. So, we obtain

$$\lim_{x \rightarrow +\infty} \frac{7/\sqrt{x} - 4/x^{3/2}}{\sqrt{1+5/x^3}} = \frac{\lim_{x \rightarrow +\infty} (7/\sqrt{x}) - \lim_{x \rightarrow +\infty} (4/x^{3/2})}{\sqrt{1 + \lim_{x \rightarrow +\infty} (5/x^3)}} = \frac{0-0}{\sqrt{1+0}} = 0$$

(H) We divide the numerator and denominator by x^2 . Note that $x^2 = \sqrt{x^4}$. We obtain

$$\lim_{x \rightarrow -\infty} \frac{3x+2/x^2}{\sqrt{(x^4-2)/x^4}} = \lim_{x \rightarrow -\infty} \frac{3x+2/x^2}{\sqrt{1-2/x^4}} = \frac{\lim_{x \rightarrow -\infty} 3x + \lim_{x \rightarrow -\infty} (2/x^2)}{\sqrt{1 - \lim_{x \rightarrow -\infty} (2/x^4)}} = \frac{\lim_{x \rightarrow -\infty} 3x + 0}{\sqrt{1-0}} = -\infty$$

49. (A) Remember that a vertical asymptote is a vertical line $x = c$ to which the graph gets closer and closer as x approaches c from the right or from the left. Hence, we obtain vertical asymptotes by setting the denominator $3x+2=0$. Thus, the only vertical asymptote is the line $x = -\frac{2}{3}$. Recall that a horizontal asymptote is a line $y = d$ to which the graph gets closer and closer as $x \rightarrow +\infty$ or $x \rightarrow -\infty$. In this case, $\lim_{x \rightarrow \pm\infty} \frac{4x-5}{3x+2} = \lim_{x \rightarrow \pm\infty} \frac{4-5/x}{3+2/x} = \frac{4}{3}$. Thus, the line $y = \frac{4}{3}$ is a horizontal asymptote both on the right and the left.

(B) $x^2 - 2x - 3 = (x-3)(x+1)$. Hence, the denominator is 0 when $x = 3$ and when $x = -1$. So those lines are the vertical asymptotes. (Observe that the numerator is not 0 when $x = 3$ and when $x = -1$.) To obtain horizontal asymptotes, we compute

$\lim_{x \rightarrow +\infty} \frac{2x+3}{\sqrt{x^2-2x-3}}$ and $\lim_{x \rightarrow -\infty} \frac{2x+3}{\sqrt{x^2-2x-3}}$. In both cases, we divide the numerator and denominator by x . The first limit becomes

$$\lim_{x \rightarrow +\infty} \frac{2+3/x}{\sqrt{(x^2-2x-3)/x^2}} = \lim_{x \rightarrow +\infty} \frac{2+3/x}{\sqrt{1-2/x-3/x^2}} = \frac{2+0}{\sqrt{1-0-0}} = 2$$

For the second limit, remember that $x = -\sqrt{x^2}$ when $x < 0$:

$$\lim_{x \rightarrow -\infty} \frac{2x+3}{\sqrt{x^2-2x-3}} = \lim_{x \rightarrow -\infty} \frac{2+3/x}{-\sqrt{(x^2-2x-3)/x^2}} = \lim_{x \rightarrow -\infty} \frac{2+3/x}{-\sqrt{1-2/x-3/x^2}} = \frac{2+0}{-\sqrt{1-0-0}} = -2$$

Hence, the horizontal asymptotes are $y = 2$ on the right and $y = -2$ on the left.

(C) The function is defined only for $x \geq 0$. There are no values $x = c$ such that $f(x)$ approaches ∞ as $x \rightarrow c$. So there are no vertical asymptotes. To find out whether there is a horizontal asymptote, we compute $\lim_{x \rightarrow +\infty} \sqrt{x+1} - \sqrt{x}$:

$$\lim_{x \rightarrow +\infty} (\sqrt{x+1} - \sqrt{x}) = \lim_{x \rightarrow +\infty} \frac{\sqrt{x+1} + \sqrt{x}}{\sqrt{x+1} + \sqrt{x}} = \lim_{x \rightarrow +\infty} \frac{(x+1) - x}{\sqrt{x+1} + \sqrt{x}} = \lim_{x \rightarrow +\infty} \frac{1}{\sqrt{x+1} + \sqrt{x}} = 0$$

Thus, $y = 0$ is a horizontal asymptote on the right.

50. As x approaches 4 from the right, $x - 4 > 0$ and, therefore, $3/(x - 4) > 0$; hence, since $3/(x - 4)$ is getting larger and larger in magnitude, $\lim_{x \rightarrow 4^+} [3/(x - 4)] = +\infty$. As x approaches 4 from the left, $x - 4 < 0$, and, therefore, $3/(x - 4) < 0$; hence, since $3/(x - 4)$ is getting larger and larger in magnitude, $\lim_{x \rightarrow 4^-} [3/(x - 4)] = -\infty$.

51. $x^2 - 7x + 12 = (x - 4)(x - 3)$. As x approaches 3 from the right, $x - 3 > 0$, and, therefore, $1/(x - 3) > 0$ and $1/(x - 3)$ is approaching $+\infty$; at the same time, $1/(x - 4)$ is approaching -1 and is negative. So, as x approaches 3 from the right, $1/(x^2 - 7x + 12)$ is approaching $-\infty$. Thus, $\lim_{x \rightarrow 3^+} \frac{1}{x^2 - 7x + 12} = -\infty$. As x approaches 3 from the left, the only difference from the case just analyzed is that $x - 3 < 0$, and, therefore, $1/(x - 3)$ approaches $-\infty$. Hence, $\lim_{x \rightarrow 3^-} \frac{1}{x^2 - 7x + 12} = +\infty$.

52. $f(x + h) = 4(x + h)^2 - (x + h) = 4(x^2 + 2xh + h^2) - x - h = 4x^2 + 8xh + 4h^2 - x - h$. Hence, $f(x + h) - f(x) = (4x^2 + 8xh + 4h^2 - x - h) - (4x^2 - x) = 8xh + 4h^2 - h$. So,

$$\frac{f(x + h) - f(x)}{h} = \frac{8xh + 4h^2 - h}{h} = 8x + 4h - 1.$$

$$\text{Hence, } \lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h} = \lim_{h \rightarrow 0} (8x + 4h - 1) = 8x - 1.$$

$$\begin{aligned} 53. \quad f(x + h) - f(x) &= \sqrt{2(x + h)} - \sqrt{2x} = (\sqrt{2(x + h)} - \sqrt{2x}) \frac{\sqrt{2(x + h)} + \sqrt{2x}}{\sqrt{2(x + h)} + \sqrt{2x}} \\ &= \frac{2(x + h) - 2x}{\sqrt{2(x + h)} + \sqrt{2x}} \\ &= \frac{2h}{\sqrt{2(x + h)} + \sqrt{2x}} \end{aligned}$$

$$\text{Hence } \lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{2}{\sqrt{2(x + h)} + \sqrt{2x}} = \frac{2}{2\sqrt{2x}} = \frac{1}{\sqrt{2x}}$$

54. Rationalize the numerator:

$$\begin{aligned}\frac{\sqrt{x^2+5}-3}{x^2-2x} \cdot \frac{\sqrt{x^2+5}+3}{\sqrt{x^2+5}+3} &= \frac{(x^2+5)-9}{x(x-2)(\sqrt{x^2+5}+3)} \\ &= \frac{x^2-4}{x(x-2)(\sqrt{x^2+5}+3)} \\ &= \frac{(x+2)(x-2)}{x(x-2)(\sqrt{x^2+5}+3)} \\ &= \frac{x+2}{x(\sqrt{x^2+5}+3)}\end{aligned}$$

We obtain $\lim_{x \rightarrow 2} \frac{x+2}{x(\sqrt{x^2+5}+3)} = \frac{4}{2(\sqrt{9}+3)} = \frac{1}{3}$.

55. (A) $f(x) = x^n \left(a_n + \frac{a_{n-1}}{x} + \cdots + \frac{a_1}{x^{n-1}} + \frac{a_0}{x^n} \right)$.

Now, $\lim_{x \rightarrow +\infty} \frac{a_n-1}{x} = \cdots = \lim_{x \rightarrow +\infty} \frac{a_1}{x^{n-1}} = \lim_{x \rightarrow +\infty} \frac{a_0}{x^n} = 0$ So the sum inside the parentheses

approaches a_n . Since $\lim_{x \rightarrow +\infty} x^n = +\infty$, $\lim_{x \rightarrow +\infty} f(x) = +\infty$.

(B) Dividing the numerator and denominator by x^n ,

$$f(x) = \frac{a_n + (a_{n-1}/x) + \cdots + (a_1/x^{n-1}) + (a_0/x^n)}{b_n + (b_{n-1}/x) + \cdots + (b_1/x^{n-1}) + (b_0/x^n)}$$

Since each a_{n-j}/x^j and b_{n-j}/x^j approaches 0, $\lim_{x \rightarrow +\infty} f(x) = a_n/b_n$.

(C) Factoring out x^n from the numerator and then dividing the numerator and denominator by x^k , $f(x)$ becomes

$$x^{n-k} \left[\frac{a_n + (a_{n-1}/x) + \cdots + (a_1/x^{n-1}) + (a_0/x^n)}{b_k + (b_{k-1}/x) + \cdots + (b_1/x^{k-1}) + (b_0/x^k)} \right]$$

As $x \rightarrow +\infty$, all the quotients a_{n-j}/x^j and b_{k-j}/x^j approach 0, and, therefore, the quantity inside the parentheses approaches $a_n/b_k > 0$. Since x^{n-k} approaches $+\infty$, $\lim_{x \rightarrow +\infty} f(x) = +\infty$.

(D) Dividing the numerator and denominator by x^k ,

$$f(x) = \frac{(a_n/x^{k-n}) + (a_{n-1}/x^{k-n+1}) + \cdots + (a_1/x^{k-1}) + (a_0/x^k)}{b_k + (b_{k-1}/x) + \cdots + (b_1/x^{k-1}) + (b_0/x^k)}$$

Since each of the quotients a_{n-j}/x^{k-n+j} and b_{k-j}/x^j approaches 0, the denominator approaches $b_k > 0$, and the numerator approaches 0. Hence, $\lim_{x \rightarrow \pm\infty} f(x) = 0$.

(E) By Question 55(B), the limit is $\frac{7}{4}$.

(F) By Question 55(C), the limit is $+\infty$.

(G) By Question 55(D), the limit is 0.

(H) Divide the numerator and denominator by $x^{2/3}$, which is essentially the “highest power of x ” in the denominator (pay attention only to the term of highest order). We obtain $\frac{3x^{1/3} - 4/x^{2/3}}{\sqrt[3]{(x^2 - 1)/x^2}} = \frac{3x^{1/3} - 4/x^{2/3}}{\sqrt[3]{1 - 1/x^2}}$.

Since $1/x^2$ approaches 0, the denominator approaches 1. In the numerator, $4/x^{2/3}$ approaches 0. Since $x^{1/3}$ approaches $+\infty$, our limit is $+\infty$. (Note that the situation is essentially the same as in Question 55(C).)

(I) Divide the numerator and denominator by x , which is essentially the highest power in the denominator (forgetting about -2 in $x^3 - 2$). We obtain $\frac{2 + 1/x}{\sqrt[3]{(x^3 - 2)/x^3}} = \frac{2 + 1/x}{\sqrt[3]{1 - 2/x^3}}$.

Since $2/x^3$ and $1/x$ approach 0, our limit is 2. (This is essentially the same situation as in Question 55(B).)

Chapter 7: Continuity

56. $f(a)$ is defined, $\lim_{x \rightarrow a} f(x)$ exists, and $\lim_{x \rightarrow a} f(x) = f(a)$.

57. (A) $x = 0$ is a point of discontinuity because $\lim_{x \rightarrow 0} f(x)$ does not exist. $x = 1$ is a point of discontinuity because $\lim_{x \rightarrow 1} f(x) \neq f(1)$ (since $\lim_{x \rightarrow 1} f(x) = 0$ and $f(1) = 2$).

(B) $f(x)$ is continuous everywhere. In particular, $f(x)$ is continuous at $x = 0$ because $f(0) = (0)^2 = 0$ and $\lim_{x \rightarrow 0} f(x) = 0$.

(C) $f(x)$ is not continuous at $x = 0$ because $\lim_{x \rightarrow 0} f(x)$ does not exist.

(D) Since $x^2 - 4 = (x - 2)(x + 2)$, $f(x) = x - 2$ if $x \neq -2$. So $f(x)$ is not continuous at $x = -2$ because $\lim_{x \rightarrow -2} f(x) \neq f(-2)$ (since $f(-2) = 0$ but $\lim_{x \rightarrow -2} f(x) = -4$). However, $x = -2$ is called a *removable discontinuity* because, if we redefine $f(x)$ at $x = -2$ by setting $f(-2) = -4$ then the new function is continuous at $x = -2$. (Compare Question 57(A).)

(E) Since $x^2 - 1 = (x - 1)(x + 1)$, $f(x) = x + 1$ wherever it is defined. However, $f(x)$ is not defined when $x = 1$, since $(x^2 - 1)/(x - 1)$ does not make sense when $x = 1$. Therefore, $f(x)$ is not continuous at $x = 1$.

(F) Since $x^2 - 9 = (x - 3)(x + 3)$, $f(x) = x + 3$ for $x \neq 3$. However, $f(x) = x + 3$ also when $x = 3$, since $f(3) = 6 = 3 + 3$. Thus, $f(x) = x + 3$ for all x , and, therefore, $f(x)$ is continuous everywhere.

(G) $f(x)$ is discontinuous at $x = 1$ because $\lim_{x \rightarrow 1} f(x)$ does not exist. $f(x)$ is continuous at $x = 2$ because $f(2) = 2 + 1 = 3$ and $\lim_{x \rightarrow 2} f(x) = 3$. Obviously $f(x)$ is continuous for all other x .

(H) See Figure A7.1. $f(x) = \frac{3(x+1)}{(x+1)(x-4)} = \frac{3}{x-4}$. $f(x)$ is discontinuous at $x = 4$ and $x = -1$ because it is not defined at those points. (However, $x = -1$ is a removable discontinuity. If we let $f(-1) = -\frac{3}{5}$, the new function is continuous at $x = -1$.) The only vertical asymptote is $x = 4$. Since $\lim_{x \rightarrow \pm\infty} \frac{3x+3}{x^2-3x-4} = 0$, the x -axis, $y = 0$, is a horizontal asymptote to the right and to the left.

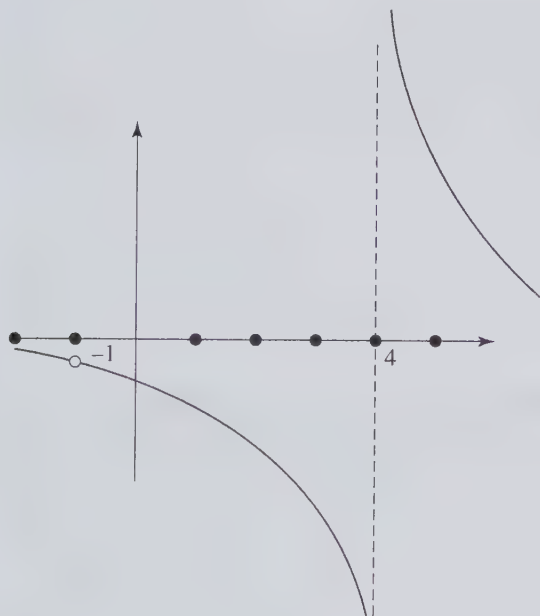


Figure A7.1

58. (A) $f(a)$ is defined, $\lim_{x \rightarrow a^-} f(x)$ exists, and $\lim_{x \rightarrow a^-} f(x) = f(a)$. (B) $f(a)$ is defined, $\lim_{x \rightarrow a^+} f(x)$ exists, and $\lim_{x \rightarrow a^+} f(x) = f(a)$.

59. At $x = 0$, $f(x)$ is not continuous on the left, since $\lim_{x \rightarrow 0^-} f(x) = 3 \neq 1 = f(0)$. At $x = 0$, $f(x)$ is continuous on the right, since $\lim_{x \rightarrow 0^+} f(x) = 1 = f(0)$. At $x = 2$, $f(x)$ is continuous neither on the left nor the right, since $\lim_{x \rightarrow 2^-} f(x) = 2$, $\lim_{x \rightarrow 2^+} f(x) = 0$, but $f(2) = 3$. At $x = 3$, $f(x)$ is continuous on the left, since $\lim_{x \rightarrow 3^-} f(x) = 2 = f(3)$. At $x = 3$, $f(x)$ is not continuous on the right since $\lim_{x \rightarrow 3^+} f(x) = 0 \neq f(3)$.

60. $f(x)$ is continuous, $f(1) = -1$, and $f(2) = 6$. Since $f(1) < 0 < f(2)$, the intermediate value theorem implies that there must be a number c in the interval $(1, 2)$ such that $f(c) = 0$.

61. $f(-4) = 0$, $f(0) = 4$, f is continuous on $[-4, 0]$, and $0 < \sqrt{7} < 4$. We must find a number c in $[-4, 0]$ such that $f(c) = \sqrt{7}$. So, $\sqrt{16 - c^2} = \sqrt{7}$, $16 - c^2 = 7$, $c^2 = 9$, $c = \pm 3$. Hence, the desired value of c is -3 .

62. (A) No. $f(0) = 0$, but $\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \frac{1}{x} = +\infty$.

(B) Yes. When continuity *over an interval* is considered, at the endpoints we are concerned with only the one-sided limit. So, although f is discontinuous at $x = 1$, the left-hand limit at 1 is 2 and $f(1) = 2$.

(C) No. The right-hand limit at $x = 1$ is $\lim_{x \rightarrow 1^+} (x - 1) = 0$, whereas $f(1) = 2$.

63. Since $\lim_{x \rightarrow 0^-} 3x^2 - 1 = -1$, the value of $cx + d$ at $x = 0$ must be -1 ; that is, $d = -1$.

Since $\lim_{x \rightarrow 1^+} \sqrt{x + 8} = 3$, the value of $cx + d$ at $x = 1$ must be 3; that is, $3 = c(1) - 1$, $c = 4$.

Chapter 8: The Derivative

64. (A) $f(x + \Delta x) = A(x + \Delta x) + B = Ax + A\Delta x + B$. Then $f(x + \Delta x) - f(x) = (Ax + A\Delta x + B) - (Ax + B) = A\Delta x$.

Hence, $\frac{f(x + \Delta x) - f(x)}{\Delta x} = A$. Thus, $f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} = \lim_{\Delta x \rightarrow 0} A = A$.

(B) $f(x + \Delta x) = 2(x + \Delta x)^2 - 3(x + \Delta x) + 5 = 2[x^2 + 2x\Delta x + (\Delta x)^2] - 3x - 3\Delta x + 5 = 2x^2 + 4x\Delta x + 2(\Delta x)^2 - 3x - 3\Delta x + 5$. So, $f(x + \Delta x) - f(x) = [2x^2 + 4x\Delta x + 2(\Delta x)^2 - 3x - 3\Delta x + 5] - (2x^2 - 3x + 5) = 4x\Delta x + 2(\Delta x)^2 - 3\Delta x = (\Delta x)(4x + 2\Delta x - 3)$.

Thus, $\frac{f(x + \Delta x) - f(x)}{\Delta x} = 4x + 2\Delta x - 3$. Hence, $f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} =$

$$\lim_{\Delta x \rightarrow 0} (4x + 2\Delta x - 3) = 4x - 3.$$

(C) $f(x + \Delta x) = (x + \Delta x)^3 = x^3 + 3x^2 \Delta x + 3x(\Delta x)^2 + (\Delta x)^3$. So, $f(x + \Delta x) - f(x) = [x^3 + 3x^2 \Delta x + 3x(\Delta x)^2 + (\Delta x)^3] - x^3 = 3x^2 \Delta x + 3x(\Delta x)^2 + (\Delta x)^3 = (\Delta x)(3x^2 + 3x \Delta x + (\Delta x)^2)$. So $\frac{f(x + \Delta x) - f(x)}{\Delta x} = 3x^2 + 3x \Delta x + (\Delta x)^2$. Thus, $f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} =$

$$\lim_{\Delta x \rightarrow 0} [3x^2 + 3x \Delta x + (\Delta x)^2] = 3x^2.$$

65. (A) $f'(x) = na_n x^{n-1} + (n-1)a_{n-1} x^{n-2} + \cdots + 2a_2 x + a_1$.

(B) $f'(x) = 35x^4 - 12x^3 + 12x + 3$.

66. (A) $D_x[f(x) + g(x)] = D_x f(x) + D_x g(x)$

$$D_x[f(x) \cdot g(x)] = f(x) \cdot D_x g(x) + g(x) \cdot D_x f(x)$$

$$D_x \left[\frac{f(x)}{g(x)} \right] = \frac{g(x) \cdot D_x f(x) - f(x) \cdot D_x g(x)}{[g(x)]^2}$$

(Note the various ways of denoting a derivative: $f'(x)$, $D_x f(x)$, $\frac{dy}{dx}$, y' .)

(B) $F'(x) = (5x^3 - 20x + 13)(24x^5 + 10x^4 - 14x + 2) + (4x^6 + 2x^5 - 7x^2 + 2x)(15x^2 - 20)$. (In such cases, do not bother to carry out the tedious multiplications unless a particular question requires it.)

(C) $G'(x) = \frac{(x^2 + 7)(3) - (3x - 2)(2x)}{(x^2 + 7)^2} = \frac{3x^2 + 21 - 6x^2 + 4x}{(x^2 + 7)^2} = \frac{-3x^2 + 4x + 21}{(x^2 + 7)^2}$.

(D) (i) $-40x^4 + 3\sqrt{3}x^2 + 4\pi x$.

(ii) $102x^{50} + 36x^{11} - 28 + \sqrt[3]{7}$.

67. (A) When $x = 3$, $f(x) = 45$. So the point is $(3, 45)$. Recall that the slope of the tangent line is the derivative $f'(x)$, evaluated for the given value of x . But $f'(x) = 12x^2 - 14x$. Hence, $f'(3) = 12(9) - 14(3) = 66$. Thus, the slope-intercept equation of the tangent line has the form $y = 66x + b$. Since the point $(3, 45)$ is on the tangent line, $45 = 66(3) + b$, and, therefore, $b = -153$. Thus, the equation is $y = 66x - 153$.

(B) The derivative is $y' = 5x^4 + 4$. Hence, the slope of the tangent line at a point $A(x_0, y_0)$ of the graph is $5x_0^4 + 4$. The line $\overleftrightarrow{AB}$ has slope $\frac{(x_0^5 + 4x_0 - 3) - 1}{x_0 - 0} = \frac{x_0^5 + 4x_0 - 4}{x_0}$.

So the line $\overleftrightarrow{AB}$ is the tangent line if and only if $(x_0^5 + 4x_0 - 4)/x_0 = 5x_0^4 + 4$. Solving, $x_0 = -1$. So there is only one point $(-1, -8)$.

68. The normal line is the line perpendicular to the tangent line. Since $y' = 3x^2 - 2x$, the slope of the tangent line at $x = 1$ is $3(1)^2 - 2(1) = 1$. Hence, the slope of the normal line is the negative reciprocal of 1, namely -1 . Thus, the required slope-intercept equation has the form $y = -x + b$. On the curve, when $x = 1$, $y = (1)^3 - (1)^2 = 0$. So the point $(1, 0)$ is on the normal line, and, therefore, $0 = -1 + b$. Thus, $b = 1$, and the required equation is $y = -x + 1$.

69. Recall the definition of the derivative: When $f(x) = 5x^4$, $f'(x) = \lim_{\Delta x \rightarrow 0} \frac{5(x + \Delta x)^4 - 5x^4}{\Delta x}$.

In particular, for $x = \frac{1}{3}$, $f'\left(\frac{1}{3}\right) = \lim_{\Delta x \rightarrow 0} \frac{5\left(\frac{1}{3} + \Delta x\right)^4 - 5\left(\frac{1}{3}\right)^4}{\Delta x}$. If we replace Δx by h in this limit, we obtain the limit to be evaluated, which is therefore equal to $f'\left(\frac{1}{3}\right)$. But $f'(x) = 20x^3$. So the value of the limit is $20\left(\frac{1}{3}\right)^3 = \frac{20}{27}$.

70. $y' = x^2 - 1$ is the slope of the tangent line. To be parallel to the line $y = 3x$ having slope 3, it also must have slope 3. Hence, $x^2 - 1 = 3$, $x^2 = 4$, $x = \pm 2$. Thus, the points are

$$\left(2, \frac{2}{3}\right) \text{ and } \left(-2, -\frac{2}{3}\right).$$

71. (A) $f(x + \Delta x) - f(x) = \{[f(x + \Delta x) - f(x)]/\Delta x\} \Delta x$. Thus,

$$\lim_{\Delta x \rightarrow 0} [f(x + \Delta x) - f(x)] = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} \cdot \lim_{\Delta x \rightarrow 0} \Delta x = f'(x) \cdot 0 = 0.$$

$$\text{Hence, } 0 = \lim_{\Delta x \rightarrow 0} [f(x + \Delta x) - f(x)] = \lim_{\Delta x \rightarrow 0} f(x + \Delta x) - \lim_{\Delta x \rightarrow 0} f(x) = \lim_{\Delta x \rightarrow 0} f(x + \Delta x) - f(x).$$

So $\lim_{\Delta x \rightarrow 0} f(x + \Delta x) = f(x)$, and f is continuous at x .

(B) Consider the function $f(x) = |x|$ at $x = 0$. Clearly, f is continuous everywhere.

However, $\frac{f(0 + \Delta x) - f(0)}{\Delta x} = \frac{|\Delta x| - 0}{\Delta x} = \frac{|\Delta x|}{\Delta x}$. When $\Delta x > 0$, $|\Delta x|/\Delta x = 1$, and, when $\Delta x < 0$, $|\Delta x|/\Delta x = -1$.

Therefore, $\lim_{\Delta x \rightarrow 0} \frac{|\Delta x|}{\Delta x} = f'(0)$ does not exist.

72. The slope of the tangent line is the derivative $f'(x) = 9x^2 - 22x - 15$. The tangent line is horizontal when and only when its slope is 0. Hence, we set $9x^2 - 22x - 15 = 0$, $(9x + 5)(x - 3) = 0$; $x = 3$ or $x = -\frac{5}{9}$. Thus, the desired points are $(3, 0)$ and $\left(-\frac{5}{9}, \frac{16384}{243}\right)$.

73. The graph (Figure A8.1), reveals a sharp point at $x = 3$, $y = 0$, where there is no unique tangent line. Thus the function is not differentiable at $x = 3$. (This can be verified in a more rigorous way by considering the Δ -definition.)

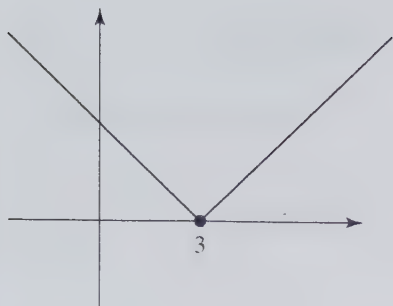


Figure A8.1

Chapter 9: The Chain Rule

$$74. (f \circ g)(x) = f(g(x)) = f(x^3) = (x^3)^2 + 2(x^3) - 5 = x^6 + 2x^3 - 5$$

$$(g \circ f)(x) = g(f(x)) = g(x^2 + 2x - 5) = (x^2 + 2x - 5)^3$$

$$75. \text{ Let } g(x) = 3x - 5 \text{ and let } f(x) = \sqrt{x}. \text{ Then } (f \circ g)(x) = f(g(x)) = f(3x - 5) = \sqrt{3x - 5}.$$

76.

$$(f \circ g)(x) = f(g(x)) = f\left(\frac{1}{x-1}\right) = \left(\frac{2}{x-1}\right) \text{ and } (g \circ f)(x) = g(f(x)) = g(2x) = \frac{1}{2x-1}.$$

$$\text{So we must solve } \frac{2}{x-1} = \frac{1}{2x-1}, \quad 4x-2 = x-1, \quad 3x=1, \quad x=\frac{1}{3}.$$

$$77. \text{ (A) } (f \circ g)'(x) = f'(g(x)) \cdot g'(x).$$

$$\text{ (B) } \frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}. \text{ Here the first occurrence of } y \text{ refers to } y \text{ as a function of } x, \text{ while}$$

the second occurrence of y (on the right side) refers to y as a function of u .

$$\text{ (C) Use the chain rule. Think of the function as a composition } (f \circ g)(x) \text{ where } f(x) = x^4 \text{ and } g(x) = x^3 - 2x^2 + 7x - 3. \text{ Then } f'(x) = 4x^3 \text{ and } g'(x) = 3x^2 - 4x + 7. \text{ Hence,}$$

$$\frac{d}{dx} (x^3 - 2x^2 + 7x - 3)^4 = 4(x^3 - 2x^2 + 7x - 3)^3 \cdot (3x^2 - 4x + 7).$$

$$\text{ (D) We can write } \frac{1}{(3x^2 + 5)^4} = (3x^2 + 5)^{-4}. \text{ Now we can use the chain rule.}$$

$$\text{Remember that } \frac{d}{dx} x^r = rx^{r-1} \text{ for any real number } r. \text{ In particular, } \frac{d}{dx} x^{-4} = -4x^{-5}. \text{ By the}$$

$$\text{chain rule, } \frac{d}{dx} (3x^2 + 5)^{-4} = -4(3x^2 + 5)^{-5} \cdot (6x) = -\frac{24x}{(3x^2 + 5)^5}.$$

(E) We can write $\sqrt{2x+7} = (2x+7)^{1/2}$. By the chain rule,

$$\frac{d}{dx}(2x+7)^{1/2} = \frac{1}{2}(2x+7)^{-1/2} \cdot (2) = (2x+7)^{-1/2} = \frac{1}{(2x+7)^{1/2}} = \frac{1}{\sqrt{2x+7}}.$$

$$\left(\text{Here, we have used } \frac{d}{dv} v^{1/2} = \frac{1}{2} v^{-1/2} \text{ and } \frac{d}{dx}(2x+7) = 2. \right)$$

(F) Use the chain rule. $\frac{d}{dx} \left(\frac{x+2}{x-3} \right)^3 = 3 \left(\frac{x+2}{x-3} \right)^2 \cdot \frac{d}{dx} \left(\frac{x+2}{x-3} \right)$. Here we must

$$\text{calculate } \frac{d}{dx} \left(\frac{x+2}{x-3} \right) \text{ by the quotient rule: } \frac{d}{dx} \left(\frac{x+2}{x-3} \right) = \frac{(x-3) \cdot 1 - (x+2) \cdot 1}{(x-3)^2} = -\frac{5}{(x-3)^2}.$$

$$\text{Hence, } \frac{d}{dx} \left(\frac{x+2}{x-3} \right)^3 = 3 \left(\frac{x+2}{x-3} \right)^2 \cdot \left[-\frac{5}{(x-3)^2} \right] = -15 \frac{(x+2)^2}{(x-3)^4}.$$

78. Think of this function as a product of $(4x^2 - 3)^2$ and $(x+5)^3$, and first apply the product rule: $\frac{d}{dx} [(4x^2 - 3)^2(x+5)^3] = (4x^2 - 3)^2 \cdot \frac{d}{dx} (x+5)^3 + (x+5)^3 \cdot \frac{d}{dx} (4x^2 - 3)^2$. By the chain rule, $\frac{d}{dx} (x+5)^3 = 3(x+5)^2 \cdot 1 = 3(x+5)^2$, and $\frac{d}{dx} (4x^2 - 3)^2 = 2(4x^2 - 3) \cdot (8x) = 16x(4x^2 - 3)$. Thus, $\frac{d}{dx} [(4x^2 - 3)^2(x+5)^3] = (4x^2 - 3)^2 \cdot 3(x+5)^2 + (x+5)^3 \cdot 16x(4x^2 - 3)$.

We can factor out $(4x^2 - 3)$ and $(x+5)^2$ to obtain $(4x^2 - 3)(x+5)^2[3(4x^2 - 3) + 16x(4x+5)] = (4x^2 - 3)(x+5)^2(12x^2 - 9 + 16x^2 + 80x) = (4x^2 - 3)(x+5)^2(28x^2 + 80x - 9)$.

$$79. \text{ By the quotient rule, } y' = \frac{(x^2+1) \frac{d}{dx} \sqrt{x-1} - \sqrt{x-1}(2x)}{(x^2+1)^2}.$$

$$\text{By the chain rule, } \frac{d}{dx} \sqrt{x-1} = \frac{d}{dx} (x-1)^{1/2} = \frac{1}{2} (x-1)^{-1/2} = \frac{1}{2\sqrt{x-1}}.$$

$$\text{Thus, } y' = \frac{(x^2+1) \frac{1}{2\sqrt{x-1}} - 2x\sqrt{x-1}}{(x^2+1)^2}.$$

When $x = 2$, $\sqrt{x-1} = 1$, and therefore, at the point $\left(2, \frac{1}{5}\right)$, $y' = -\frac{3}{50}$. Hence, a point-slope equation of the tangent line is $y - \frac{1}{5} = -\frac{3}{50}(x - 2)$. Solving for y , we obtain the slope-intercept equation $y = -\frac{3}{50}x + \frac{8}{25}$.

80. $y = (x^2 + 16)^{1/2}$. Hence, by the chain rule,

$$y' = \frac{1}{2}(x^2 + 16)^{-1/2} \cdot \frac{d}{dx}(x^2 + 16) = \frac{1}{2}(x^2 + 16)^{-1/2} \cdot (2x) = x(x^2 + 16)^{-1/2} = \frac{x}{\sqrt{x^2 + 16}}.$$

At the point $(3, 5)$, $\sqrt{x^2 + 16} = 5$, and, therefore, $y' = \frac{3}{5}$. This is the slope tangent line. Hence, the slope of the normal line is $-\frac{5}{3}$, and a point-slope equation for it is $y - 5 = -\frac{5}{3}(x - 3)$. Solving for y , we obtain the slope-intercept equation $y = -\frac{5}{3}x + 10$.

81. $\frac{dy}{dz} = \frac{dy}{dx} \cdot \frac{dx}{dz} = 3x^2 \cdot (6z) = 18zx^2 = 18z(3z^2 + 5)^2.$

82. By the chain rule, $H'(x) = F'(G(x)) \cdot G'(x)$. Hence, $H'(3) = F'(G(3)) \cdot G'(3) = F'(7) \cdot 6 = 2 \cdot 6 = 12$.

83. $F(x) = (1 + x^2)^{4/3}$. By the chain rule,

$$F'(x) = \frac{4}{3}(1 + x^2)^{1/3} \cdot \frac{d}{dx}(1 + x^2) = \frac{4}{3}(1 + x^2)^{1/3} \cdot (2x) = \frac{8}{3}x\sqrt[3]{1 + x^2}.$$

84. We are asked to find the value of dy/dt when $t = 4$. $dy/dx = 3x^2 - 3 = 3(x^2 - 1)$, and $dx/dt = 1/(4\sqrt{t})$.

Hence, $\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt} = \frac{3(x^2 - 1)}{4\sqrt{t}}$ when $t = 4$, $x = 4$, and $\frac{dy}{dt} = 3(16 - 1)/(4 \cdot 2) = \frac{45}{8}$ per unit of time.

Chapter 10: Trigonometric Functions and Their Derivatives

85. Consider a circle with a radius of one unit (Figure A10.1). Let the center be C , and let CA and CB be two radii for which the intercepted arc $\widehat{AB}$ of the circle has length 1. Then the central angle $\angle ABC$ has a measure of one radian.

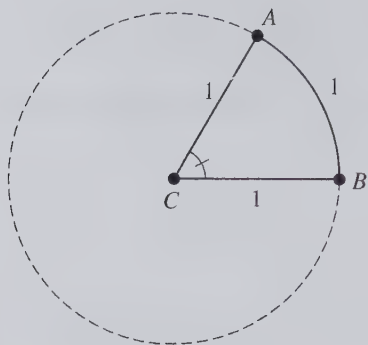


Figure A10.1

86. 2π radians is the same as 360 degrees. Hence, 1 radian $= 180/\pi$ degrees, and 1 degree $= \pi/180$ radians. So if an angle has a measure of D degrees and R radians, then $D = (180/\pi) R$ and $R = (\pi/180)D$.

87. We use the formula $R = (\pi/180)D$. Hence $30^\circ = \pi/6$ radians, $45^\circ = \pi/4$ radians, $60^\circ = \pi/3$ radians, $90^\circ = \pi/2$ radians, $120^\circ = 2\pi/3$ radians, $135^\circ = 3\pi/4$ radians, $180^\circ = \pi$ radians, $270^\circ = 3\pi/2$ radians, $360^\circ = 2\pi$ radians.

88. We use the formula $D = (180/\pi) R$. Thus, $3\pi/5$ radians $= 108^\circ$ and $5\pi/6$ radians $= 150^\circ$.

89. The arc length s , the radius r , and the central angle θ (measured in radians) are related by the equation $s = r\theta$. In this case, $r = 10$ inches and $\theta = \pi/5$. Hence, $s = 2\pi$ inches.

90. See Figure A10.2. $\pi/3$ radians $= 60^\circ$, and the minus sign indicates that a 60° rotation is to be taken in the clockwise direction. (Positive angles correspond to counter-clockwise rotations.)

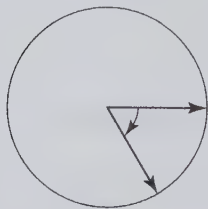


Figure A10.2

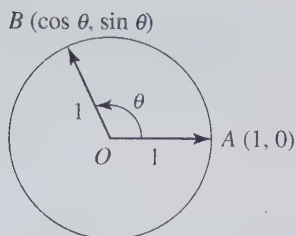


Figure A10.3

91. Refer to Figure A10.3. Place an arrow $\overrightarrow{OA}$ of unit length so that its initial point O is the origin of a coordinate system and its endpoint A is $(1, 0)$. Rotate $\overrightarrow{OA}$ about the point O through an angle with radian measure θ . Let $\overrightarrow{OB}$ be the final position of the arrow after the rotation. Then $\cos \theta$ is defined to be the x -coordinate of B , and $\sin \theta$ is defined to be the y -coordinate of B .

92.

θ	$\sin \theta$	$\cos \theta$
0	0	1
$\pi/6$	$1/2$	$\sqrt{3}/2$
$\pi/4$	$\sqrt{2}/2$	$\sqrt{2}/2$
$\pi/3$	$\sqrt{3}/2$	$1/2$
$\pi/2$	1	0
π	0	-1
$3\pi/2$	-1	0
2π	0	1

Notice that $9\pi/4 = 2\pi + \pi/4$, and the sine and cosine functions have a period of 2π ; that is, $\sin(\theta + 2\pi) = \sin \theta$ and $\cos(\theta + 2\pi) = \cos \theta$. Hence, $\sin(9\pi/4) = \sin(\pi/4) = \sqrt{2}/2$ and $\cos(9\pi/4) = \cos(\pi/4) = \sqrt{2}/2$.

93. (A) In general, $\cos(-\theta) = \cos \theta$. Hence, $\cos(-\pi/6) = \cos(\pi/6) = \sqrt{3}/2$. (B) In general, $\sin(-\theta) = -\sin \theta$. Hence, $\sin(-\pi/6) = -\sin(\pi/6) = -\frac{1}{2}$. (C) $2\pi/3 = \pi/2 + \pi/6$. We use the identity $\cos(\theta + \pi/2) = -\sin \theta$. Thus, $\cos(2\pi/3) = -\sin(\pi/6) = -\frac{1}{2}$. (D) We use the identity $\sin(\theta + \pi/2) = \cos \theta$. Thus, $\sin(2\pi/3) = \cos(\pi/6) = \sqrt{3}/2$.

94. We use the values calculated in Question 92 to draw Figure A10.4.

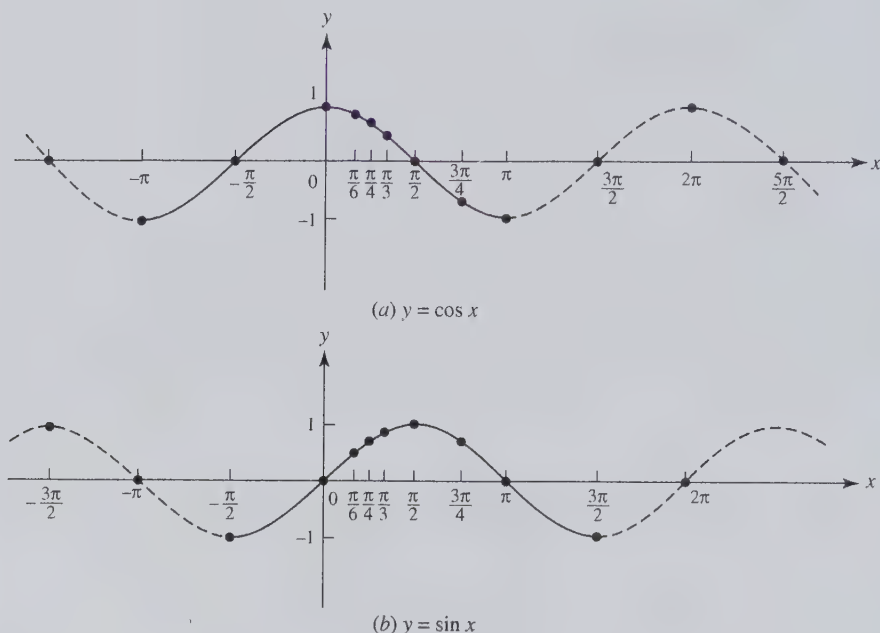


Figure A10.4

95. Because $\cos 3(x + 2\pi/3) = \cos(3x + 2\pi) = \cos 3x$, the function is of period $p = 2\pi/3$. Hence, the length of each wave is $2\pi/3$. The number f of waves over an interval of length 2π is 3. (In general, this number f , called the *frequency* of the function, is given by the equation $f = 2\pi/p$.) The graph is indicated in Figure A10.5.

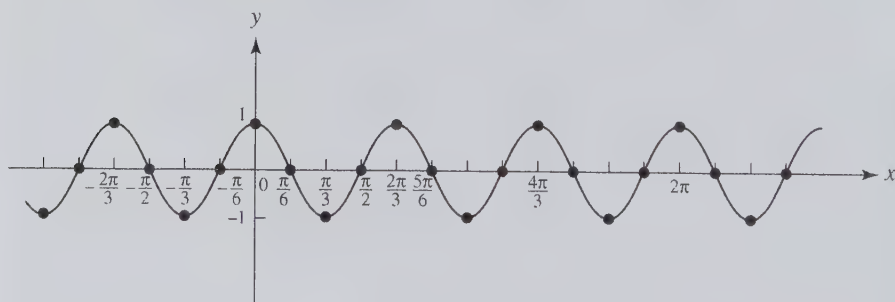


Figure A10.5

96. The period $p = \pi/2$. (In general, $p = 2\pi/b$, where b is the coefficient of x .) The coefficient 1.5 is the *amplitude*, the greatest height above the x -axis reached by points of the graph. The graph looks like Figure A10.6.

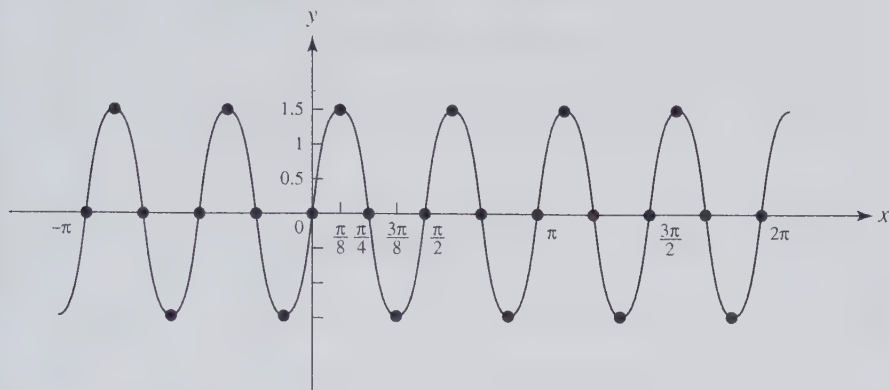


Figure A10.6

$$97. \text{ (A) } \lim_{x \rightarrow 0} \frac{\sin 3x}{x} = \lim_{x \rightarrow 0} 3 \frac{\sin 3x}{3x} = 3 \cdot \lim_{x \rightarrow 0} \frac{\sin 3x}{3x} = 3 \cdot \lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 3 \cdot 1 = 3.$$

$$\begin{aligned} \text{ (B) } \lim_{x \rightarrow 0} \frac{\sin 2x}{\sin 3x} &= \lim_{x \rightarrow 0} \left(\frac{\sin 2x}{2x} \cdot \frac{3x}{\sin 3x} \cdot \frac{2}{3} \right) \\ &= \lim_{x \rightarrow 0} \frac{\sin 2x}{2x} \cdot \lim_{x \rightarrow 0} \frac{3x}{\sin 3x} \cdot \frac{2}{3} \\ &= \lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} \cdot \lim_{\theta \rightarrow 0} \frac{\theta}{\sin \theta} \cdot \frac{2}{3} = 1 \cdot 1 \cdot \frac{2}{3} = \frac{2}{3}. \end{aligned}$$

$$\begin{aligned} \text{ (C) } \lim_{x \rightarrow 0} \frac{1 - \cos x}{x} &= \lim_{x \rightarrow 0} \frac{1 - \cos x}{x} \cdot \frac{1 + \cos x}{1 + \cos x} \\ &= \lim_{x \rightarrow 0} \frac{1 - \cos^2 x}{x(1 + \cos x)} \\ &= \lim_{x \rightarrow 0} \frac{\sin^2 x}{x(1 + \cos x)} = \lim_{x \rightarrow 0} \frac{\sin x}{x} \cdot \lim_{x \rightarrow 0} \frac{\sin x}{1 + \cos x} = 1 \cdot \frac{0}{2} = 0. \end{aligned}$$

98. (A) By the identity $\sin(u+v) = \sin u \cos v + \cos u \sin v$, $\sin(x+\Delta x) = \sin x \cos(\Delta x) + \cos x \sin(\Delta x)$. Hence, $\sin(x+\Delta x) - \sin x = \sin x[\cos(\Delta x) - 1] + \cos x \sin(\Delta x)$, and

$$\frac{\sin(x+\Delta x) - \sin x}{\Delta x} = \sin x \cdot \frac{\cos(\Delta x) - 1}{\Delta x} + \cos x \cdot \frac{\sin(\Delta x)}{\Delta x}.$$

$$\begin{aligned}
 \text{Thus, } \frac{d}{dx}(\sin x) &= \lim_{\Delta x \rightarrow 0} \frac{\sin(x + \Delta x) - \sin x}{\Delta x} \\
 &= \sin x \cdot \lim_{\Delta x \rightarrow 0} \frac{\cos(\Delta x) - 1}{\Delta x} + \cos x \cdot \lim_{\Delta x \rightarrow 0} \frac{\sin(\Delta x)}{\Delta x} \\
 &= (\sin x)(0) + (\cos x)(1) = \cos x.
 \end{aligned}$$

Here, we have used $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$ and $\lim_{\theta \rightarrow 0} \frac{\cos \theta - 1}{\theta} = 0$ (Question 97(C)).

$$\begin{aligned}
 \text{(B) By the identity } \cos x &= \sin\left(\frac{\pi}{2} - x\right), \frac{d}{dx}(\cos x) = \frac{d}{dx}\left[\sin\left(\frac{\pi}{2} - x\right)\right] = \\
 \cos\left(\frac{\pi}{2} - x\right) \cdot \frac{d}{dx}\left(\frac{\pi}{2} - x\right) & \text{ [chain rule]} = \sin x \cdot (-1) \left[\sin x = \cos\left(\frac{\pi}{2} - x\right)\right] = -\sin x.
 \end{aligned}$$

(C) $\sin 3x$ is a composite function of $3x$ and the sine function. By the chain rule and the fact that $\frac{d}{dx}(\sin x) = \cos x$, $\frac{d}{dx}(\sin 3x) = \cos 3x \cdot \frac{d}{dx}(3x) = \cos 3x \cdot 3 = 3 \cos 3x$.

(D) $\cos^2 x = (\cos x)^2$. Hence, by the chain rule, $\frac{d}{dx}(\cos^2 x) = 2(\cos x) \cdot \frac{d}{dx}(\cos x) = 2 \cos x \cdot (-\sin x) = -2 \sin x \cos x = -\sin 2x$.

$$\begin{aligned}
 \text{(E) } \sqrt{\sin x} &= (\sin x)^{1/2}. \text{ By the chain rule, } \frac{d}{dx}(\sqrt{\sin x}) = \frac{1}{2}(\sin x)^{-1/2} \cdot \frac{d}{dx}(\sin x) = \\
 \frac{1}{2\sqrt{\sin x}} \cdot \cos x &= \frac{\cos x}{2\sqrt{\sin x}}.
 \end{aligned}$$

99. The slope of the tangent line is the derivative y' . By the chain rule, since $\sin^2 x = (\sin x)^2$, $y' = 2(\sin x) \cdot \frac{d}{dx}(\sin x) = 2 \sin x \cos x$. When $x = \pi/3$, $\sin x = \sqrt{3}/2$ and $\cos x = \frac{1}{2}$. So $y' = 2 \cdot \sqrt{3}/2 \cdot \frac{1}{2} = \sqrt{3}/2$. At the point where $x = \pi/3$, $y = (\sqrt{3}/2)^2 = \frac{3}{4}$. So a point-slope equation of the tangent line is $y - \frac{3}{4} = (\sqrt{3}/2)(x - \pi/3)$.

100. (A) Remember that $\tan x = \sin x / \cos x$ and $\sec x = 1 / \cos x$. By the quotient rule,

$$\begin{aligned}
 \frac{d}{dx}(\tan x) &= \frac{d}{dx}\left(\frac{\sin x}{\cos x}\right) = \frac{(\cos x) \frac{d}{dx}(\sin x) - (\sin x) \frac{d}{dx}(\cos x)}{(\cos x)^2} \\
 &= \frac{(\cos x)(\cos x) - (\sin x)(-\sin x)}{(\cos x)^2} = \frac{\cos^2 x + \sin^2 x}{(\cos x)^2} \\
 &= \frac{1}{(\cos x)^2} = \sec^2 x.
 \end{aligned}$$

(B) Note that $\tan(\pi/3) = \sin(\pi/3)/\cos(\pi/3) = (\sqrt{3}/2)/\frac{1}{2} = \sqrt{3}$, and $\sec(\pi/3) = 1/\cos(\pi/3) = 1/\frac{1}{2} = 2$. By the chain rule, $y = 2(\tan x) \cdot \frac{d}{dx}(\tan x) = 2(\tan x)(\sec^2 x)$. Thus, when $x = \pi/3$, $y' = 2\sqrt{3} \cdot 4 = 8\sqrt{3}$, so the slope of the tangent line is $8\sqrt{3}$. Hence, a point-slope equation of the tangent line is $y - 3 = 8\sqrt{3}(x - \pi/3)$.

(C) By the chain rule,

$$\begin{aligned}\frac{d}{dx}(\sec x) &= \frac{d}{dx}(\cos x)^{-1} = -(\cos x)^{-2} \cdot \frac{d}{dx}(\cos x) = -\frac{1}{\cos^2 x} \cdot (-\sin x) \\ &= \frac{\sin x}{\cos^2 x} = \frac{1}{\cos x} \cdot \frac{\sin x}{\cos x} = \sec x \tan x.\end{aligned}$$

101. By the chain rule, $y' = 3 \left[2 \sec x \cdot \frac{d}{dx}(\sec x) \right] = 3(2 \sec x \cdot \sec x \cdot \tan x) = 6 \sec^2 x \tan x$.

So the slope of the tangent line is $y' = 6 \left(\frac{4}{3} \right) (\sqrt{3}/3) = 8\sqrt{3}/3$. Hence, the slope of the normal line is the negative reciprocal $-\sqrt{3}/8$. Thus, a point-slope equation of the normal line is $y - 4 = -(\sqrt{3}/8)(x - \pi/6)$.

102. $\frac{\tan x}{x} = \frac{\sin x}{\cos x} \cdot \frac{1}{x} = \frac{\sin x}{x} \cdot \frac{1}{\cos x}$. Hence, $\lim_{x \rightarrow 0} \frac{\tan x}{x} = \lim_{x \rightarrow 0} \frac{\sin x}{x} \cdot \lim_{x \rightarrow 0} \frac{1}{\cos x} = 1 \cdot \frac{1}{1} = 1$.

103. A line is horizontal when and only when its slope is 0. The slope of the tangent line is $y' = D_x(\sec x) = \sec x \tan x$. Hence, we must solve $\sec x \tan x = 0$. Since $\sec x = 1/\cos x$, $\sec x$ is never 0. Hence, $\tan x = 0$. But, since $\tan x = \sin x/\cos x$, $\tan x = 0$ is equivalent to $\sin x = 0$. The latter occurs when and only when $x = n\pi$ for some integer n .

Chapter 11: Rolle's Theorem, the Mean Value Theorem, and the Sign of the Derivative

104. If f is continuous over a closed interval $[a, b]$ and differentiable on the open interval (a, b) , and if $f(a) = f(b) = 0$, then there is at least one number c in (a, b) such that $f'(c) = 0$.

105. $f(x)$ is clearly differentiable everywhere, and $f(-1) = f(3) = 0$. Hence, Rolle's theorem applies. $f'(x) = 2x - 2$. Setting $f'(x) = 0$, we obtain $x = 1$. Thus, $f'(1) = 0$ and $-1 < 1 < 3$.

106. There is a discontinuity at $x = 1$, since $\lim_{x \rightarrow 1} f(x)$ does not exist. Hence, Rolle's theorem does not apply.

107. $f(x)$ is differentiable within $(0, 8)$ but not at 0. However, it is continuous at $x = 0$ and, therefore, throughout $[0, 8]$. Also, $f(0) = f(8) = 0$. Hence, Rolle's theorem applies. $f'(x) = 2/3\sqrt[3]{x} - 2/3(\sqrt[3]{x})^2$. Setting $f'(x) = 0$, we obtain $x = 1$, which is between 0 and 8.

108. (A) If $f(x)$ is continuous on the closed interval $[a, b]$ and differentiable on the open interval (a, b) , then there is a number c in (a, b) such that $f'(c) = \frac{f(b) - f(a)}{b - a}$.

(B) $f'(x) = 2$. Hence, the mean value theorem applies. Note that $\frac{f(4) - f(1)}{4 - 1} = \frac{11 - 5}{4 - 1} = 2$. Thus, we can take c to be any point in $(1, 4)$.

(C) $f'(x) = 6x - 5$, and the mean value theorem applies. Setting $6c - 5 = \frac{f(5) - f(2)}{5 - 2} = \frac{51 - 3}{3} = 16$, we find $c = \frac{21}{6} = \frac{7}{2}$, which lies between 2 and 5.

(D) $f(x)$ is continuous for $x \geq 0$ and differentiable for $x > 0$. Thus, the mean value theorem is applicable.

$f'(x) = \frac{3}{4\sqrt[4]{x}}$. Setting $\frac{3}{4\sqrt[4]{c}} = \frac{f(16) - f(0)}{16 - 0} = \frac{8 - 0}{16} = \frac{1}{2}$, we find $c = \frac{81}{16}$, which lies between 0 and 16.

109. Assume $a < u < v < b$. Then the mean value theorem applies to $f(x)$ on the closed interval (u, v) . So for some c between u and v , $f'(c) = [f(v) - f(u)]/(v - u)$. Hence, $f(v) - f(u) = f'(c)(v - u)$. Since $u < v$, $v - u > 0$. By hypothesis, $f'(c) > 0$. Hence, $f(v) - f(u) > 0$, and $f(v) > f(u)$. Thus, $f(x)$ is increasing in (a, b) .

110. (A) $f'(x) = 3$. Hence, $f(x)$ is increasing everywhere.

(B) $f'(x) = 2x - 4$. Since $2x - 4 > 0 \Leftrightarrow x > 2$, $f(x)$ is increasing when $x > 2$. Similarly, since $2x - 4 < 0 \Leftrightarrow x < 2$, $f(x)$ is decreasing when $x < 2$.

(C) $f'(x) = 3x^2 - 18x + 15 = 3(x - 5)(x - 1)$. The key points are $x = 1$ and $x = 5$. $f'(x) > 0$ when $x > 5$, $f'(x) < 0$ for $1 < x < 5$, and $f'(x) > 0$ when $x < 1$. Thus, $f(x)$ is increasing when $x < 1$ or $x > 5$, and it is decreasing when $1 < x < 5$.

(D) $f(x)$ is defined for $x \neq 0$. $f'(x) = 1 - (1/x^2)$. Hence, $f'(x) < 0 \Leftrightarrow 1 < 1/x^2$, which is equivalent to $x^2 < 1$. Hence, $f(x)$ is decreasing when $-1 < x < 0$ or $0 < x < 1$, and it is increasing when $x > 1$ or $x < -1$.

111. $f(0) = -4 < 0$, and $f(1) = 2 > 0$. Hence, by the intermediate value theorem, $f(x) = 0$ for some x between 0 and 1. $f'(x) = 15x^2 - 4x + 3$. By the quadratic formula, we see that $f'(x)$ has no real roots and is, therefore, always positive. Hence, $f(x)$ is an increasing function and, thus, can take on the value 0 at most once.

112. Let $f(x) = x^3 + 2x - 5$. Since $f(0) = -5 < 0$ and $f(2) = 7 > 0$, the intermediate value theorem tells us that there is a root of $f(x) = 0$ between 0 and 2. Since $f'(x) = 3x^2 + 2 > 0$ for all x , $f(x)$ is an increasing function and, therefore, can assume the value 0 at most once. Hence, $f(x)$ assumes the value 0 exactly once.

113. (A) Let $a < x \leq b$. The mean value theorem applies to $f(x)$ on the interval $[a, x]$. Hence, there exists a c between a and x such that $f'(c) = \frac{f(x) - f(a)}{x - a}$. Since $f'(c) = 0$, $f(x) = f(a)$. Hence, $f(x)$ has the value $f(a)$ throughout the interval.

(B) Let $h(x) = f(x) - g(x)$. Then $h'(x) = 0$ for all x in $[a, b]$. By Question 113(A), there is a constant K such that $h(x) = K$ for all x in $[a, b]$. Hence, $f(x) = g(x) + K$ for all x in $[a, b]$.

Chapter 12: Higher-Order Derivatives and Implicit Differentiation

114. $y' = 3\pi x^2 - 7$, $y'' = 6\pi x$, $y''' = 6\pi$, and $y^{(n)} = 0$ for $n \geq 4$.

115. $y = (3+x)^{-1}$.

$$y' = -(3+x)^{-2} = -\frac{1}{(3+x)^2}.$$

$$y'' = 2(3+x)^{-3} = \frac{2}{(3+x)^3}.$$

$$y''' = -6(3+x)^{-4} = -\frac{6}{(3+x)^4}.$$

The general pattern is $y^{(n)} = (-1)^n (n!) (3+x)^{-(n+1)} = (-1)^n (n!) \frac{1}{(3+x)^{n+1}}$.

116. $y' = \cos x$, $y'' = -\sin x$, $y''' = -\cos x$, $y^{(4)} = \sin x$, and then the pattern of these four functions keeps on repeating.

117. By implicit differentiation, $2x + 2yy' = 0$, $y' = -x/y$. By the quotient rule,

$$y'' = -\frac{y \cdot 1 - x \cdot y'}{y^2} = -\frac{y - x(-x/y)}{y^2} = -\frac{y^2 + x^2}{y^3} = -\frac{a^2}{y^3}.$$

118. Use implicit differentiation. (*) $2x + 2(xy' + y) + 6yy' = 0$. When $y = 1$, the original equation yields $x^2 + 2x + 3 = 2$, $x^2 + 2x + 1 = 0$, $(x + 1)^2 = 0$, $x + 1 = 0$, $x = -1$.

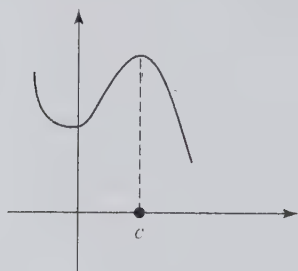
Substitute -1 for x and 1 for y in (*), which results in $-2 + 2(-y' + 1) + 6y' = 0$; so, $y' = 0$ when $y = 1$. To find y'' , first simplify (*) to $x + xy' + y + 3yy' = 0$, and then differentiate implicitly to get $1 + (xy'' + y') + y' + 3(yy'' + y'y') = 0$. In this equation, substitute -1 for x , 1 for y , and 0 for y' , which results in $1 - y'' + 3y'' = 0$, $y'' = -\frac{1}{2}$.

119. Use implicit differentiation to get $2b^2x + 2a^2yy' = 0$, $y' = -(b^2/a^2)(x/y)$. Now differentiate by the quotient rule.

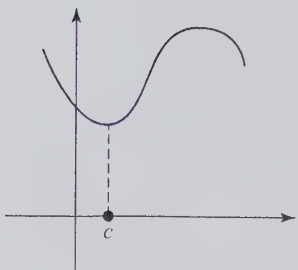
$$\begin{aligned} y'' &= -\frac{b^2}{a^2} \left(\frac{y - xy'}{y^2} \right) = -\frac{b^2}{a^2} \frac{y - x[-(b^2/a^2)(x/y)]}{y^2} \\ &= -\frac{b^2}{a^2} \frac{a^2 y^2 + b^2 x^2}{a^2 y^3} = -\frac{b^2}{a^2} \frac{a^2 y^2 + b^2 x^2}{a^2 y^3} = -\frac{b^4}{a^2 y^3}. \end{aligned}$$

Chapter 13: Maxima and Minima

120. If $f'(c) = 0$ and $f''(c) < 0$, then $f(x)$ has a relative maximum at c . (See Figure A13.1(a).) If $f'(c) = 0$ and $f''(c) > 0$, then $f(x)$ has a relative minimum at c . (See Figure A13.1(b).) If $f'(c) = 0$ and $f''(c) = 0$, we cannot draw any conclusions at all.



(a)



(b)

Figure A13.1

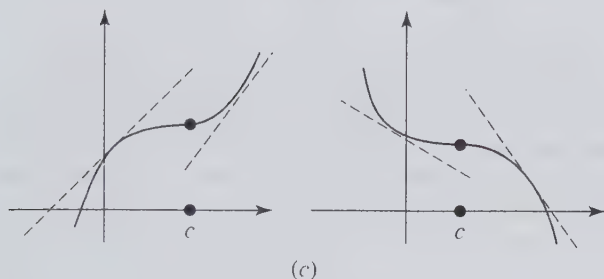
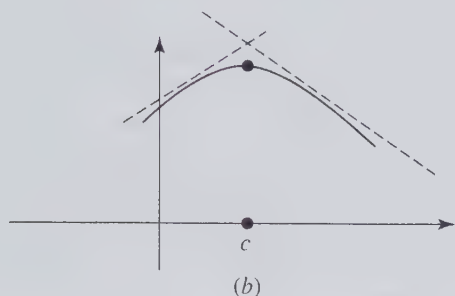
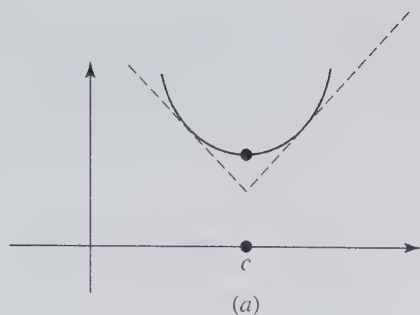


Figure A13.2

121. Assume $f'(c) = 0$. If f' is negative to the left of c and positive to the right of c —the case $\{-, +\}$ —then f has a relative minimum at c . (See Figure A13.2(a).) If f' is positive to the left of c and negative to the right of c —the case $\{+, -\}$ —then f has a relative maximum at c . (See Figure A13.2(b).) If f' has the same sign to the left and to the right of c — $\{+, +\}$ or $\{-, -\}$ —then f has an inflection point at c . (See Figure A13.2(c).)

122. (A) Recall that a critical number is a number c such that $f'(c)$ is defined and either $f'(c) = 0$ or $f'(c)$ does not exist. Now $f'(x) = -2 + 2x$. So we set $-2 + 2x = 0$. Hence, the only critical number is $x = 1$. But $f''(x) = 2$. In particular, $f''(1) = 2 > 0$. Hence, by the second-derivative test, $f(x)$ has a relative minimum at $x = 1$.

(B) $f'(x) = 3x^2 - 10x - 8 = (3x + 2)(x - 4)$. Hence, the critical numbers are $x = 4$ and $x = -\frac{2}{3}$. Now $f''(x) = 6x - 10$. So $f''(4) = 14 > 0$, and, by the second-derivative test, there is a relative minimum at $x = 4$. Similarly, $f''\left(-\frac{2}{3}\right) = -14$; therefore, there is a relative maximum at $x = -\frac{2}{3}$.

$$(C) f'(x) = x \cdot 3(x-1)^2 + (x-1)^3 = (x-1)^2(3x+x-1) = (x-1)^2(4x-1).$$

So the critical numbers are $x = 1$ and $x = \frac{1}{4}$. Now $f''(x) = (x-1)^2 \cdot 4 + 2(x-1)$

$$(4x-1) = 2(x-1)[2(x-1) + 4x-1] = 2(x-1)(6x-3) = 6(x-1)(2x-1). \text{ Thus,}$$

$$f''\left(\frac{1}{4}\right) = 6\left(-\frac{3}{4}\right)\left(-\frac{1}{2}\right) = \frac{9}{4} > 0, \text{ and, therefore, by the second-derivative test, there is a}$$

relative minimum when $x = \frac{1}{4}$. On the other hand, $f''(1) = 6 \cdot 0 \cdot 1 = 0$, and, therefore, the

second-derivative test, is inapplicable. Let us use the first-derivative test, $f'(x) = (x-1)^2(4x-1)$. For $x \neq 1$, $(x-1)^2$ is positive. Since $4x-1$ has the value 3 when $x=1$, $4x-1 > 0$ just to the left and to the right of 1. Hence, $f'(x)$ is positive both on the left and on the right of $x=1$, and this means that we have the case $\{+, +\}$. By the first-derivative test, there is an inflection point at $x=1$.

(D) $f'(x) = \cos x - 1$. The critical numbers are the solutions of $\cos x = 1$, and these are the numbers $x = 2\pi n$ for any integer n . Now $f''(x) = -\sin x$. So $f''(2\pi n) = -\sin(2\pi n) = -0 = 0$, and, therefore, the second-derivative test is inapplicable. Let us use the first-derivative test. Immediately to the left and right of $x = 2\pi n$, $\cos x < 1$, and, therefore, $f'(x) = \cos x - 1 < 0$. Hence, the case $\{-, -\}$ holds, and there is an inflection point at $x = 2\pi n$.

$$123. f'(x) = \frac{2}{3}(x-1)^{-1/3} = \frac{2}{3}[1/(x-1)^{1/3}]. \text{ There are no values of } x \text{ for which } f'(x) =$$

0, but $x = 1$ is a critical number, since $f'(1)$ is not defined. Try the first-derivative test

(which is also applicable when $f'(c)$ is not defined). To the left of $x = 1$, $(x-1)$ is negative, and, therefore, $f'(x)$ is negative. To the right of $x = 1$, $(x-1)$ is positive, and, therefore, $f'(x)$ is positive. Thus, the case $\{-, +\}$ holds, and there is a relative minimum at $x = 1$.

124. (A) Find all the critical numbers of $f(x)$ in $[a, b]$. List all these critical numbers, $c_1, c_2, \dots$, and add the endpoints a and b to the list. Calculate $f(x)$ for each x in the list. The largest value thus obtained is the maximum value of $f(x)$ on $[a, b]$, and the minimal value thus obtained is the minimal value of $f(x)$ on $[a, b]$.

x	a	b	c_1	c_2	$\dots$	c_n
$f(x)$	$f(a)$	$f(b)$	$f(c_1)$	$f(c_2)$	$\dots$	$f(c_n)$

(B) $f'(x) = 8x - 7$. Solving $8x - 7 = 0$, we find the critical $x = \frac{7}{8}$, which lies in the interval. So we list $\frac{7}{8}$ and the endpoints -2 and 3 in a table and calculate the corresponding values $f(x)$. The absolute maximum 33 is assumed at $x = -2$. The absolute minimum $-\frac{1}{16}$ is assumed at $x = \frac{7}{8}$.

x	-2	3	$\frac{7}{8}$
$f(x)$	33	18	$-\frac{1}{16}$

(C) $f'(x) = 12x^2 - 16x = 4x(3x - 4)$. So the critical numbers are $x = 0$ and $x = \frac{4}{3}$. But $x = \frac{4}{3}$ does not lie in the interval. Hence, we list only 0 and the endpoints -1 and 1 and calculate the corresponding values of $f(x)$. So the absolute maximum 1 is achieved at $x = 0$, and the absolute minimum -11 is achieved at $x = -1$.

x	-1	1	0
$f(x)$	-11	-3	1

(D) $f'(x) = 4x^3 - 6x^2 - 2x - 4 = 2(2x^3 - 3x^2 - x - 2)$. We first search for roots of $2x^3 - 3x^2 - x - 2$ by trying integral factors of the constant term 2 . It turns out that $x = 2$ is a root. Dividing $2x^3 - 3x^2 - x - 2$ by $x - 2$, we obtain the quotient $2x^2 + x + 1$. By the quadratic formula, the roots of the latter are $x = (-1 \pm \sqrt{-7})/4$, which are not real. Thus, the only critical number is $x = 2$. So, listing 2 and the endpoints 0 and 4 , we calculate the corresponding values of $f(x)$. Thus, the absolute maximum 99 is attained at $x = 4$, and the absolute minimum -9 at $x = 2$.

x	0	4	2
$f(x)$	3	99	-9

$$(E) f'(x) = \frac{(x+2)(3x^2) - x^3}{(x+2)^2} = \frac{3x^3 + 6x^2 - x^3}{(x+2)^2} = \frac{2x^3 + 6x^2}{(x+2)^2} = \frac{2x^2(x+3)}{(x+2)^2}.$$

Thus, the critical numbers are $x = 0$ and $x = -3$. However, $x = -3$ is not in the given interval. So we list 0 and the endpoints -1 and 1 . The absolute maximum $\frac{1}{3}$ is assumed at $x = 1$, and the absolute minimum -1 is assumed at $x = -1$.

x	0	-1	1
$f(x)$	0	-1	$\frac{1}{3}$

$$\begin{aligned} \text{(E)} \quad x &= 4 \sin \frac{2\pi}{3} \cos \frac{\pi}{3} = 4 \left(\frac{\sqrt{3}}{2} \right) \left(\frac{1}{2} \right) = \sqrt{3}, \quad y = 4 \sin \frac{2\pi}{3} \sin \frac{\pi}{3} = 4 \left(\frac{\sqrt{3}}{2} \right) \left(\frac{\sqrt{3}}{2} \right) = 3, \\ z &= 4 \cos \frac{2\pi}{3} = 4 \left(\frac{1}{2} \right) = 2. \end{aligned}$$

421. $x^2 + y^2 + z^2 = \rho^2$ and $z = \rho \cos \phi$. Thus, we get $\rho^2 + 6\rho \cos \phi = 0$. Then $\rho = 0$ or $\rho + 6 \cos \phi = 0$. Since the origin is the only solution of $\rho = 0$, and the origin also lies on $\rho + 6 \cos \phi = 0$, $\rho + 6 \cos \phi = 0$ is the desired equation.

422. Since $r = \rho \sin \phi = 3$, this is the right circular cylinder with radius 3 and the z -axis as axis of symmetry.

Chapter 42: Partial Derivatives

423. First, consider y constant. Then, differentiating with respect to x , we obtain $f_x = 12x^2 - 6y^2x + 2$. If we keep x fixed and differentiate with respect to y , we get $f_y = -6x^2y + 3$.

$$424. \text{ Let } f(x, y) = \begin{cases} \frac{xy}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}.$$

$$f_x(0, 0) = \lim_{\Delta x \rightarrow 0} \frac{f(\Delta x, 0) - f(0, 0)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{0 - 0}{\Delta x} = 0.$$

$$f_y(0, 0) = \lim_{\Delta y \rightarrow 0} \frac{f(0, \Delta y) - f(0, 0)}{\Delta y} = \lim_{\Delta y \rightarrow 0} \frac{0 - 0}{\Delta y} = 0.$$

By Question 418(D), $f(x, y)$ is discontinuous at the origin.

425. (A) If (x_0, y_0) is inside an open disk throughout which f_{xy} and f_{yx} exist, and if f_{xy} and f_{yx} are continuous at (x_0, y_0) , then $f_{xy}(x_0, y_0) = f_{yx}(x_0, y_0)$. Similar conditions ensure equality for $n \geq 3$ partial differentiations, regardless of the order in which the derivatives are taken.

$$\text{(B)} \quad f_x = 6xy - 2y, f_{xy} = 6x - 2, f_y = 3x^2 - 2x + 10y, f_{yx} = 6x - 2.$$

$$426. \text{ (A)} \quad f_x = e^x \cos y, f_{xy} = -e^x \sin y, f_y = -e^x \sin y, f_{yx} = -e^x \sin y.$$

$$\text{(B)} \quad f_x = 6x - 2y, f_{xy} = -2, f_y = -2x + 15y^2, f_{yx} = -2.$$

$$\text{(C)} \quad f_x = 2x \cos y + y^2 \cos x, f_{xy} = -2x \sin y + 2y \cos x, f_y = -x^2 \sin y + 2y \sin x, f_{yx} = -2x \sin y + 2y \cos x.$$

$$427. \text{ (A)} \quad f_x = 12x^3 - 6x^2y^2, f_y = -4x^3y + 7, f_{xx} = 36x^2 - 12xy^2, f_{yy} = -4x^3,$$

$$f_{xy} = -12x^2y, f_{yx} = -12x^2y.$$

$$(B) f_x = ye^{xy}y^2 + \frac{1}{y} = e^{xy}y^3 + \frac{1}{y}, f_y = 2e^{xy}y + xe^{xy}y^2 - \frac{x}{y^2} = e^{xy}(2y + xy^2) - \frac{x}{y^2},$$

$$f_{xx} = e^{xy}y^3(y) = e^{xy}y^4, f_{yy} = e^{xy}(2 + 2xy) + xe^{xy}(2y + xy^2) + \frac{2x}{y^3}$$

$$= e^{xy}(2 + 4xy + x^2y^2) + \frac{2x}{y^3},$$

$$f_{xy} = 3e^{xy}y^2 + xe^{xy}y^3 - \frac{1}{y^2}, f_{yx} = e^{xy}(y^2) + ye^{xy}(2y + xy^2) - \frac{1}{y^2}$$

$$= 3e^{xy}y^2 + xe^{xy}y^3 - \frac{1}{y^2}.$$

$$(C) f_x = 2xy - 2z, f_y = x^2 + 2yz, f_z = y^2 - 2x, f_{xy} = 2x, f_{yx} = 2x, f_{xz} = -2, f_{zx} = -2, f_{yz} = 2y, f_{zy} = 2y.$$

$$428. \text{ Let } f(x, y) = \begin{cases} xy \left(\frac{x^2 - y^2}{x^2 + y^2} \right) & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}.$$

$$\text{Then } f_x(0, y) = \lim_{\Delta x \rightarrow 0} \frac{f(\Delta x, y) - f(0, y)}{\Delta x} = \lim_{\Delta x \rightarrow 0} y \frac{(\Delta x)^2 - y^2}{(\Delta x)^2 + y^2} = -y$$

$$\text{and } f_y(x, 0) = \lim_{\Delta y \rightarrow 0} \frac{f(x, \Delta y) - f(x, 0)}{\Delta y} = \lim_{\Delta y \rightarrow 0} x \frac{x^2 - (\Delta y)^2}{x^2 + (\Delta y)^2} = x.$$

$$\text{Consequently, } f_{xy}(0, 0) = \lim_{\Delta y \rightarrow 0} \frac{f_x(0, \Delta y) - f_x(0, 0)}{\Delta y} = \lim_{\Delta y \rightarrow 0} \frac{-\Delta y - 0}{\Delta y} = -1$$

$$\text{and } f_{yx}(0, 0) = \lim_{\Delta x \rightarrow 0} \frac{f_y(\Delta x, 0) - f_y(0, 0)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{\Delta x - 0}{\Delta x} = 1.$$

Thus, $f_{xy}(0, 0) \neq f_{yx}(0, 0)$. (The conditions of Question 425(A) are not met by this function.)

$$429. \frac{\partial z}{\partial x} = ae^{ax} \sin ay, \frac{\partial^2 z}{\partial x^2} = a^2 e^{ax} \sin ay, \frac{\partial z}{\partial y} = ae^{ax} \cos ay, \frac{\partial^2 z}{\partial y^2} = -a^2 e^{ax} \sin ay.$$

$$430. \partial z / \partial x = e^{-t} \cos x, \partial^2 z / \partial x^2 = -e^{-t} \sin x, \partial z / \partial y = -e^{-t} \sin y, \partial^2 z / \partial y^2 = -e^{-t} \cos y, \\ \partial z / \partial t = -e^{-t} (\sin x + \cos y) = -e^{-t} \sin x - e^{-t} \cos y = \partial^2 z / \partial x^2 + \partial^2 z / \partial y^2.$$

$$431. f_x = g'(x)h(y), f_{xy} = g'(x)h'(y), f_y = g(x)h'(y), f_{yx} = g'(x)h'(y).$$

$$432. \text{ (A) } f_x = 3(x+at)^2, \quad f_{xx} = 6(x+at), \quad f_t = 3(x+at)^2(a) = 3a(x+at)^2, \\ f_{tt} = 6a(x+at)(a) = 6a^2(x+at) = a^2 f_{xx}.$$

$$\text{ (B) } f_x = \cos(x+at), \quad f_{xx} = -\sin(x+at), \quad f_t = a \cos(x+at), \\ f_{tt} = -a^2 \sin(x+at) = a^2 f_{xx}.$$

$$\text{ (C) } f_x = e^{x-at}, \quad f_{xx} = e^{x-at}, \quad f_t = -ae^{x-at}, \quad f_{tt} = a^2 e^{x-at} = a^2 f_{xx}.$$

$$433. \text{ (A) } \frac{du}{dt} = f_{x_1} \frac{dx_1}{dt} + \cdots + f_{x_n} \frac{dx_n}{dt}.$$

(B) By the formula of Question 433(A),

$$\frac{du}{dt} = \frac{\partial u}{\partial x} \frac{dx}{dt} + \frac{\partial u}{\partial y} \frac{dy}{dt} + \frac{\partial u}{\partial z} \frac{dz}{dt} = 2x \cos t - 4y e^t + 3z^2(3) \\ = 2 \sin t \cos t - 4e^t(e^t) + 9(3t)^2 = \sin 2t - 4e^{2t} + 81t^2.$$

$$\text{ (C) } \frac{\partial w}{\partial u} = \frac{\partial \Phi}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial \Phi}{\partial y} \frac{\partial y}{\partial u} + \frac{\partial \Phi}{\partial z} \frac{\partial z}{\partial u}; \quad \frac{\partial w}{\partial v} = \frac{\partial \Phi}{\partial x} \frac{\partial x}{\partial v} + \frac{\partial \Phi}{\partial y} \frac{\partial y}{\partial v} + \frac{\partial \Phi}{\partial z} \frac{\partial z}{\partial v}.$$

$$434. \quad V = lwh. \quad \frac{dV}{dt} = \frac{\partial V}{\partial l} \frac{dl}{dt} + \frac{\partial V}{\partial w} \frac{dw}{dt} + \frac{\partial V}{\partial h} \frac{dh}{dt} = wh(2) + lh(-1) + lw(2) \\ = 5(3)(2) + 10(3)(-1) + 10(5)(2) = 30 - 30 + 100 = 100 \text{ ft}^3/\text{s}.$$

435. (A) Differentiate $f(tx, ty) = t^2 f(x, y)$ with respect to t . By the chain rule,

$$f_x(tx, ty) \frac{\partial(tx)}{\partial t} + f_y(tx, ty) \frac{\partial(ty)}{\partial t} = nt^{n-1} f(x, y). \text{ Hence, } f_x(tx, ty)(x) + f_y(tx, ty)(y) =$$

$nt^{n-1} f(x, y)$. Let $t = 1$. Then $xf_x(x, y) + yf_y(x, y) = nf(x, y)$. (A similar result holds for functions f of more than two variables.)

(B) $f(x, y)$ is homogeneous of degree 3, since $f(tx, ty) = (tx)(ty)^2 + (tx)^2(ty) - (ty)^3 = t^3(xy^2 + x^2y - y^3) = t^3 f(x, y)$. So we must show that $xf_x + yf_y = 3f$. $f_x = y^2 + 2xy$, $f_y = 2xy + x^2 - 3y^2$. Hence, $xf_x + yf_y = x(y^2 + 2xy) + y(2xy + x^2 - 3y^2) = xy^2 + 2x^2y + 2xy^2 + x^2y - 3y^3 = 3(xy^2 + x^2y - y^3) = 3f(x, y)$.

(C) $f(x, y)$ is homogeneous of degree 1, since $f(tx, ty) = \sqrt{(tx)^2 + (ty)^2} = t\sqrt{x^2 + y^2} = tf(x, y)$ for $t > 0$. We must check that $xf_x + yf_y = \sqrt{x^2 + y^2}$. But $f_x = x/\sqrt{x^2 + y^2}$ and $f_y = y/\sqrt{x^2 + y^2}$. Thus, $xf_x + yf_y = (x^2/\sqrt{x^2 + y^2}) + (y^2/\sqrt{x^2 + y^2}) = (x^2 + y^2)/\sqrt{x^2 + y^2} = \sqrt{x^2 + y^2}$.